

INTEGRATED DESIGN AND DEPLOYMENT OF SOLAR PV SYSTEM FOR INFRASTRUCTURE APPLICATIONS

**A Project Report Submitted in partial fulfilment of the requirements
for the award of the degree**

Of

BACHELOR OF TECHNOLOGY

IN

ELECTRICAL AND ELECTRONICS ENGINEERING

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**DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING
SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)**

Accredited by NBA and by NAAC with 'A+' Grade | Approved by AICTE, New Delhi

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RVS Knowledge District, Chittoor-517127, AP

2022-2026



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Certificate

This is to certify that this Project report entitled " **Integrated Design and Deployment of Solar PV System for Infrastructure Applications**" is a bonafide work done by the following students under my guidance and supervision, in partial fulfillment of the requirements for the award of the degree of Bachelor in Electrical and Electronics Engineering in Sri Venkateswara College of Engineering and Technology(A), Chittoor during the academic year 2025 – 2026.

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ABSTRACT

The increasing demand for electrical energy and the environmental impacts associated with conventional fossil-fuel-based power generation have necessitated the adoption of renewable energy technologies. Among the available renewable resources, solar photovoltaic (PV) systems have emerged as a viable and sustainable solution, particularly for urban infrastructure applications in regions with high solar potential such as India. Tamil Nadu, especially Chennai, receives significant annual solar irradiation, making it suitable for large-scale solar PV deployment.

This project focuses on the integrated design and simulation of a grid-Connected solar photovoltaic system for a residential township located in Chennai, Tamil Nadu. The township consists of multiple residential towers with defined load requirements, wherein 20% of the daily energy demand is proposed to be met through solar PV generation. Detailed manual design calculations are carried out for PV array sizing, inverter selection, and battery sizing for an equivalent standalone configuration. The electrical characteristics of the PV array, including I–V and P–V curves, are modeled using MATLAB Simscape.

The performance of the proposed system is simulated and validated using industry-standard software tools such as PVsyst. A comparative analysis between grid-Connected and standalone PV system configurations is performed in terms of energy yield, system losses, and performance ratio. The impact of partial shading caused by nearby structures and vegetation is also analyzed. Furthermore, AI/ML-based techniques are employed to forecast daily solar energy generation, enhancing the reliability and operational planning of the PV system. The results demonstrate the technical feasibility and effectiveness of grid-Connected solar PV systems for urban infrastructure applications.

Chapter I

INTRODUCTION

1.1 Introduction

1.1.1 Necessity of Renewable Energy

The improvement of human living standards, industrial growth, and economic expansion all depend on energy. Global electricity demand has significantly increased as a result of urbanization, industrialization, and the rapid expansion of technological infrastructure. Coal, oil, and natural gas are examples of fossil fuels that have traditionally played a significant role in power generation. Despite the fact that these conventional energy sources have supported industrial progress for decades, their extensive use has resulted in serious environmental issues like air pollution, climate change, and the emission of greenhouse gases.

In addition, fossil fuel resources are finite and their continuous exploitation raises concerns regarding long-term energy security. These issues have encouraged governments, researchers, and policymakers worldwide to explore sustainable energy alternatives. Renewable energy sources such as solar, wind, hydro, and biomass provide environmentally friendly solutions capable of reducing carbon emissions while ensuring long-term energy availability.

Due to its abundance, wide geographic availability, and scalability, solar energy has emerged as one of the most promising renewable energy solutions. With the development of photovoltaic technologies and power electronics, solar power technologies have become increasingly appealing because they enable the generation of clean electricity without emitting harmful emissions.

1.1.2 Importance of Solar Photovoltaic Systems

Using semiconductors, solar photovoltaic (PV) technology converts sunlight directly into electricity. Unlike conventional power plants, PV systems do not involve fuel combustion or moving mechanical parts, making them environmentally sustainable and operationally reliable.

PV systems can be installed in a variety of configurations such as rooftop installations, ground-mounted solar farms, and integrated building systems. Their modular design allows flexible expansion depending on energy demand. In recent years, improvements in photovoltaic module technologies, inverter performance, and power electronics have significantly enhanced overall system efficiency.

Several studies have demonstrated that PV systems can effectively reduce grid dependency and operational electricity costs while contributing to environmental sustainability [2], [4], [10]. Research also indicates that careful system design and performance evaluation using simulation tools can optimize energy yield and minimize system losses [5], [6]. Additionally, modern photovoltaic technologies such as mono-crystalline, poly-crystalline, and silicon hetero-junction modules provide improved efficiency under varying climatic conditions [1].

1.1.3 Solar Energy Potential in India

India receives a lot of solar radiation throughout the year because it is in the tropical region. The country's average solar insolation is between 4 and 7 kWh/m²/day, making it ideal for installing solar power on a large scale.

Various regions in India, particularly southern states such as Tamil Nadu, Andhra Pradesh, and Karnataka, exhibit strong solar potential due to favorable climatic conditions and high solar irradiance levels. Solar resource assessment studies using PV simulation tools show that both rooftop and ground-mounted PV installations can generate a lot of energy each year [5].

In recent years, government initiatives such as the National Solar Mission, renewable energy incentives, and net-metering policies have accelerated solar PV adoption across the country. Consequently, India has emerged as one of the fastest-growing solar energy markets globally.

1.2 Solar Photovoltaic Systems

Solar PV systems can be broadly categorized based on their operational configuration and grid-Connection.

1.2.1 Grid-Connected PV Systems

Grid-Connected photovoltaic systems are directly linked to the utility grid. These systems generate electricity during daylight hours and supply excess power to the grid when production exceeds the local load demand. During nighttime or low-irradiance conditions, electricity is drawn from the grid to meet energy requirements.

The major advantages of grid-Connected systems include:

- Elimination of large battery storage systems
- Lower installation and maintenance costs
- Higher system efficiency
- Simplified system architecture

Several researchers have analyzed the design optimization and techno-economic feasibility of grid-Connected rooftop PV systems using simulation tools such as PVsyst [2], [10], [13]. Important performance indicators including Performance Ratio (PR), Capacity Utilization Factor (CUF), and annual energy generation are commonly used to evaluate system efficiency [6].

Furthermore, integration strategies combining solar PV with conventional grid infrastructure have been explored to improve grid stability and manage intermittency issues [9].

1.2.2 Standalone PV Systems

Standalone photovoltaic systems operate independently from the utility grid and are typically used in remote areas where grid access is unavailable or unreliable. These systems require battery storage to supply electricity during non-sunshine hours.

Designing a standalone PV system requires careful consideration of several factors, including:

- Load demand assessment
- Calculating battery capacity
- Limits on discharge depth
- Autonomy period requirements

Research conducted by Al-Sarraj and Yigit [1] investigated standalone PV systems utilizing silicon hetero-junction (HJT) modules, demonstrating improved efficiency under high-temperature climatic conditions. Similarly, studies on battery-integrated PV farm configurations show that energy storage systems improve overall energy utilization efficiency and system reliability [8], [12].

Although standalone PV systems provide energy independence, they involve higher initial costs due to battery installation and periodic replacement requirements.

1.2.3 Comparative Overview

Comparative analysis between grid-Connected and standalone PV systems highlights distinct operational and economic characteristics.

Grid-Connected systems generally offer:

- Higher overall efficiency
- Lower lifecycle cost
- Less complicated maintenance

On the other hand, standalone systems provide:

- Complete energy independence
- Applicability to areas that are far away or isolated
- Protection against grid outages and instability

Grid-Connected PV systems are more financially viable in urban environments, according to economic studies [1, 2], whereas standalone systems are preferable in remote locations without dependable grid infrastructure.

1.3 Need for Simulation and Forecasting

1.3.1 Importance of Performance Evaluation

Numerous technical and environmental factors influence a photovoltaic system's performance. These include:

- The intensity of the sun's rays • Ambient and module temperature

- Shaded environments • Inverter efficiency
- System losses and wiring inefficiencies

Accurate evaluation of these parameters is essential for ensuring system reliability, financial feasibility, and long-term sustainability of solar energy projects.

1.3.2 Role of Simulation Tools such as PVsyst

PVsyst is one of the most widely used software platforms for photovoltaic system design and energy yield estimation. It enables detailed modelling of system components, meteorological conditions, shading effects, and electrical losses.

Several studies have confirmed the effectiveness of PVsyst for predicting annual energy production and system losses with high accuracy [3], [5], [6], [10]. Advanced modelling techniques that incorporate temperature and irradiance-dependent parameters further improve simulation accuracy [7].

Validation studies comparing simulated outputs with real solar plant data indicate that PVsyst provides reliable performance estimates and is widely used for feasibility analysis and system design.

1.3.3 Role of MATLAB and Simscape Modeling

MATLAB and Simscape provide powerful modelling environments for analysing the electrical characteristics of photovoltaic systems. These platforms enable detailed mathematical modelling of PV modules based on semiconductor physics.

Through MATLAB-based simulations, researchers can generate I–V and P–V characteristics of PV modules under varying irradiance and temperature conditions. This kind of modeling aids in:

- Understanding nonlinear PV behavior
- Investigating the effects of partial shading
- Validating PV system performance

Studies have shown that MATLAB-based modelling significantly improves the accuracy of PV performance analysis and enables deeper investigation of shading-related losses in PV arrays [11], [14].

1.4 Literature Survey

[1] Ahmad Imad Al-Sarraj and Fatih Yigit, “Modelling the Use of PVsyst Software for a Stand-Alone PV Solar System ‘Off-Grid’ with Batteries by Utilizing Silicon Hetero-Junction Technology (HJT) Panels in Iraq/Basra,” 2023 Middle East and North Africa Solar Conference (MENA-SC), 2023.

This paper presents the modelling of a stand-alone solar photovoltaic system using silicon hetero-junction panels under high-temperature climatic conditions. The study highlights that advanced PV module technology combined with simulation tools improves system efficiency and contributes to reduction in carbon emissions.

[2] Z. Serat, S. A. Z. Fatemi and S. Shirzad, "Design and Economic Analysis of On-Grid Solar Rooftop PV System Using PVsyst Software," 2023 International Conference, 2023.

This research focuses on the design and economic analysis of a grid-Connected rooftop solar PV system using PVsyst. The results demonstrate significant annual energy generation with a favourable payback period and return on investment.

[3] Y. Siregar, Y. "Optimization Design and Simulating Solar PV System Using PVsyst Software," by Hutahuruk and Suherman, International Conference, 2020.

The study investigates optimization of solar PV system placement and configuration through simulation analysis. It shows that proper module orientation and layout planning can enhance annual energy yield and overall system performance.

[4] S. Sharma, C. P. L. Kurian and P. IEEE Conference Publication, 2018: Paragond, "Solar PV System Design Using PVsyst: A Case Study of an Academic Institute."

The design and feasibility of institutional building photovoltaic systems are the subject of this paper. The research evaluates system performance under varying load conditions and tilt angles to achieve improved carbon footprint reduction.

[5] C. P. P. Kandasamy K. Prabu and Niruba, "Solar Potential Assessment Using PVsyst Software," 2013 International Conference on Green Computing, Communication and Conservation of Energy (ICGCE), Chennai, India, 2013, pp. 667–672.

The research presents solar resource assessment and performance evaluation of a large-scale grid-Connected PV system using simulation software. The findings indicate that analysing radiation levels and system losses is essential for determining project feasibility.

[6] P. N. Yadav. Kumar and S. S. Chandel, "Simulation and Performance Analysis of a 1 kWp Photovoltaic System Using PVsyst," IEEE Conference, 2015.

By examining the availability of solar radiation and the performance ratio, this study examines the simulation and performance characteristics of a small-scale photovoltaic system. The results confirm the viability of solar PV systems for distributed energy applications.

[7] K. J. T. Sauer. C. Roessler and W. Hansen, IEEE Journal of Photovoltaics, "Modeling the Irradiance and Temperature Dependence of Photovoltaic Modules in PVsyst," 2015.

The paper develops mathematical modelling techniques to analyse the influence of irradiance and temperature on photovoltaic module output. Improved model accuracy enables better prediction of energy yield and supports financial planning of solar projects.

[8] O. M. Akeyo, V. N. Rallabandi Jewell and D. M. Ionel, "The Design and Analysis of Large Solar PV Farm Configurations with DC Connected Battery Systems," IEEE Conference, 2020.

This research proposes a configuration of large solar PV farms integrated with DC-Connected battery storage systems. The results demonstrate improved utilization of clipped energy and enhancement of overall plant capacity.

[9] A. Al Omari presented at the 2017 IEEE Conference on "Integrating Solar PV with the Electricity Grid Through Conventional Power Plants."

In order to supply internal loads effectively, the study investigates the integration of photovoltaic systems with conventional power plants. Simulation results show attractive economic returns and smooth grid interaction without major technical constraints.

[10] B. K. Nallamothu, R. S. and Janga R. Pendem, "Design and Analysis of Grid-Connected Solar PV System Using PVsyst Software," presented at the 2024 IEEE Students Conference on Engineering and Systems (SCES) in Prayagraj, India, in 2024.

This paper presents the design and performance analysis of a grid-Connected solar photovoltaic system using PVsyst simulation software. In order to determine the feasibility of a system, the research looks at the availability of solar radiation, system losses, and overall energy production.

[11] A. P. Behera and Mohanty, "A Comparative Study on Grid-Tied PV System Using PVsyst Software & MATLAB," 2024 IEEE Third International Conference on Power Electronics, Intelligent Control and Energy Systems (ICPEICES), Delhi, India, 2024, pp. 526–531.

This study presents a comparative performance analysis of grid-Connected solar photovoltaic systems using PVsyst and MATLAB simulation platforms. The findings demonstrate that system performance evaluation accuracy is enhanced by integrating multiple simulation tools.

[12] M. H. Radhi, E. J. Mahdi and A. K. Mftwol, "Design and Performance Analysis of Solar PV System Size 2.56 kWp," 2019 4th Scientific International Conference Najaf (SICN), Iraq, 2019, pp. 70–73.

Solar radiation, the effects of temperature, the operation of the inverter, and the integration of the batteries are all examined in this study to present the design and performance evaluation of a 2.56 kWp solar photovoltaic system. The research shows that proper system design improves energy yield and overall efficiency.

[13] B. P. Mohanty and M. Srivalli, "Optimization and Design of Grid Connected Rooftop Solar Power Plant under Various Operating Conditions," 2020 International Conference on Computational Intelligence for Smart Power System and Sustainable Energy (CISPSSE), India, 2020.

The design and optimization of a rooftop solar photovoltaic power plant that is Connected to the grid and operates under a variety of conditions is the primary focus of this research. The research shows that loss minimization and computational modelling improve energy yield and operational efficiency.

[14] H. Patel and V. Agarwal, "MATLAB-Based Modeling to Study the Effects of Partial Shading on PV Array Characteristics," IEEE Transactions on Energy Conversion, vol. 23, no. 1, pp. 302–310, Mar. 2008.

This study analyses the impact of partial shading on photovoltaic array performance using MATLAB-based modelling. The results show that shading significantly reduces output power and causes multiple maximum power points, making proper array design important.

[15] Y. Miaji and M. The Middle East and North Africa Solar Conference (MENA-SC), Dubai, UAE, 2023, Miaji, "The Sunny Intelligence: A Novel Proposal for Combining the AI with Solar Power for Excellent Output Power Forecasting."

This paper proposes an innovative approach to integrate Artificial Intelligence techniques with solar photovoltaic systems for improving output power forecasting accuracy. The study discusses predictive modelling methods to enhance solar energy utilization and operational efficiency.

[16] E. S. Brunette, R. C. Flemmer and C. L. Flemmer, "A Review of Artificial Intelligence," 4th International Conference on Autonomous Robots and Agents (ICARA), Wellington, New Zealand, 2009, pp. 385–392.

The concepts of artificial intelligence, such as intelligent agents, machine learning, neural processing, and decision-making logic, are reviewed in this paper. It highlights the potential of AI in complex engineering systems for automation and predictive tasks.

1.5 Research Gap

Although extensive research exists on photovoltaic system design, simulation modelling, and battery integration, most studies focus on individual aspects of PV system development. Limited research has addressed a comprehensive framework integrating:

- PV system sizing by hand • MATLAB-based photovoltaic modelling
- PVsyst simulation validation
- Comparative analysis of grid-Connected and standalone systems
- Partial shading analysis
- Artificial intelligence and machine learning-based energy forecasting

For extensive residential infrastructure in the Indian climate. This study aims to address this gap by developing an integrated design, simulation, validation, and forecasting approach for solar PV systems.

1.6 Objectives of the Study

The primary objectives of this research are:

1. To create a grid-Connected solar PV system that is technically optimized for a Chennai residential township.
2. To perform detailed manual sizing calculations for PV system components.
3. To use MATLAB/Simscape to model the characteristics of photovoltaic arrays.
4. To use pVsyst simulation to verify the performance of the system.
5. To compare grid-Connected and standalone PV system configurations.
6. To investigate how partial shading affects the performance of PV systems.
7. To implement Artificial Intelligence and Machine Learning techniques for solar energy forecasting.

Chapter II

METHODOLOGY AND DESIGN CALCULATIONS

2.1 Project Description

A newly constructed residential township located in Chennai; Tamil Nadu is considered for the proposed solar PV system design. The township consists of 10 residential towers, each comprising 30 flats, resulting in a total of 300 flats. Each flat has a Connected load of 5 kW with an average daily usage of 6 hours.

The objective of the project is to design a solar PV system capable of supplying 20% of the total daily energy demand of the township. The same PV array design is considered for both:

- Grid-Connected solar PV system
- Standalone solar PV system with battery storage

The standalone system additionally includes a battery bank to supply power during non-sunlight hours.

2.2 Assumptions and Design Data

The following assumptions and system specifications are considered for the design:

PV Module Specifications (at STC)

- Maximum Power, P_{max} = **650 W**
- Voltage at Maximum Power, V_{mp} = **37.7 V**
- Current at Maximum Power, I_{mp} = **17.27 A**
- Open Circuit Voltage, V_{oc} = **45.5 V**
- Short Circuit Current, I_{sc} = **18.35 A**
- Module Efficiency = **20.9%**

Battery Specifications (for Standalone System)

- Battery Voltage = **12 V**
- Battery Efficiency = **85%**
- Depth of Discharge (DoD) = **50%**

Other Assumptions

- Average Peak Sun Hours (PSH) for Chennai = **5.5 hours/day**
- Overall system losses (cables, inverter, temperature, dust, mismatch) = **20%**
- Number of autonomy days (standalone system) = **2 days**

2.3 Load Calculation

2.3.1 Energy Consumption per Flat

Connected load = 5 kW

Average usage = 6 hours/day

Daily energy consumption per flat:

$$E_{flat} = 5 \times 6$$

$$E_{flat} = 30 \text{ kWh/day}$$

2.3.2 Energy Consumption per Tower

Each tower contains 30 flats.

$$E_{tower} = 30 \times 30$$

$$E_{tower} = 900 \text{ kWh/day}$$

2.3.3 Total Energy Consumption of Township

Total towers = 10

$$E_{total} = 10 \times 90$$

$$E_{total} = 9000 \text{ kWh/day}$$

2.3.4 Required Solar Contribution

The PV system is designed to supply 20% of total energy demand.

$$E_{solar} = 0.2 \times 9000$$

$$E_{solar} = 1800 \text{ kWh/day}$$

Thus, the solar PV system must generate 1800 kWh/day.

2.4 Solar PV System Capacity Design

2.4.1 Required PV Capacity

PV capacity required without losses:

$$P_{required} = \frac{E_{solar}}{PSH}$$

$$P_{required} = \frac{1800}{5.5} = 327.27 \text{ kW}$$

Considering 20% system losses:

$$P_{design} = \frac{327.27}{0.8}$$

$$P_{design} = 409.1 \text{ kW}$$

Therefore, the selected PV plant capacity ≈ 410 kW.

2.5 PV Array Size calculations

2.5.1 Number of PV Modules

$$N_{modules} = \frac{P_{plant}}{P_{module}}$$

$$N_{modules} = \frac{410000}{650}$$

$$N_{modules} \approx 631$$

Thus, 630 modules are selected for practical configuration.

2.6 Series–Parallel Configuration of PV Array

2.6.1 Number of Modules in Series

Desired string voltage ≈ 700 V

$$N_s = \frac{700}{37.7}$$

$$N_s \approx 18$$

Therefore, 18 modules in series per string

2.6.2 String Voltage

Operating voltage of string:

$$V_{string} = 18 \times 37.7$$

$$V_{string} = 678.6 \text{ V}$$

Open circuit voltage:

$$V_{oc,string} = 18 \times 45.5$$

$$V_{oc,string} = 819 \text{ V}$$

2.6.3 Number of Parallel Strings

$$N_p = \frac{630}{18}$$

$$N_p \approx 35$$

2.6.4 Total Array Short Circuit Current

$$I_{array} = N_p \times I_{sc}$$

$$I_{array} = 35 \times 18.35$$

$$I_{array} = 642.25 \text{ A}$$

Final PV Array Configuration

18 modules in series $\times$ 35 parallel strings

Total modules = **630**

Total plant capacity $\approx$ **409.5 kW**

2.7 Inverter Size calculations (Grid-Connected System)

Typical DC/AC oversizing ratio = 1.15

$$P_{inverter} = \frac{P_{PV}}{1.15}$$

$$P_{inverter} = \frac{410}{1.15}$$

$$P_{inverter} \approx 356 \text{ kW}$$

Selected inverter capacity: 350–360 kW grid-Connected inverter

2.8 Battery Bank Size calculation (Standalone System Only)

2.8.1 Energy Required for Autonomy

Autonomy period = **2 days**

$$E_{autonomy} = 1800 \times 2$$

$$E_{autonomy} = 3600 \text{ kWh}$$

2.8.2 Corrected Battery Energy

Considering **Depth of Discharge and battery efficiency**

$$E_{battery} = \frac{3600}{0.5 \times 0.85}$$

$$E_{battery} = 8470.6 \text{ kWh}$$

2.8.3 Battery Bank Capacity

Assume battery bank voltage = **240 V**

$$Q_{Ah} = \frac{8,470,600}{240}$$

$$Q_{Ah} = 35,294 \text{ Ah}$$

2.8.4 Number of Batteries

Using **12 V, 200 Ah batteries**

Batteries in series

$$N_s = \frac{240}{12}$$

$$N_s = 20$$

Parallel strings

$$N_p = \frac{35,294}{200}$$

$$N_p \approx 177$$

Total Battery Requirement

$$N_{total} = 20 \times 177$$

$$N_{total} = 3540 \text{ batteries}$$

Table: 2.1 Final Design Summary of Grid-Connected and Standalone Solar PV Systems

Parameter	Grid-Connected System	Standalone System
Total Township Load	9000 kWh/day	9000 kWh/day
Solar Energy Contribution	1800 kWh/day (20%)	1800 kWh/day (20%)
Peak Sun Hours	5.5 hours	5.5 hours
System Losses	20 %	20 %
PV Plant Capacity	≈ 410 kW	≈ 410 kW
Total PV Modules	630	630
PV Array Configuration	18 Series $\times$ 35 Parallel	18 Series $\times$ 35 Parallel
Inverter Rating	350–360 kW	Charge Controller + Battery Inverter
Battery Autonomy	—	2 Days
Battery Bank Voltage	—	240 V
Total Batteries Required	—	3540 (12V, 200Ah)

Chapter III

MODELING, SIMULATION AND PERFORMANCE ANALYSIS OF PV SYSTEM

3.1 Introduction

To ensure technical viability, energy dependability, and long-term operational performance, the design and implementation of a solar photovoltaic (PV) system require in-depth modelling and simulation. While preliminary system sizing can be carried out using analytical calculations, accurate evaluation of system behaviour under varying environmental conditions requires advanced simulation tools.

Solar PV systems operate under continuously changing environmental parameters such as solar irradiance, ambient temperature, and shading conditions. These factors significantly influence the electrical output characteristics of PV modules and the overall efficiency of the system. Consequently, prior to actual implementation, simulation is crucial for predicting system performance, estimating energy yield, identifying various losses, and assessing operational reliability.

In this work, both grid-Connected and standalone photovoltaic systems are modelled and simulated for a residential township located in Chennai, Tamil Nadu. The region receives high solar radiation throughout the year, making it suitable for solar power generation.

This study uses two major simulation platforms:

- MATLAB/Simulink – used for mathematical modelling of the photovoltaic cell and to analyze its electrical characteristics.
- PVsyst – used for system-level simulation including loss analysis, shading evaluation, and annual energy production estimation.

3.2 Mathematical Modelling of PV Cell

The photovoltaic cell is modelled using the single-diode equivalent circuit model, which is widely used to represent the nonlinear electrical behaviour of solar cells.

The model consists of the following components:

- Photocurrent source representing the light-generated current
- Diode representing the p-n junction characteristics
- Series resistance representing internal electrical losses
- Shunt resistance representing leakage currents

The equivalent circuit diagram of the single diode PV model is illustrated in Figure 3.1.

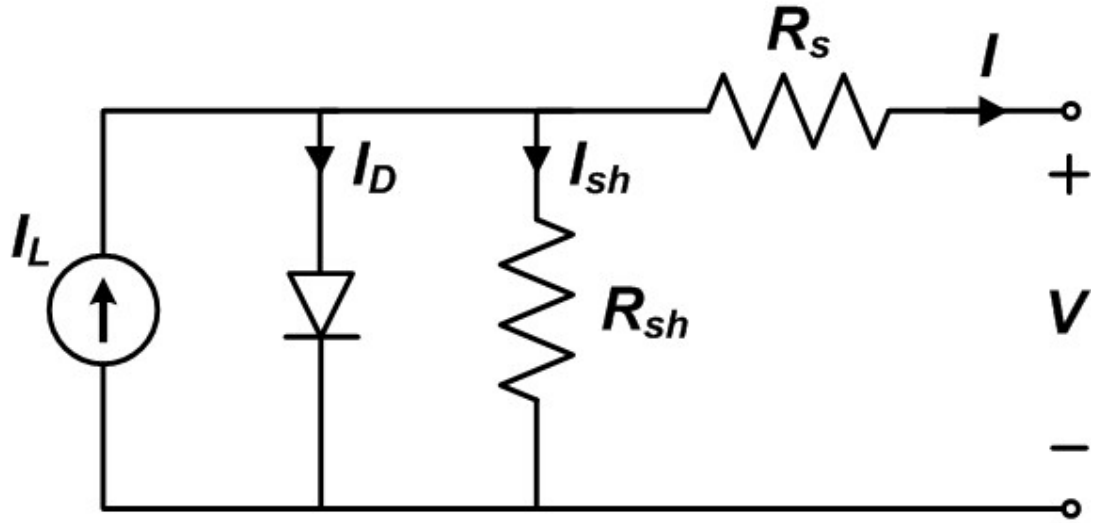


Figure: 3.1 Single Diode Equivalent Circuit of PV Cell

The output current of the photovoltaic cell can be expressed as:

$$I = I_L - I_0 \left[\exp \left(\frac{V + IR_s}{nV_T} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \quad 3.1$$

The five parameters in this equation are primary to all single diode equivalent circuit models:

- I_L : light current (A)
- I_0 : diode reverse saturation current (A)
- R_s : series resistance (Ω)
- R_{sh} : shunt resistance (Ω)
- n : diode ideality factor (unitless)

This mathematical model describes the nonlinear relationship between the output current and voltage of a PV cell.

The model is implemented in MATLAB/Simulink to simulate the electrical behaviour of photovoltaic modules under different operating conditions.

3.3 Maximum Power Point Tracking Technique

Voltage–current relationships in solar photovoltaic modules are not linear. There is a specific operating point at which the output power is at its highest for each irradiance and temperature condition. The Maximum Power Point (MPP) is the operating point in question. To ensure that the PV system always operates at this optimal point, Maximum Power Point Tracking (MPPT) techniques are used.

In this study, the Perturb and Observe (P&O) algorithm is used for MPPT control during simulation.

The following is how the Perturb and Observe method works:

1. The PV array's operating voltage is slightly altered.
2. The output power of the PV module is measured.
3. The operating point moves toward the maximum power point if the power increases after a perturbation, and the perturbation continues in the same direction.
4. The direction of the perturbation is reversed if the power decreases.

The MPPT controller continuously adjusts the operating voltage of the PV array so that the system operates at maximum efficiency.

This method is widely used due to its:

- Simple implementation
- Low complexity of computation
- High reliability in practical systems

3.4 Modelling of PV Array Using MATLAB/Simulink

3.4.1 MATLAB/Simulink Model Description

The photovoltaic array is modelled in MATLAB/Simulink using the previously described single-diode model. The model consists of the following major components:

- The solar modules are represented by a PV array block.
- Irradiance input source
- Temperature input source
- Controller for MPPT
- Electrical load or inverter interface

PV performance can be analyzed using the simulation under various environmental conditions.

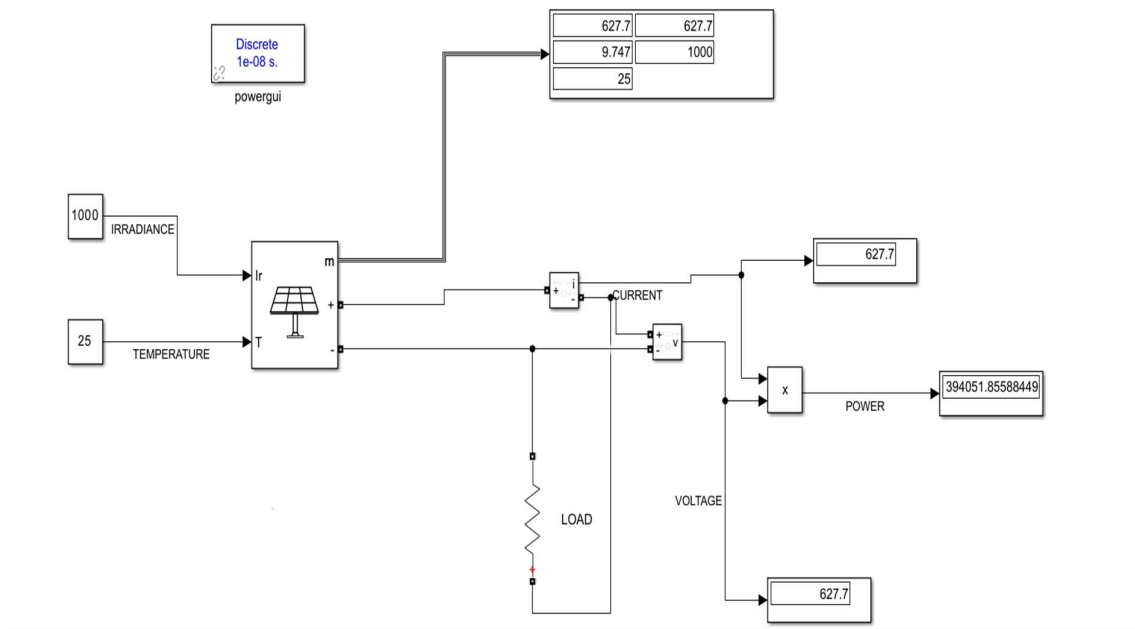


Figure: 3.2 – MATLAB/Simulink PV Model

3.4.2 Simulation Results

The simulation provides important electrical characteristics of the PV array, including:

- Current–Voltage (I–V) characteristics
- Power–Voltage (P–V) characteristics
- Output power variation with irradiance

The I–V curves show that the output current increases with increasing solar irradiance, while the open-circuit voltage remains relatively constant.

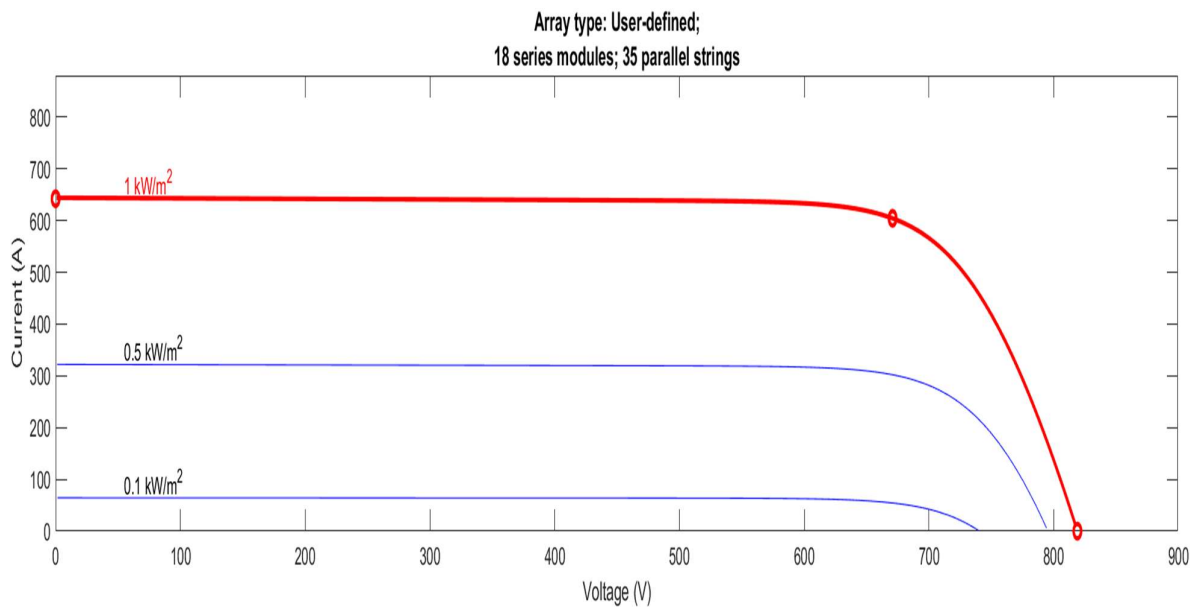


Figure: 3.3 V–I Characteristics of PV Array Under Different Irradiance Levels.

Similarly, the P–V curves indicate that the maximum power point shifts depending on irradiance and temperature variations.

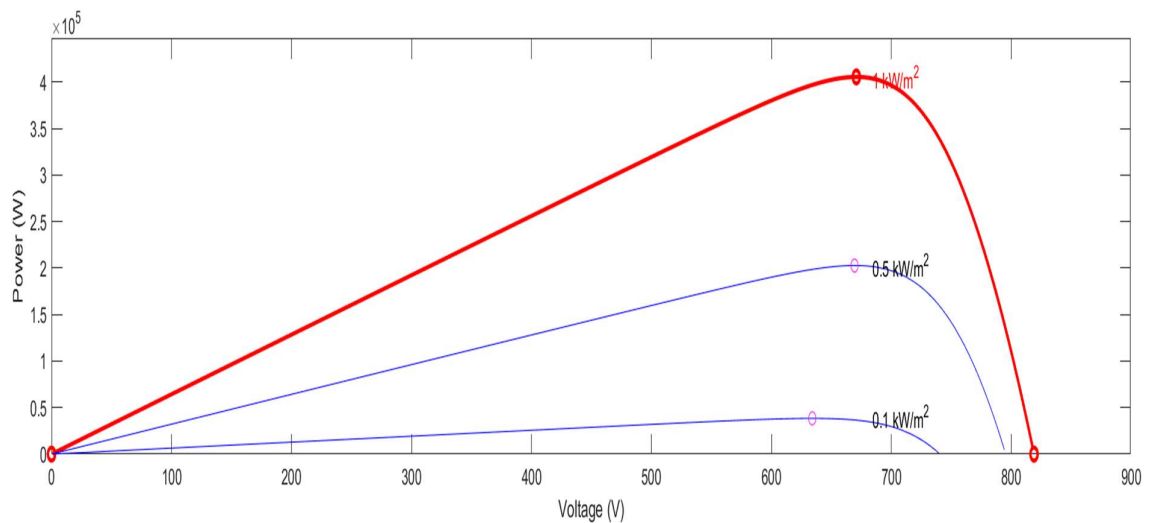


Figure: 3.4 P–V Characteristics of PV Array Under Different Irradiance Level.

These results validate the nonlinear electrical behaviour of photovoltaic modules.

3.5 Simulation of Grid-Connected PV System Using PVsyst

A block diagram that includes the PV array, inverter, protection devices, and grid interconnection is used to depict the photovoltaic system's overall configuration and power flow. The simulation report that is attached to Appendix D serves as a reference and contains the in-depth system performance analysis as well as the simulation results for grid-connected conditions with and without shading.

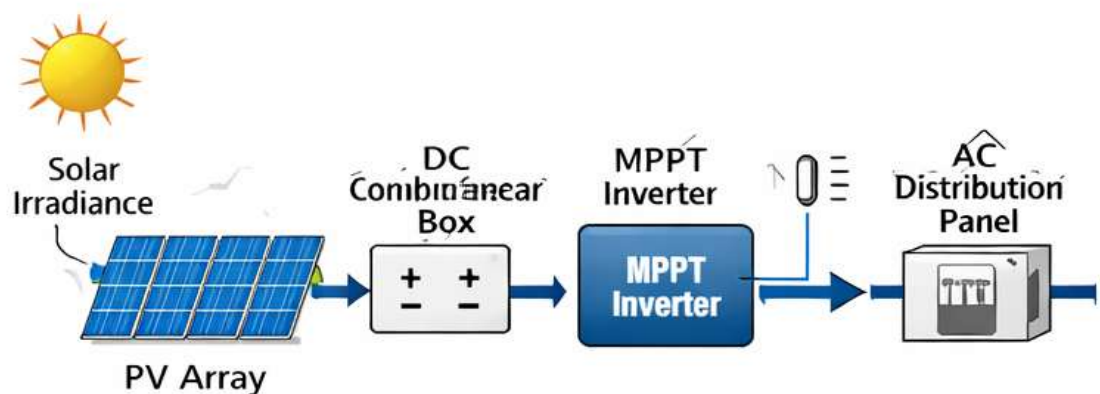


Figure: 3.5 Block diagram of Grid Connected Solar PV System

3.5.1 Meteorological Data

Chennai receives strong solar radiation throughout the year. The annual global horizontal irradiation is approximately 1853.9 kWh/m²/year.

Table: 3.1 Monthly Solar Irradiation Data – Chennai

Month	Global Horizontal Irradiation (kWh/m ² /month)	Diffuse Horizontal Irradiation (kWh/m ² /month)
January	144.2	66.3
February	156.4	60.5
March	189.7	74.8
April	184.7	81.6
May	182.6	91.3
June	162.3	84.5
July	158.0	91.5
August	157.1	89.7
September	153.7	78.1
October	134.9	74.4
November	112.4	62.0
December	117.8	63.0
Annual Total	1853.9	917.8

3.5.2 PV Module Specifications

The manufacturer's datasheet serves as the source for the parameters of the photovoltaic module used in the simulations. These include the maximum power point, temperature coefficients, open-circuit voltage, short-circuit current, and rated power. When deciding between a module's series or parallel configuration, the desired system voltage and power capacity are taken into account. Table 3.2 contains the in-depth specifications for PV modules, and Appendix A contains the manufacturer's datasheet for your perusal.

Table: 3.2 PV Module Parameters

Parameter	Specification
Manufacturer	AE Solar
Module Type	AE 650ME-132
Rated Power (Pmax)	650 W
Open Circuit Voltage (Voc)	45.5 V
Short Circuit Current (Isc)	18.32 A

Parameter	Specification
Voltage at Maximum Power (V _{mp})	37.5 V
Current at Maximum Power (I _{mp})	17.35 A
Module Efficiency (%)	22.34 %
Temperature Coefficient of Power (%/°C)	-0.34

3.5.3 Inverter and Battery Specifications

The rated power, DC input voltage range, and conversion efficiency of the inverter are all specified for the photovoltaic system that is connected to the grid. The battery's type, nominal voltage, capacity, and allowable discharge depth are all specified in the standalone configuration. System dependability, energy storage capacity, and overall efficiency are all significantly impacted by these parameters. Table 3.3 and Table 3.4 provide a summary of the battery and inverter specifications, respectively. The battery datasheet is included in Appendix C for reference, while the inverter datasheet is provided in Appendix B.

Table: 3.3 Grid Connected Inverter Specifications

Parameter	Value
Inverter Type	Grid-Connected
Rated Power (kW)	360
Maximum DC Voltage (V)	800
MPPT Voltage Range (V)	510 V
Inverter Efficiency (%)	97.80 %

Table: 3.4 Battery Specifications

Parameter	Value
Battery Type	Narada,Lead-acid
Nominal Voltage (V)	12.0 V
Capacity (Ah / kWh)	200 Ah
Depth of Discharge (%)	50 %
Battery Efficiency (%)	97.0 %

3.5.4 Grid-Connected PV System Without Shading

The grid-Connected photovoltaic system is modelled in this configuration without taking shading effects into account. The system layout includes PV modules, an inverter, and grid interconnection. Parameters such as tilt angle, azimuth orientation, and inverter sizing are defined within PVsyst. The

annual and monthly energy generation results obtained from the simulation are presented in Table 3.5, while the corresponding is illustrated in Figure 3.6

Table: 3.5 Energy Output Summary – Grid Connected PV System Without Shading

Parameter	Value
Global Incident Irradiation (kWh/m ² /year)	1885.9
Effective Irradiation on Collectors (kWh/m ² /year)	1843.6
Array Nominal Energy (kWh/year)	654583
Available Energy at Inverter Output (kWh/year)	635493
Energy Injected into Grid (kWh/year)	635493
Performance Ratio (%)	82.30

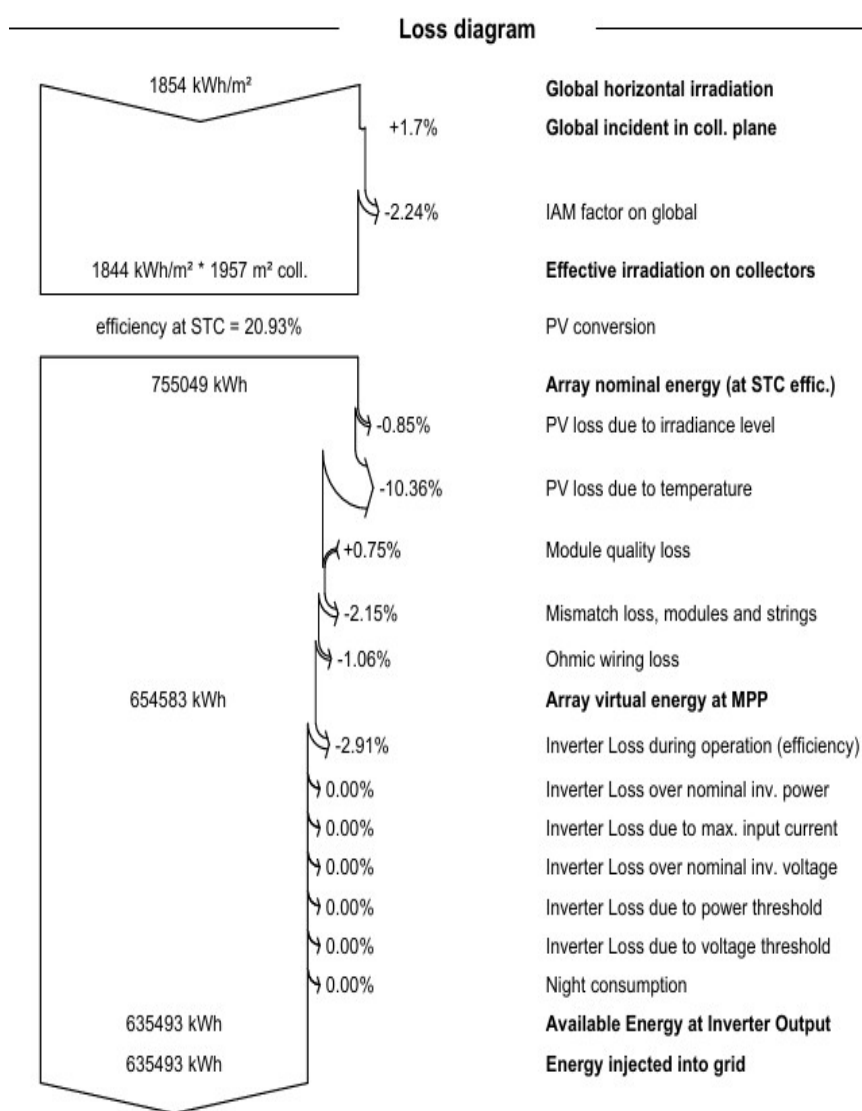


Figure: 3.6 Loss Diagram – Grid System Without Shading

3.5.5 Grid-Connected PV System with Shading

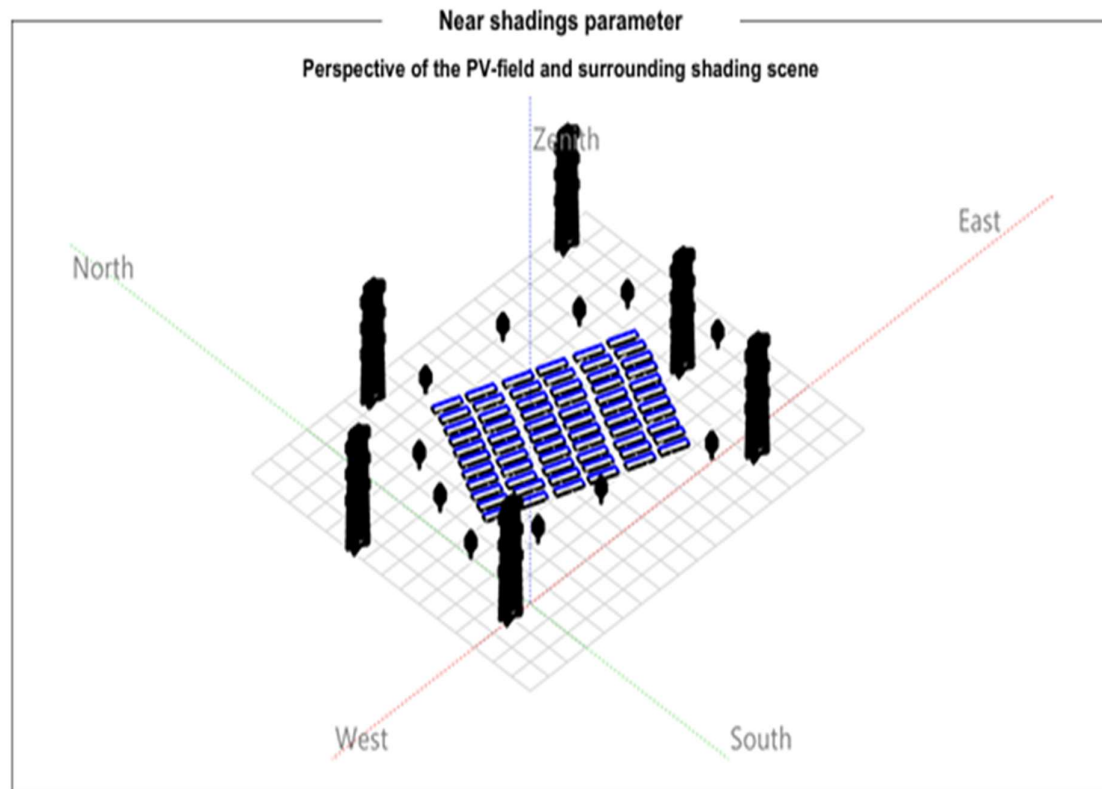


Figure: 3.7 Near Shading Scene Created in PVSyst for Grid-Connected PV System

To evaluate the influence of shading, a shading scene is created in PVSyst by introducing nearby obstacles such as buildings and structural elements around the photovoltaic array. The shading profile and its impact on incident solar irradiance are carefully analyzed in the simulation environment. Some photovoltaic (PV) modules receive less irradiance because some sunlight is blocked, which causes mismatch losses and lower overall energy generation. The resulting decrease in energy yield and performance ratio due to near-shading losses is observed from the simulation outputs. The shading loss distribution and energy flow pattern of the grid-Connected photovoltaic system under shaded conditions are depicted in Figure 3.8, and the corresponding annual energy output results are summarized in Table 3.6.

According to the simulation results, shading reduces the effective irradiation that reaches the collectors and, as a result, the amount of energy that is available at the inverter output. However, the generated power can still be continuously exported to the utility grid without compromising supply reliability because the system is Connected to the grid. Due to the increased optical and electrical losses brought about by shading, the performance ratio suffers a slight decline. Thus, the grid-Connected configuration demonstrates stable operation even under moderate shading conditions, although a measurable reduction in annual energy injection and system efficiency is observed.

Table: 3.6 Energy Output Summary – Grid Connected PV System with Shading

Parameter	Value
Global Incident Irradiation (kWh/m ² /year)	1885.9
Near Shading Loss (%)	2.29
Effective Irradiation on Collectors (kWh/m ² /year)	1804.9
Array Virtual Energy at MPP (kWh/year)	641208
Energy Injected into Grid (kWh/year)	622518
Performance Ratio (%)	80.61 %

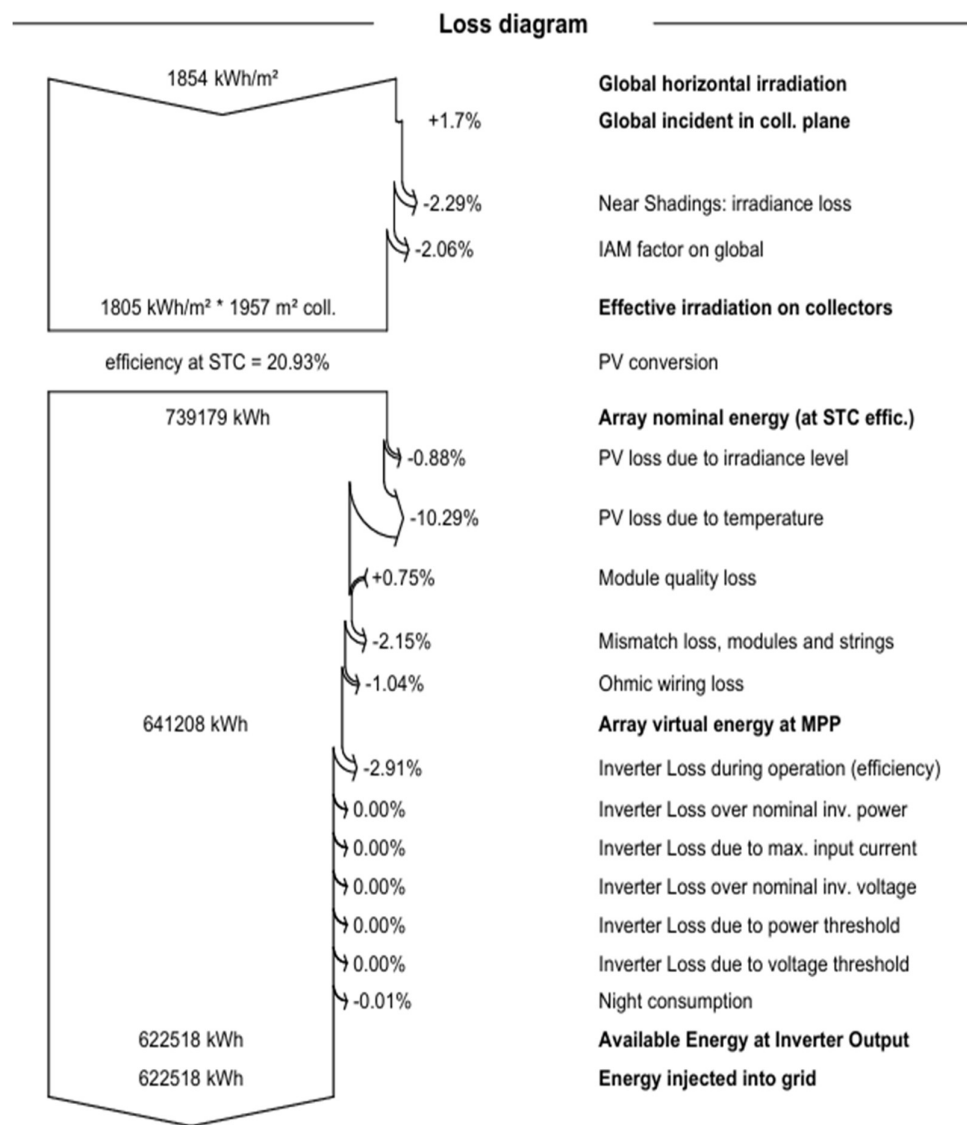


Figure: 3.8 Loss Diagram – Grid System with Shading

3.6 Simulation of Standalone PV System Using PVsyst

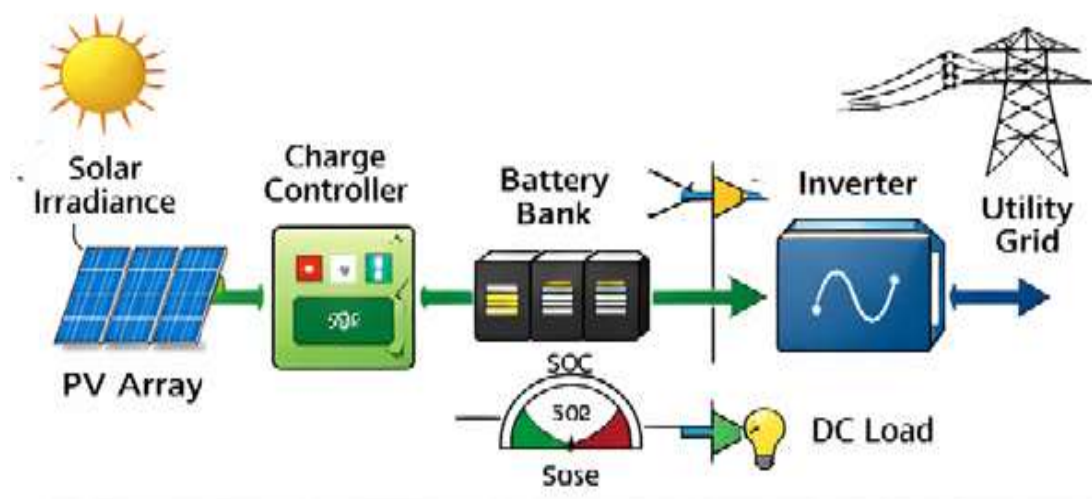


Figure: 3.9 Block diagram of standalone Solar PV System

A block diagram depicting the PV array, charge controller, battery bank, inverter, and load is used to illustrate the configuration of the standalone photovoltaic system. The simulation report that is attached to Appendix D serves as a reference and contains detailed performance analysis and simulation results for both operating conditions without and with shading.

3.6.1 Standalone PV System Without Shading

The standalone photovoltaic system is simulated without shading to assess its performance under ideal operating conditions. PV modules, a battery bank, a charge controller, and a load profile that has already been established make up the system. The behavior of the battery's state of charge and energy balance are revealed by the simulation results. The monthly energy distribution and battery performance are presented in Table 3.7 and Figure 3.10, respectively.

Table :3.7 Energy Balance – Standalone PV System Without Shading

Parameter	Value
PV Energy Generated (kWh/year)	626300
Energy Supplied to Load (kWh/year)	657000
Excess / Unused Energy (kWh/year)	12145

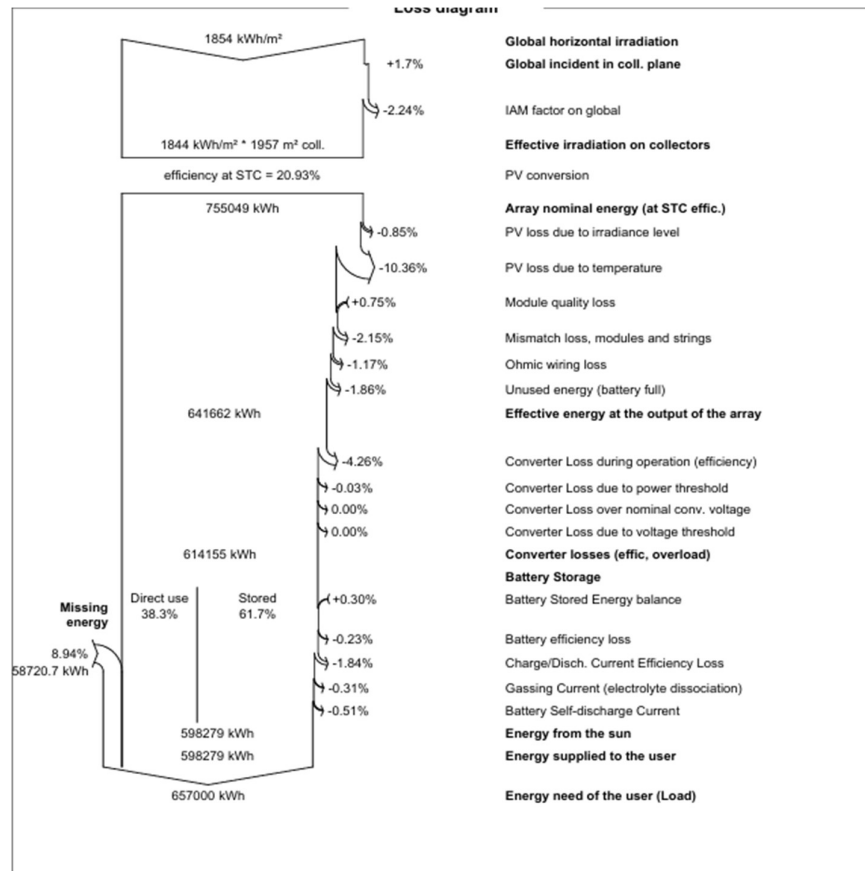


Figure: 3.10 Loss Diagram – Standalone Without Shading

3.6.2 Standalone PV System with Shading

In this case, shading conditions are incorporated into the standalone system model to analyse their effect on system performance. Shading leads to reduced PV output, which subsequently affects battery charging and load supply. Figure 3.12 depicts the variation in battery state of charge under shading conditions and summarizes the energy availability results in Table 3.8.

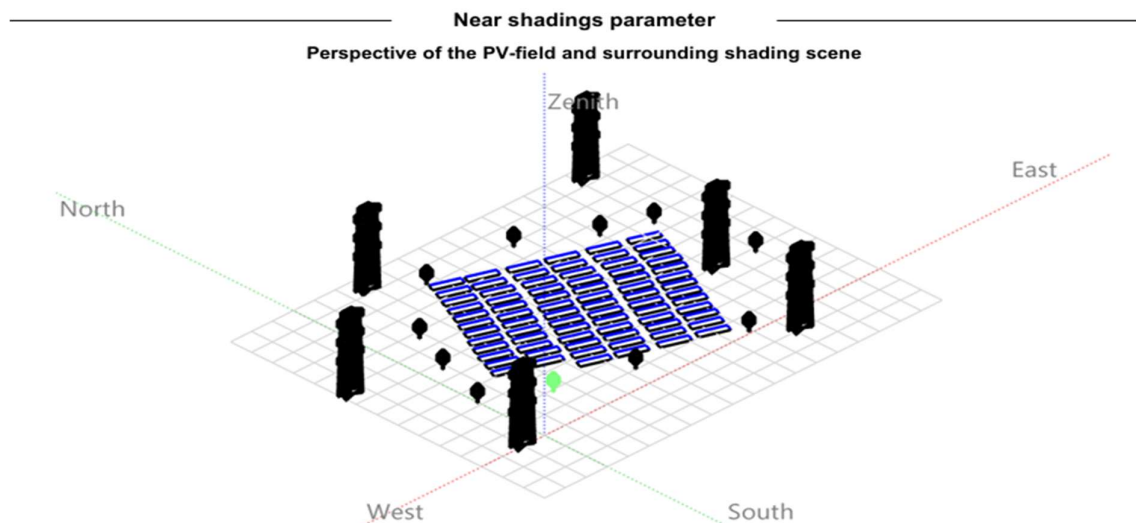


Figure: 3.11 Near Shading Scene Created in PVsyst for Standalone PV System

Table :3.8 Energy Balance – Standalone PV System with Shading

Parameter	Value
PV Energy Generated (kWh/year)	613087
Energy Supplied to Load (kWh/year)	657000
Excess / Unused Energy (kWh/year)	7225

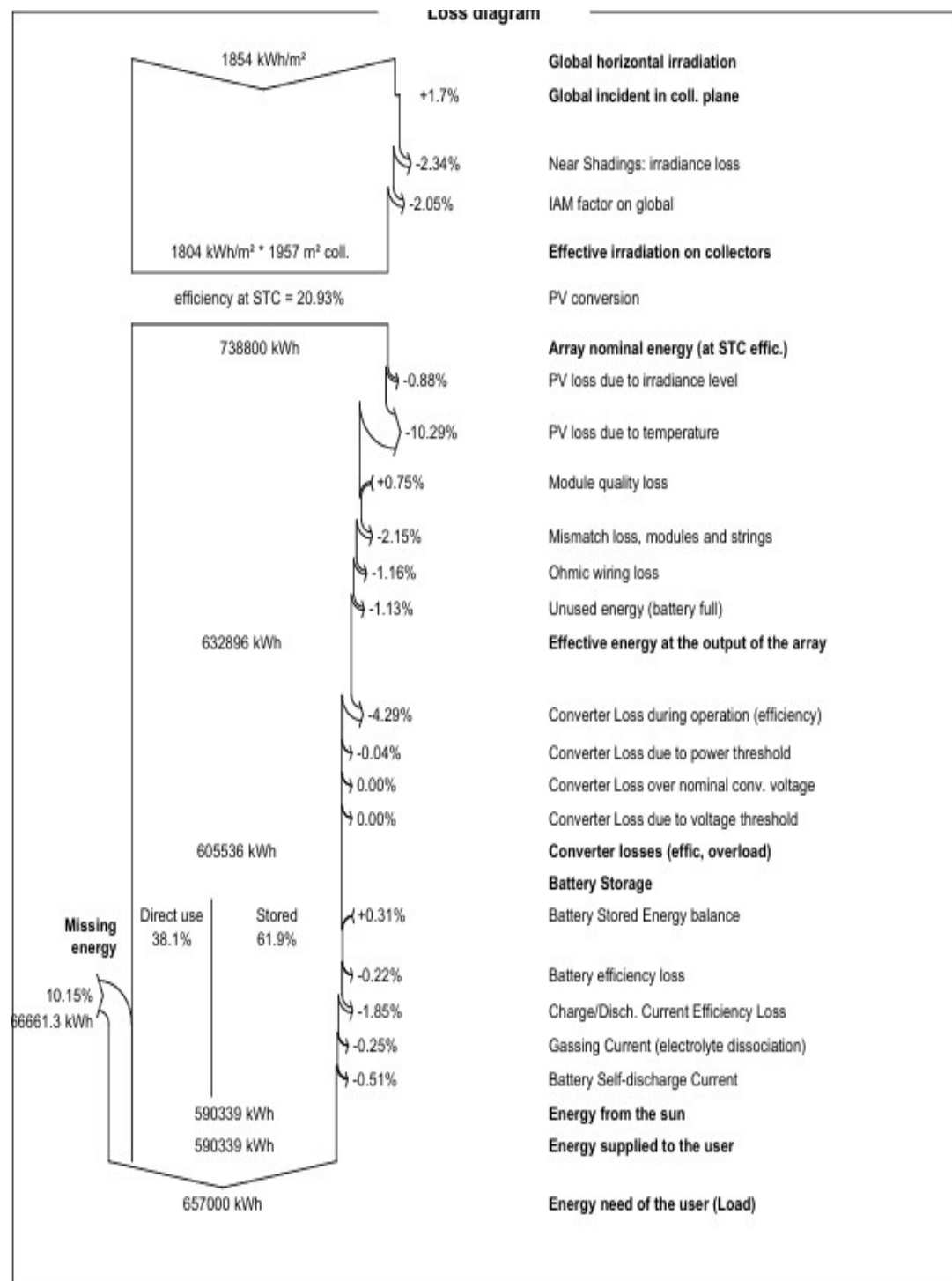


Figure: 3.12 Loss Diagram – Standalone with Shading

3.7 Comparative Analysis

3.7.1 Grid Vs Standalone

I. Energy Yield Comparison

Under unobstructed conditions, the grid-Connected PV system contributes 635489 kWh annually to the utility grid, whereas the standalone PV system generates 626300 kWh annually. The standalone configuration, in contrast to the grid-Connected system, requires energy storage and battery management to meet load demand. The monthly energy output comparison for both systems is summarized in Table 3.9, and the overall annual energy trend is illustrated in Figure 3.13.

Table: 3.9 Monthly Energy Output Comparison of Grid-Connected and Standalone PV Systems

Month	Grid-Connected Energy (kWh)	Standalone PV Energy (kWh)
January	55620	54882
February	57891	57261
March	65663	65078
April	60228	59563
May	57205	56469
June	50111	49280
July	49581	48715
August	51243	50357
September	52314	51446
October	48450	47600
November	42017	41229
December	45164	44420
Annual Total	635,489	626300

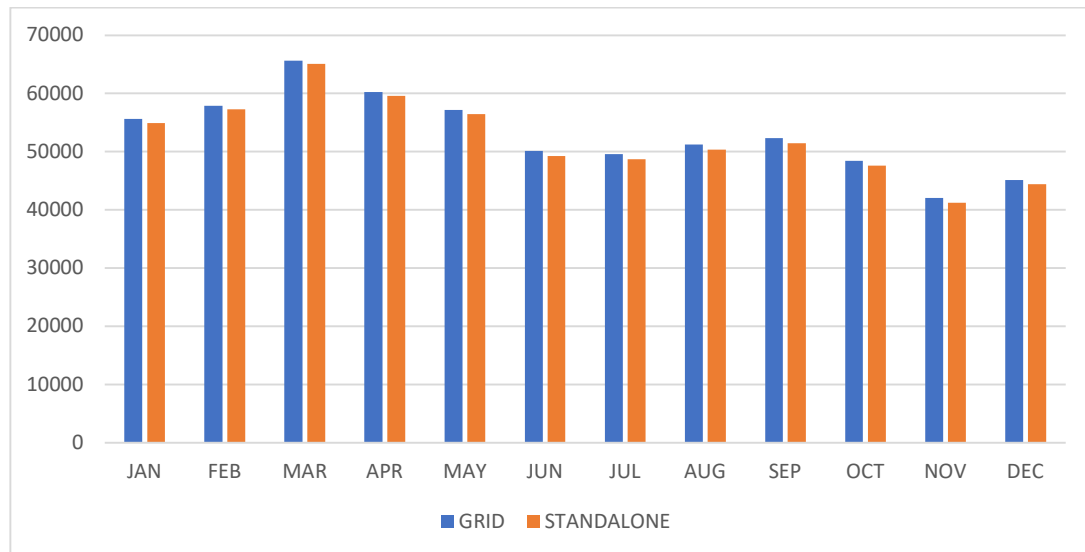


Figure: 3.13 Monthly energy comparison of grid-Connected and standalone PV systems

3.7.2 Impact of Shading on System Performance

Shading affects both grid-Connected and standalone PV systems; however, its impact is more pronounced in standalone configurations due to their reliance on battery storage. Battery charging is limited when PV generation is reduced in shaded conditions, which can result in lower energy availability for the load and an increased risk of energy deficit. The month-wise impact of shading on energy yield is summarized in Table 3.10, and the corresponding trend is illustrated in Figure 3.14.

Table: 3.10: Monthly Energy Output Comparison Under Shaded Conditions

Month	Grid-Connected Energy (kWh)	Standalone PV Energy (kWh)
January	55703	53321
February	58356	55912
March	66534	63893
April	61078	58571
May	57941	55458
June	50692	48306
July	50055	47692
August	51827	49416
September	52861	50450
October	48862	46565
November	42149	40099
December	45150	43071
Annual Total	641208	612754

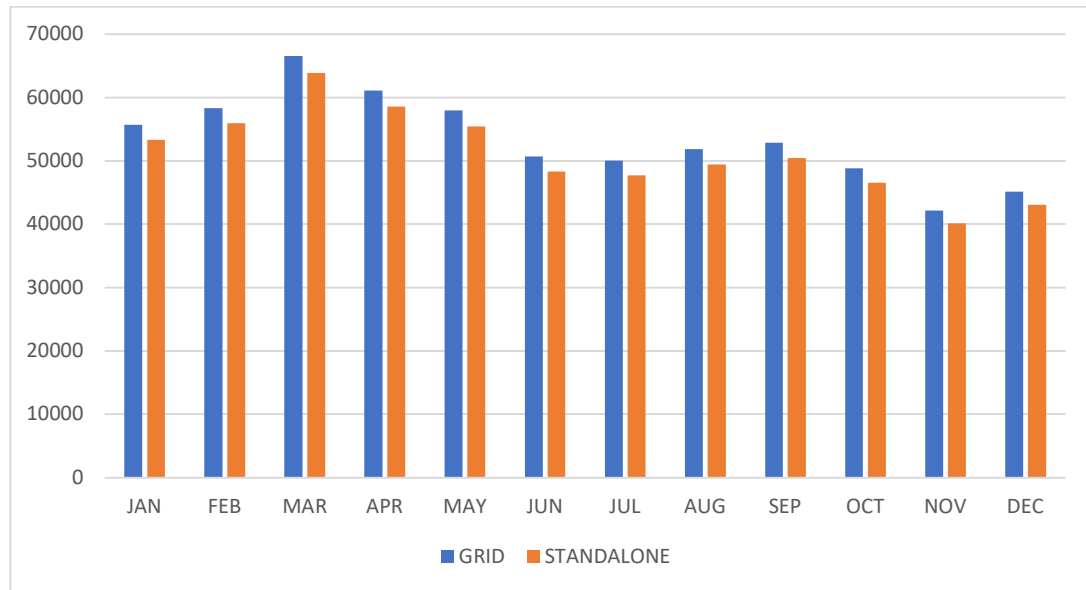


Figure: 3.14 Monthly Energy Yield Comparison – Grid-Connected Vs Standalone PV Systems Under Shading

Chapter IV

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING–BASED ENERGY PREDICTION

4.1 Introduction

Environmental and operational factors, such as solar irradiance, ambient temperature, module temperature, wind speed, system configuration, and partial shading, have a significant impact on the generation of photovoltaic (PV) energy. PV output is not affected by these parameters on their own; rather, they interact in a nonlinear way. The relationship becomes even more complex in standalone PV systems because the delivered energy depends not only on the available solar input but also on battery charging and discharging behavior, load demand, and system constraints. Although the output of grid-Connected systems is comparatively more stable, significant fluctuations in energy production are still caused by variations in the atmosphere and shading. While conventional analytical techniques are useful for comprehending the behavior of PV systems, they frequently fail to account for the cumulative effect of multiple variables under actual operating conditions. Simulation-based methods such as PVsyst provide accurate performance estimation, but they do not directly learn from patterns in historical or simulated data. To overcome these limitations, this chapter adopts Artificial Intelligence (AI) and Machine Learning (ML) techniques for energy prediction.

AI and ML are well suited for this problem because they can learn complex nonlinear relationships from data without requiring an explicit mathematical equation for every operating condition. PV energy output can be predicted using input variables like irradiance, temperature, wind speed, and array temperature once the models have been trained. This makes the approach highly useful for forecasting, performance analysis, system monitoring, and decision-making.

The advantage of AI/ML in PV energy prediction is that it can:

1. capture nonlinear dependencies among weather and system parameters,
2. deal with variations brought on by changing environmental conditions and shading,
3. compare predictive performance across different PV configurations, and provide fast estimation once the model is trained.

In this chapter, supervised regression-based machine learning models are developed for both grid-Connected and standalone PV systems under shaded and non-shaded conditions. The models are trained using simulation data obtained from PVsyst and processed in MATLAB. The MATLAB implementation used for preprocessing, training, validation, and result visualization is provided in Appendix E as a reference.

4.2 Objective of AI & ML-Based Energy Prediction

Using selected array-level and meteorological parameters, this chapter aims to develop and evaluate machine learning models that accurately predict PV system energy output. Using statistical performance metrics, the aim of the study is to compare model performance for grid-Connected and standalone PV systems, as well as to examine how shading affects prediction accuracy.

4.3 Data Source and System Classification

The datasets used for AI&ML modeling are obtained from PV system simulations performed using PVsyst software and further processed in the MATLAB environment. For both grid-Connected and standalone PV systems, hourly simulation data representing realistic operating conditions is taken into consideration. Separate datasets are maintained for shaded and non-shaded conditions to ensure a consistent comparative analysis.



Scan the QR code to access the hourly simulation dataset used for the AI&ML model development

4.4 Input and Output Variable Selection

4.4.1 Input Variables

The same set of input variables is used for all machine learning models and system configurations to ensure uniformity and fair comparison.

The selected input parameters are:

- Global Incident Irradiance on the Plane of Array (GlobInc)
- Horizontal Irradiance Around the World (GlobHor)
- Ambient Temperature (T_{Amb})
- Wind speed (WindVel)
- Temperature of the PV array (TArray)

Since these parameters have a direct impact on the PV array's electrical and thermal behavior, they are suitable predictors for energy output modeling.

4.4.2 Target Variables

The target variable differs based on the PV system configuration, as summarized in Table 4.1.

Table: 4.1 Target Variables for AI & ML Energy Prediction

System Type	Target Variable
Grid-Connected	E_Grid
Standalone	E_solar

For the grid-Connected PV system, the energy injected into the utility grid (E_Grid) is considered as the output parameter. For the standalone PV system, the energy produced by solar (E_solar) is used as the prediction target.

4.5 Data Preprocessing

The dataset is pre-processed prior to model development to enhance data quality and model dependability. Among the preprocessing steps are:

1. removal of non-operational conditions such as zero irradiance periods,
2. normalization of input features to bring all variables to a common scale,
3. division of the dataset into training and testing subsets.

Normalization improves model convergence and prevents any single feature from dominating the learning process due to scale differences. The data are split using an 80:20 hold-out ratio, where 80% of the data are used for training and 20% for testing.

4.6 Machine Learning Models Implemented

This chapter uses five regression-based machine learning models to predict PV energy output. Different levels of nonlinear modeling capability and learning strategies are represented by these models. Importantly, this study is not based on rule-based logic. No fixed decision rules are manually defined for predicting energy output. The models, on the other hand, learn patterns directly from data. As a result, the strategy is adaptable, data-driven, and suitable for PV systems operating in a variety of environments.

4.6.1 Regression Tree

The Regression Tree model predicts energy output by recursively partitioning the input feature space based on decision rules derived from the selected input variables. This model provides interpretability and captures nonlinear relationships between environmental parameters and PV energy generation

4.6.2 Random Forest

The Random Forest model is implemented as an ensemble of regression trees using bootstrap aggregation. By combining predictions from multiple trees, the model enhances robustness and reduces overfitting, particularly under fluctuating irradiance and shading conditions.

4.6.3 Support Vector Regression

Support Vector Regression (SVR) employs kernel-based learning to model nonlinear relationships between the input variables and PV energy output. Standardization of input features is used to boost prediction accuracy and numerical stability.

4.6.4 Gaussian Process Regression

Gaussian Process Regression (GPR) is a probabilistic regression technique that models PV energy output while accounting for uncertainty in the input data. Smooth and nonlinear variations in PV system behavior are suitable for this strategy.

4.6.5 Neural Network

A Neural Network regression model is developed to capture complex nonlinear dependencies between meteorological inputs and PV energy output. To reduce prediction error, supervised learning is used to train the model.

4.7 AI/ML Implementation Block Diagram

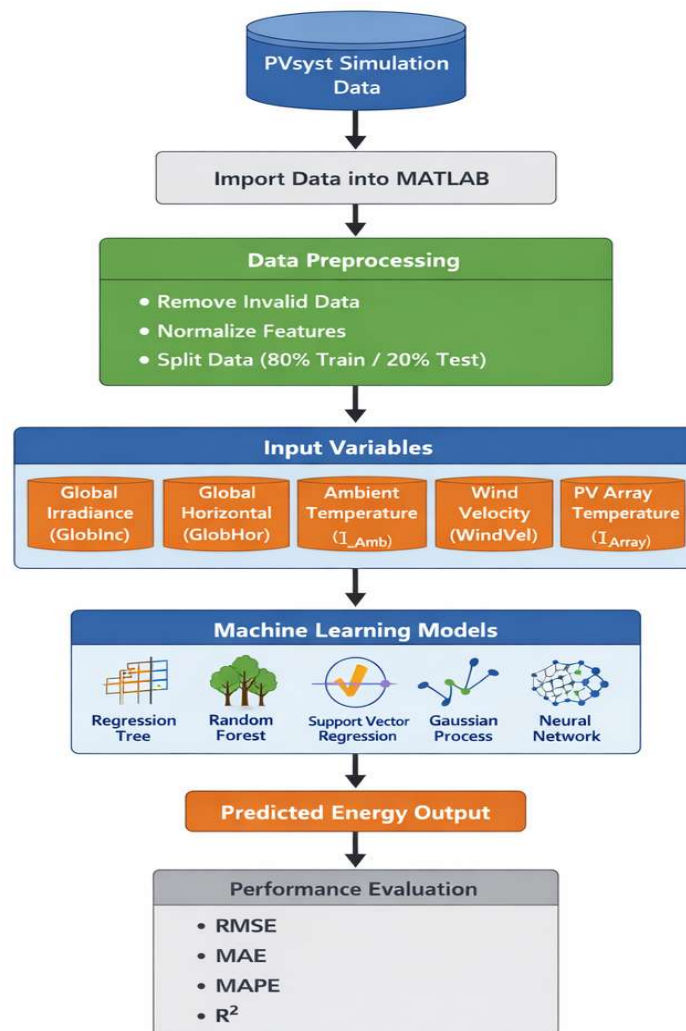


Figure: 4.1 AI&ML based PV energy prediction framework block diagram

4.8 Model Training and Validation Approach

For each PV configuration, each machine learning model is independently evaluated using the same input features. The training and testing process follows an 80:20 split.

The workflow is as follows:

1. In MATLAB, import the PVsyst simulation dataset.
2. Select the relevant input variables.
3. Define the target variable based on the type of system.
4. Normalize the dataset.
5. Split the data into training and testing subsets.
6. Train the selected regression models.

7. Test the trained models using unseen data.
8. Compare the results using standard statistical metrics.

Model performance is evaluated fairly and consistently across all configurations with this method.

4.9 Performance Evaluation Metrics

The prediction accuracy of each machine learning model is evaluated using the following statistical metrics:

- Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad 4.1$$

- Mean Absolute Error (MAE)

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad 4.2$$

- Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{100}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{y_i} \quad 4.3$$

- Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad 4.4$$

where y_i represents the actual energy output, $\hat{y}_i$ denotes the predicted output obtained from the machine learning model, $\bar{y}$ is the mean of the actual output values, and N is the total number of samples

4.10 Results and Discussion

4.10.1 Grid-Connected PV System

The prediction performance of the machine learning models for the grid-Connected PV system under non-shaded conditions is presented in Table 4.2, while the results for the shaded condition are summarized in Table 4.3. Figures 4.2 and 4.3 compare the RMSE, MAE, MAPE, and R^2 values to show how the models perform in comparison. The Gaussian Process Regression model performed best for the non-shaded grid-Connected PV system (Table 4.2), with a R^2 value of 1.0000, MAE of 0.16, MAPE of 0.99%, and the lowest RMSE of 0.16. The Neural Network model also performed exceptionally well, with RMSE of 0.73, MAE of 0.54, MAPE of 0.76%, and R^2 of 0.9999. SVR, on the other hand, had

RMSE of 6.88, MAE of 5.75, and MAPE of 27.78 percent, all of which indicate a lower prediction accuracy than the other models.

Table: 4.2 Performance Metrics for Grid-Connected PV System (Without Shading)

Model	RMSE	MAE	MAPE (%)	R ²
Regression Tree	1.93	1.43	2.38	0.9996
Random Forest	2.20	1.46	2.29	0.9995
SVR	6.88	5.75	27.78	0.9948
Gaussian Process	0.16	0.16	0.99	1.0000
Neural Network	0.73	0.54	0.76	0.9999

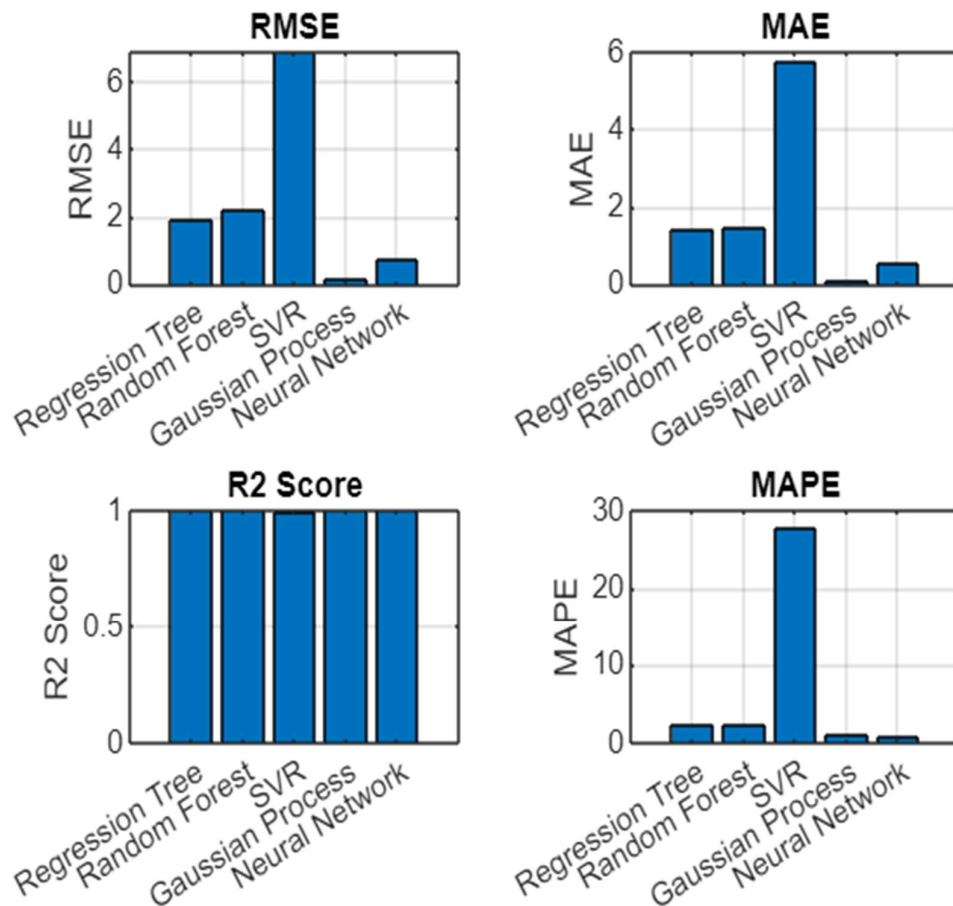


Figure: 4.2 Performance Comparison of Machine Learning Models for Grid-Connected PV System (Without Shading)

For the shaded grid-Connected PV system (Table 4.3), the Gaussian Process Regression model again delivered the most accurate predictions, with RMSE of 0.16, MAE of 0.11, MAPE of 1.28%, and R² of 1.0000. The Neural Network model remained highly effective, obtaining RMSE of 0.43, MAE of 0.35, MAPE of 1.43%, and R² of 1.0000. Among all models, SVR continued to perform the worst, with RMSE of 6.38 and MAPE of 30.14%.

Table: 4.3 Performance Metrics for Grid-Connected PV System (With Shading)

Model	RMSE	MAE	MAPE (%)	R ²
Regression Tree	2.33	1.69	2.92	0.9994
Random Forest	2.37	1.65	3.09	0.9994
SVR	6.38	5.26	30.14	0.9955
Gaussian Process	0.16	0.11	1.28	1.0000
Neural Network	0.43	0.35	1.43	1.0000

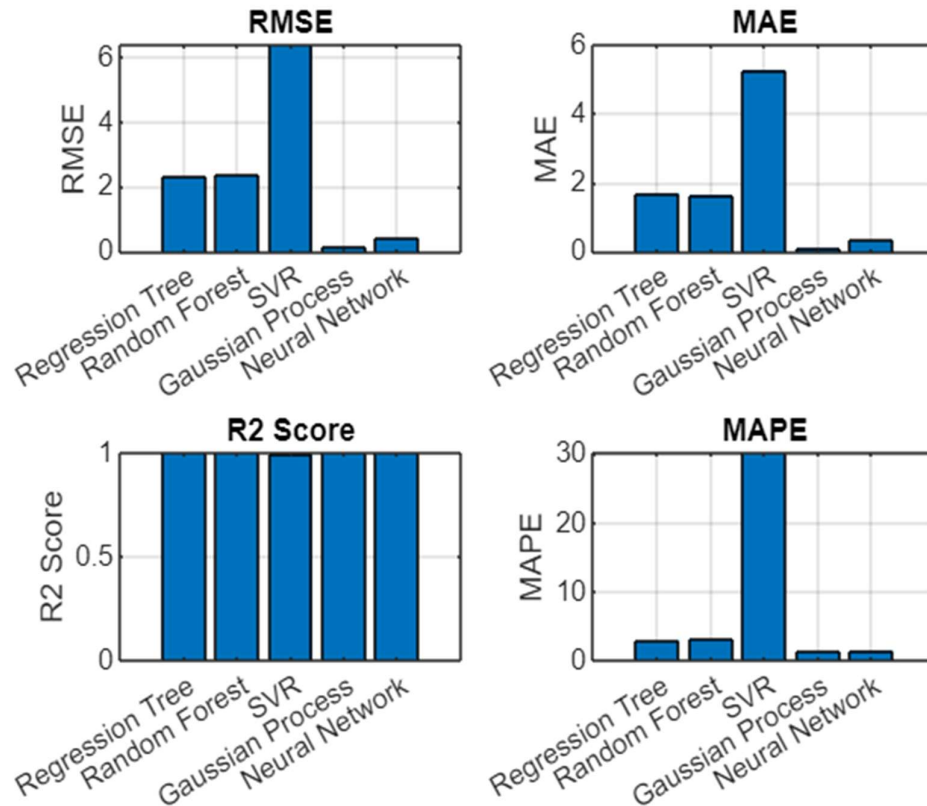


Figure: 4.3 Performance Comparison of Machine Learning Models for Grid-Connected PV System (With Shading)

Overall, the results show that SVR has the largest prediction error, while Gaussian Process Regression and Neural Network consistently offer the highest prediction accuracy for both shaded and non-shaded conditions. This demonstrates that the Gaussian Process Regression model is the most reliable for forecasting the performance of the grid-Connected PV system.

4.10.2 Standalone PV System

Under both non-shaded and shaded conditions, the standalone PV system's machine learning models are evaluated for their performance. The results for the non-shaded scenario are summarized in Table 4.4, while shaded condition results are presented in Table 4.5. Graphical comparisons are shown in Figure 4.4 and Figure 4.5.

Table :4.4 Performance Metrics for Standalone PV System (Without Shading)

Model	RMSE	MAE	R ²	MAPE (%)
Regression Tree	24.44	5.55	0.9357	204.17
Random Forest	19.09	5.20	0.9607	217.01
SVR	21.36	8.64	0.9508	279.26
Gaussian Process	20.11	5.98	0.9564	242.29
Neural Network	20.54	6.35	0.9546	243.75

For the non-shaded standalone system (Table 4.4), the models show relatively higher RMSE and MAPE values compared to grid-Connected systems. This is primarily because standalone systems operate load-dependently, and even for small absolute deviations, periods of low energy demand cause percentage-based error metrics like MAPE to rise. Among the models, Gaussian Process Regression and Neural Network demonstrate relatively better predictive performance, achieving lower RMSE and MAE values, and higher R² compared to other models. SVR exhibits the largest errors and lower prediction reliability.

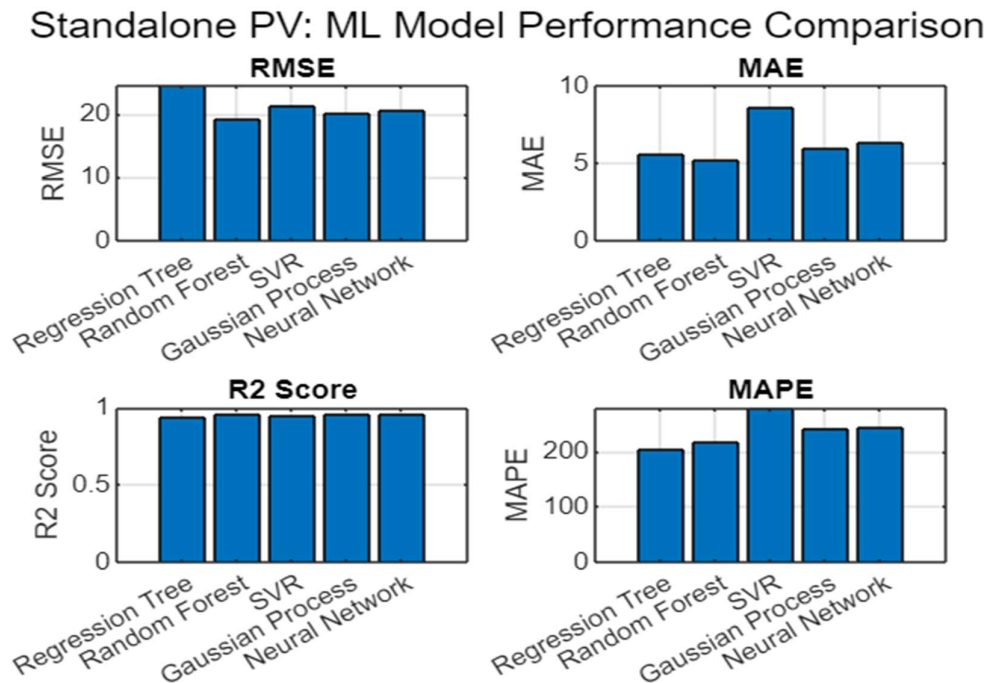


Figure: 4.4 Performance Comparison of Machine Learning Models for Standalone PV System (Without Shading)

Under shaded conditions (Table 4.5), prediction errors increase further due to the combined effects of shading and variable load. RMSE and MAPE values are higher, and R² values decrease, reflecting the difficulty of capturing the nonlinear behavior of standalone PV output. Despite these challenges, Gaussian Process Regression and Neural Network maintain superior performance relative to

the other models, highlighting their robustness even under constrained and nonlinear operating conditions.

Table :4.5 Performance Metrics for Standalone PV System (With Shading)

Model	RMSE	MAE	R ²	MAPE (%)
Regression Tree	26.27	5.78	0.9240	297.10
Random Forest	22.27	5.23	0.9454	283.31
SVR	22.87	8.03	0.9424	319.59
Gaussian Process	22.23	4.95	0.9456	294.04
Neural Network	22.33	4.92	0.9451	288.78

Standalone PV: ML Model Performance Comparison

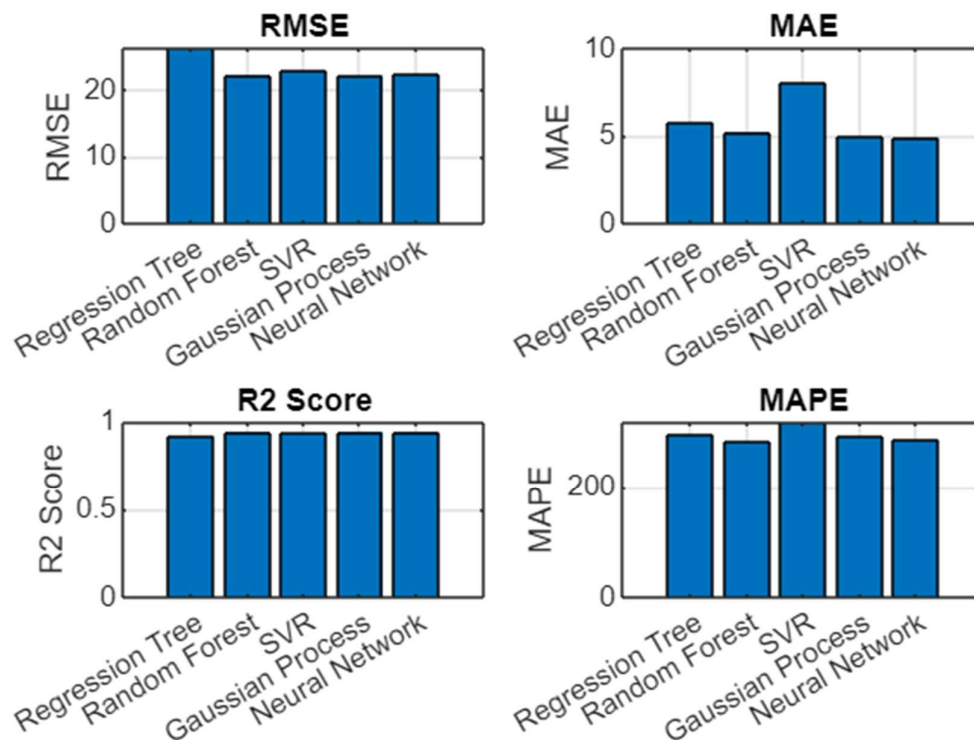


Figure: 4.5 Performance Comparison of Machine Learning Models for Standalone PV System (With Shading)

Compared to Grid-Connected Systems: Because of their load dependence and limited system flexibility, stand-alone PV systems naturally have higher prediction errors. However, the fact that Gaussian Process and Neural Network models consistently outperform other methods demonstrates that they are able to provide relatively trustworthy predictions even in difficult standalone situations.

4.11 MATLAB Implementation Details

MATLAB's built-in regression functions were used to implement all AI and machine learning models. For clarity and completeness, the full MATLAB source codes covering data preprocessing, model training, validation, and result visualization are provided in Appendix E.

The simulation dataset obtained from PVsyst was imported into MATLAB and pre-processed using normalization and filtering techniques to ensure model efficiency and accuracy. The implementation workflow is summarized as follows:

1. Import Dataset: Into the MATLAB workspace, load the PVsyst simulation dataset.
2. Feature Selection: Find relevant input features like the temperature of the array, ambient temperature, wind speed, and solar irradiance.
3. Define Target Variable:
4. Grid-Connected System: Energy injected into the grid (E_Grid)
5. Standalone System: Energy supplied to the load (E_solar)
6. Data Normalization: Scale the dataset to improve convergence during model training.
7. Train-Test Split: Divide the dataset into training (80%) and testing (20%) subsets.
8. Model Training: Train Support
9. 7, the Regression Tree, and other regression models. Neural Network, Gaussian Process Regression, and Vector Regression (SVR).
10. Measure model accuracy using RMSE, MAE, MAPE, and R2 metrics for performance evaluation.

Finally, the trained models were tested on unseen data to evaluate their predictive capability and generalization performance, ensuring that the models can reliably forecast PV system output under both grid-Connected and standalone configurations.

Chapter V

RESULTS AND DISCUSSIONS

5.1 Introduction

The simulation and machine learning results that were obtained for the solar photovoltaic (PV) system in various operating configurations are discussed and analyzed in depth in this chapter. Under shaded and unshaded conditions, the study evaluates the performance of grid-Connected and standalone PV systems. The results include the electrical behavior of the PV array, system-level energy generation, loss mechanisms, and the predictive accuracy of artificial intelligence and machine learning models.

The analysis is performed using outputs generated from MATLAB/Simulink simulations, PVsyst system modeling, and machine learning models implemented in MATLAB. Interpretation of the obtained results, identification of key performance trends, and evaluation of the efficiency of various system configurations and prediction models are the goals of this chapter. The results analysis is organized into three major sections:

1. PV array electrical characteristics obtained from MATLAB simulation
2. PVsyst is used to evaluate the performance of grid-Connected and standalone PV systems.
3. Energy prediction accuracy using machine learning models

The comparative analysis provides insights into system efficiency, energy losses due to shading, and the predictive capability of AI-based models.

5.2 Analysis of PV Array Electrical Characteristics

5.2.1 Current–Voltage Characteristics

The current–voltage characteristics of the photovoltaic array obtained from MATLAB simulation illustrate the nonlinear behavior of the PV module under different irradiance conditions. As solar irradiance increases, the output current of the PV module increases significantly, while the open-circuit voltage shows relatively small variation.

Under higher irradiance levels, the PV array produces larger short-circuit currents, indicating improved power generation capability. Conversely, at lower irradiance levels, the current decreases, resulting in reduced power output.

The simulated curves demonstrate that the voltage approaches its maximum value when the current approaches zero, and that the maximum current occurs when the PV module operates close to the short-circuit condition. The expected behavior of photovoltaic devices based on semiconductors is confirmed by this nonlinear property. The findings also confirm that temperature variations have a greater impact on the open-circuit voltage than irradiance does on the magnitude of the current.

5.2.2 Power–Voltage Characteristics

The PV module's output power varies depending on the terminal voltage, as shown by the power–voltage characteristics. There is a distinct maximum power point (MPP) in each curve, where the PV

array generates the most electrical power. As irradiance increases, the peak of the P–V curve shifts upward, indicating higher power output. The position of the MPP changes slightly with temperature variations, emphasizing the importance of maximum power point tracking (MPPT) in PV systems.

The simulation results confirm that the PV array operates most efficiently at the MPP, and any deviation from this operating point results in reduced power output.

5.3 Grid-Connected PV System Performance Analysis

5.3.1 System Performance Without Shading

Throughout the simulation period, the grid-Connected PV system without shading demonstrates stable and effective operation. The system receives a global incident irradiation of 1885.9 kWh/m²/year, resulting in effective collector irradiation of 1843.6 kWh/m²/year.

The PV array generates an annual nominal energy output of approximately 654,583 kWh, and after accounting for system losses such as inverter conversion losses, wiring losses, and temperature effects, the available energy at the inverter output is 635,493 kWh/year.

The system's 82.30% overall performance ratio indicates a high level of system efficiency. For large-scale photovoltaic installations, a performance ratio above 80% is generally regarded as excellent. PV modules can operate near their optimal power point for the majority of their operating time due to the absence of shading, maximizing energy generation.

5.3.2 System Performance with Shading

The PV system simulation's performance suffers a measurable drop in energy output when shading conditions are included. The near-shading loss of 2.29 percent results in a decrease in the PV array's effective energy generation. The annual energy injected into the grid decreases from 635,493 kWh/year to 622,518 kWh/year, representing an annual energy loss of approximately 12,971 kWh.

The performance ratio also decreases slightly from 82.30% to 80.61%, indicating the influence of shading on overall system efficiency.

Despite this reduction, the grid-Connected system continues to operate reliably because the grid acts as an energy buffer. When production exceeds demand, the generated power can be exported to the grid to ensure system operation.

5.3.3 Loss Distribution Analysis

The PVSyst loss diagram provides a detailed breakdown of energy losses occurring within the PV system. The primary sources of losses include:

1. Optical losses due to reflection and incidence angle effects
2. losses of heat caused by high module temperatures
3. Wiring losses within the DC and AC circuits
4. Conversion losses in an inverter

5. Losses in shading caused by nearby obstructions

Temperature-related losses account for a significant portion of the system's total loss. This is because photovoltaic cells produce less voltage when the module temperature rises. The shading losses, although relatively small in percentage terms, can have a noticeable impact on annual energy yield, particularly when shading occurs during peak irradiance periods.

5.4 Standalone PV System Performance Analysis

5.4.1 Standalone System Without Shading

The standalone photovoltaic system is designed to operate independently from the utility grid and relies on battery storage to meet load demand. The PV system generates approximately 626,300 kWh/year of electrical power in unobstructed conditions. The generated energy is distributed among three primary components:

- Energy supplied directly to the load
- Energy stored in the battery bank
- Excess energy that cannot be utilized

The system performance shows that the battery charging and discharging processes introduce additional losses compared with the grid-Connected system. Even though the standalone system is able to support the load, its overall efficiency is reduced due to storage-related losses.

5.4.2 Standalone System with Shading

Under shaded conditions, the PV energy generation decreases to approximately 613,087 kWh/year, resulting in reduced battery charging energy and lower available energy for load supply.

The reduction in PV energy generation leads to a greater dependence on battery storage. Since the standalone system does not have grid support, the effect of shading is more severe than in the grid-Connected case.

This result highlights the increased sensitivity of standalone PV systems to shading conditions, as the absence of grid support limits the system's ability to compensate for energy shortfalls.

5.5 Comparative Analysis of PV System Configurations

5.5.1 Grid System Comparison

The comparison between shaded and unshaded grid-Connected PV systems reveals that shading results in a reduction of approximately 12,971 kWh/year in annual energy output.

Although this reduction appears moderate, the cumulative energy loss over the lifetime of the system could be significant. Therefore, in order to minimize the effects of shading, proper system design, including optimal panel placement and obstacle avoidance, is necessary.

5.5.2 Standalone System Comparison

The standalone PV system experiences a reduction of approximately 13,547 kWh/year in PV energy generation under shaded conditions. Additionally, the battery charging cycle is directly affected by the decreased PV output, which results in a decrease in stored energy and an increase in the energy deficit. The results demonstrate that standalone systems are more vulnerable to shading because they rely entirely on local energy generation and storage.

5.5.3 Grid-Connected Vs Standalone System Performance

The operational behavior of grid-Connected PV systems and standalone PV systems differ significantly. The grid-Connected system offers:

1. Higher operational stability
2. Capacity to export energy indefinitely
3. diminished reliance on battery storage
4. Lower storage-related energy losses

In contrast, the standalone system requires energy storage and careful load management. Battery charging and discharging losses significantly reduce overall system efficiency.

Furthermore, standalone systems are more susceptible to environmental variations, including shading and irradiance fluctuations.

5.6 Machine Learning Prediction Results Analysis

5.6.1 Grid-Connected PV System Prediction

The machine learning models developed for the grid-Connected PV system demonstrate excellent prediction accuracy. Among the evaluated models, Gaussian Process Regression (GPR) and Neural Network models exhibit the highest predictive performance.

For the non-shaded grid-Connected case, the Gaussian Process model achieves the best results with RMSE = 0.16, MAE = 0.16, MAPE = 0.99%, and $R^2 = 1.0000$. With RMSE = 0.73, MAE = 0.54, MAPE = 0.76 percent, and $R^2 = 0.9999$, the Neural Network model also exhibits significant predictive power.

In contrast, the Support Vector Regression model has significantly higher prediction errors, indicating that it is not as effective at capturing the dataset's nonlinear relationships.

Again, the Gaussian Process model performs best under shaded conditions with RMSE = 0.16, MAE = 0.11, MAPE = 1.28 percent, and $R^2 = 1.0000$. With RMSE = 0.43, MAE = 0.35, MAPE = 1.43 percent, and $R^2 = 1.0000$, the neural network continues to be extremely accurate.

Table :5.1 Performance Metrics for Grid-Connected PV System

Model	Without Shading RMSE	Without Shading MAE	Without Shading MAPE (%)	Without Shading R²	With Shading RMSE	With Shading MAE	With Shading MAPE (%)	With Shading R²
Regression Tree	1.93	1.43	2.38	0.9996	2.33	1.69	2.92	0.9994
Random Forest	2.20	1.46	2.29	0.9995	2.37	1.65	3.09	0.9994
SVR	6.88	5.75	27.78	0.9948	6.38	5.26	30.14	0.9955
Gaussian Process	0.16	0.16	0.99	1.0000	0.16	0.11	1.28	1.0000
Neural Network	0.73	0.54	0.76	0.9999	0.43	0.35	1.43	1.0000

Gaussian Process Regression (GPR) is the best-performing model in the table because it shows the lowest RMSE, MAE, and MAPE values and the highest R² value, indicating the highest prediction accuracy among all models. Therefore, GPR provides the most reliable energy generation forecasting for the photovoltaic system compared to the other machine learning models.

5.6.2 Standalone PV System Prediction

The prediction performance for the standalone PV system is noticeably lower than that of the grid-Connected system. Higher RMSE and MAPE values are observed across most machine learning models.

The primary reason for this behavior is the complex interaction between PV generation, battery storage, and load demand. Standalone PV systems, in contrast to grid-Connected systems, operate in sporadic conditions, making accurate forecasting more difficult. Despite these challenges, the Neural Network and Gaussian Process Regression models still show relatively better performance than the other models. Prediction errors in SVR are still increasing.

Table :5.2 Performance Metrics for Standalone PV System

Model	Without Shading RMSE	Without Shading MAE	Without Shading R²	Without Shading MAPE (%)	With Shading RMSE	With Shading MAE	With Shading R²	With Shading MAPE (%)
Regression Tree	24.44	5.55	0.9357	204.17	26.27	5.78	0.9240	297.10
Random Forest	19.09	5.20	0.9607	217.01	22.27	5.23	0.9454	283.31
SVR	21.36	8.64	0.9508	279.26	22.87	8.03	0.9424	319.59
Gaussian Process	20.11	5.98	0.9564	242.29	22.23	4.95	0.9456	294.04
Neural Network	20.54	6.35	0.9546	243.75	22.33	4.92	0.9451	288.78

The Random Forest model is the best overall model for the standalone PV system because it has the highest prediction accuracy among all models and the lowest RMSE (19.09) and R² (0.9607) when there is no shading.

5.7 Discussion of Key Findings

The study's findings offer a number of significant insights into energy prediction and performance of photovoltaic systems. First, the MATLAB simulation results confirm the nonlinear electrical characteristics of photovoltaic modules and highlight the importance of operating at the maximum power point.

Second, PVsyst simulations demonstrate that shading has a measurable impact on annual energy generation, although grid-Connected systems can tolerate moderate shading without severe performance degradation.

Third, standalone PV systems are significantly more sensitive to environmental variations due to their dependence on battery storage and local energy generation.

Last but not least, when applied to grid-Connected systems operating in stable conditions, machine learning techniques demonstrate a significant amount of potential for predicting PV energy generation.

Conclusions

A comprehensive approach to the design, simulation, and performance evaluation of solar photovoltaic (PV) systems for a Chennai, Tamil Nadu, residential township is presented in this study. Under shaded and unshaded conditions, MATLAB/Simscape was used for electrical modeling, and PVsyst was used for system-level simulations. Grid-Connected and standalone PV systems were also taken into consideration. Manual calculations were performed to size the PV array, inverter, and battery bank, ensuring that around 20% of the township's daily energy demand is met reliably with appropriate safety margins.

The simulation results confirmed the nonlinear electrical behavior of PV modules. The I–V characteristics demonstrated that temperature has an effect on the open-circuit voltage and that irradiance has a significant impact on current. The maximum power point (MPP) that was clearly visible in the P–V curves demonstrated the significance of MPPT for efficiency. Grid-Connected systems achieved a performance ratio above 82% under unshaded conditions, with minimal impact from moderate shading. Conversely, standalone systems were more sensitive to shading and environmental fluctuations due to dependence on battery storage and load matching, resulting in noticeable energy deficits during shaded periods.

The study also integrated Artificial Intelligence and Machine Learning (AI & ML) techniques to predict energy generation. Gaussian Process Regression (GPR) and Neural Network models delivered the highest prediction accuracy, particularly for grid-Connected systems. Support Vector Regression (SVR) showed comparatively lower accuracy, highlighting the importance of selecting suitable models for nonlinear PV datasets. Standalone systems exhibited higher prediction errors due to the complex interaction between PV generation, battery operation, and load demand.

Future Scope

The proposed system can be developed as an integrated web-based energy prediction and monitoring platform supported by mobile application access and IoT-based sensing infrastructure. In this approach, real-time meteorological parameters such as solar irradiance, ambient temperature, wind speed, PV module temperature, and array voltage/current are continuously collected using sensors and transmitted to a centralized platform. These data inputs are utilized by AI and machine learning models to perform automated energy forecasting, improve predictive accuracy, detect system faults, and enable adaptive control strategies. Solar energy utilization optimization, system monitoring, and historical performance visualization are all made easier for residents and system administrators by the web platform. Additionally, a user-friendly mobile application provides real-time insights into energy production trends, forecasted generation, battery state monitoring, load scheduling support, and shading detection alerts, thereby enhancing operational awareness and enabling intelligent energy management.

By effectively capturing complex temporal patterns and nonlinear dependencies in photovoltaic energy generation, advanced artificial intelligence and deep learning techniques like LSTM, CNN, transformer architectures, and hybrid physical–data driven models can be used to expand the scope of the proposed work in the future, thereby increasing forecasting precision and system dependability. Additionally, lifecycle costs, payback times, levelized cost of electricity (LCOE), and potential reductions in carbon emissions can all be evaluated as part of a comprehensive techno-economic and environmental assessment to aid in the selection of large-scale residential solar installations. Furthermore, integration of smart grid technologies and advanced energy storage solutions such as lithium-ion or hybrid battery systems, along with intelligent control algorithms, can enable efficient demand response management, peak load shaving, enhanced grid stability, and optimized bidirectional power flow within modern distributed energy systems.

Future research can focus on optimizing shading effects and panel placement using advanced simulation tools and AI-based algorithms to reduce shading losses and improve PV array layout efficiency in urban environments. For standalone and community-scale applications, AI-driven energy management can also make hybrid renewable systems that combine solar photovoltaic (PV) with wind or microhydro power more reliable.

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APPENDIX

Appendix A PV Module Datasheet



AESOLAR

since 2003



AURORA

HIGH ENERGY YIELD
RELIABILITY
DURABILITY

30
years

Performance
Warranty

15
years

Product
Warranty

640W-660W

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CELLS

210
mm

CELL
SIZE



LID
RESISTANT



PID
RESISTANT



SALT CORROSION
RESISTANT



SAND
RESISTANT



AMMONIA
RESISTANT

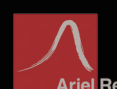


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IEC 62716 (Ammonia corrosion)
IEC 61701 (Salt mist corrosion)
IEC 60068 (Sand and dust)
IEC 62804 (PID resistance)



AE ME-132 Series 640W-660W



MONO-CRYSTALLINE PV MODULES
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Mechanical and design specification

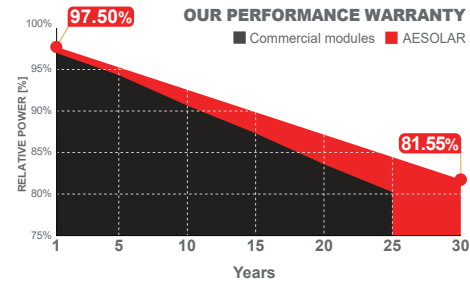
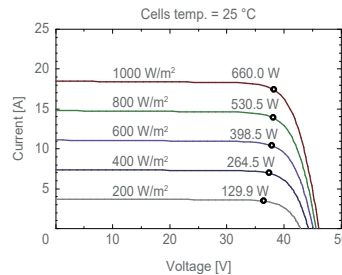
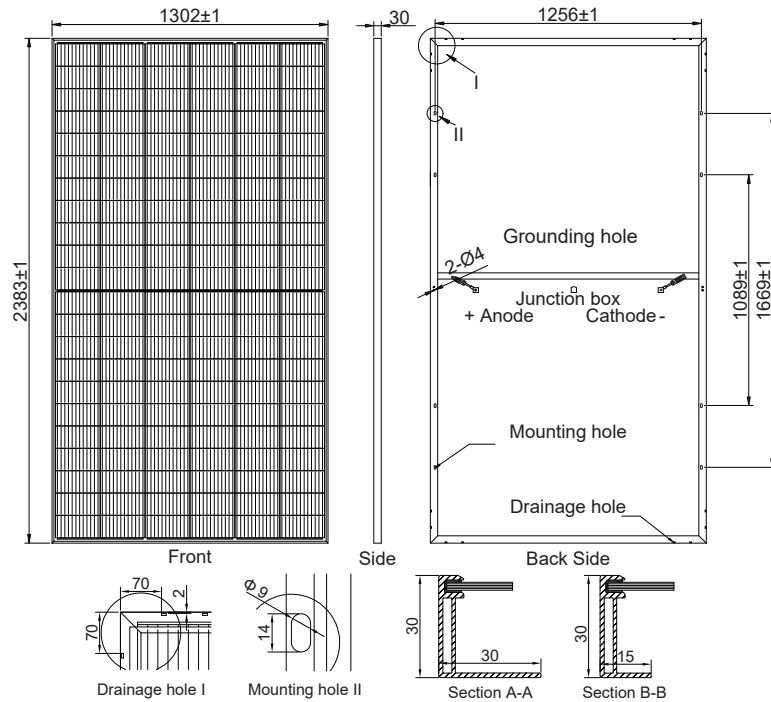
Cell type	Gallium-doped Mono c-Si PERC, Half-cut cells, 210 mm
No. of cells	132
Front cover	3.2 mm glass, high transmission, AR coated, tempered
Encapsulation	EVA
Back cover	White backsheet
Junction box	IP68 rated, 3 Bypass Diodes
Frame	30 mm anodized Aluminium alloy
Cable	1 x 4 mm ² , 350 mm length or customized
Connectors	MC 4 / MC 4 compatible
Dimension	2383 mm x 1302 mm x 30 mm
Weight	31.5 kg
Hail resistance	Max. Ø 25 mm at 23 m/s
Wind load	2400 Pa or 244 kg/m ²
Snow load	5400 Pa or 550 kg/m ²

Packaging information

Packaging configuration	36 pcs / pallet
Loading capacity	612 pcs / 40 HQ
Size / Pallet	1350 mm x 1140 mm x 2500 mm (Upright)
Weight	1167 kg / pallet

Temperature ratings

Operating temperature	(°C)	-40 to +85
Temp.coefficient of P_{max}	(%/°C)	-0.34
Temp.coefficient of V_{oc}	(%/°C)	-0.25
Temp.coefficient of I_{sc}	(%/°C)	0.040
Nom. operating temp. NOCT	(°C)	43 ± 2



Electrical specifications (STC*):		AE640ME-132	AE645ME-132	AE650ME-132	AE655ME-132	AE660ME-132
Nominal Max. Power	P_{max} (Wp)	640	645	650	655	660
Maximum operating voltage	V_{MPP} (V)	37.00	37.20	37.40	37.60	37.80
Maximum operating current	I_{MPP} (A)	17.30	17.34	17.38	17.42	17.46
Open-circuit voltage	V_{oc} (V)	45.10	45.30	45.50	45.70	45.90
Short-circuit current	I_{sc} (A)	17.74	18.25	18.32	18.38	18.43
Module efficiency	η (%)	20.63	20.79	20.95	21.11	21.27
Power tolerance	(W)	0~+5				
Maximum system Voltage	(V)	1500				
Maximum series fuse rating	(A)	30				

*STC: Standard test conditions (Irradiance 1000 W/m², Cell temperature 25°C and air mass of AM1.5), measurement tolerance P_{max} : ±3%

Electrical specifications (NMOT*):		AE640ME-132	AE645ME-132	AE650ME-132	AE655ME-132	AE660ME-132
Nominal Max. Power	P_{max} (Wp)	480	484	488	492	496
Maximum operating voltage	V_{MPP} (V)	34.70	34.90	35.10	35.30	35.50
Maximum operating current	I_{MPP} (A)	13.84	13.87	13.90	13.94	13.97
Open-circuit voltage	V_{oc} (V)	42.00	42.20	42.40	42.60	42.80
Short-circuit current	I_{sc} (A)	14.19	14.60	14.66	14.70	14.74

*NMOT: Normal Module Operating Temperature (Irradiance 800 W/m², Ambient temperature 20°C, air mass of AM1.5 and wind speed of 1 m/s)

The specifications and characteristics contained in this datasheet may deviate slightly from our actual products due to the product developments and uncertainty of measurement devices. The specifications included in the datasheet are subject to change without prior notice.

Appendix B Inverter Datasheet

SolarMax 330TS-SV / 360TS-SV

The all-round solution for large-scale PV installations.



20
More than
20 years Swiss Quality
and Experience

 **SolarMax**[®]
+ SWISS QUALITY

Modular and flexible

The SolarMax TS-SV inverters that can be directly connected to the medium-voltage grid are the best choice for PV power plants. Depending on the size of your system, an inverter system consists of up to four SolarMax TS-SV inverters (up to 1.44 MW), a master control unit, and a medium-voltage transformer. The TS-SV inverters meet both current and future requirements applicable to modern PV inverters and they are attractive because of their great flexibility, maximum energy yields, and excellent services such as the MaxControl service package.



Maximum pay-back

The TS-SV inverters are manufactured to a high standard of quality and the consistent use of film capacitors and intelligent monitoring of the power semiconductors make them especially durable and dependable. With a Euro efficiency of up to 97.4 % and an MPP efficiency of 99.9%, the TS-SV inverters maximise the yield of any solar installation. Our competent technical support team looks forward to helping you develop the best layout for your PV power plant.



Swiss Quality

For us, quality has always had the highest priority: all TS-SV series inverters are tested in compliance with current product safety standards and bear the TÜV mark "Type Approved". The development, the production, and all quality controls of the TS-SV inverters are conducted in Switzerland. This is something you can rely on.

Components

TS-SV Master Control Unit

The Master Control Unit (MCU), the central control and operation unit equipped with a display, enables as many as four TS-SV inverters to be controlled and monitored. As an option the MCU can be fitted with a MaxWeb data logger to easily integrate the installation into a MaxComm network.



Generator connection boxes

MaxConnect plus

The MaxConnect series devices are generator terminal boxes for SolarMax central inverters. They are designed for combining the individual cables of your solar generator. Thanks to single-cable monitoring the devices can detect defects and shadings prematurely.

With integrated cable protections and over-voltage conductors, as well as an integrated high-performance circuit breaker the MaxConnect plus is comprehensively secure. Connection is implemented using terminal clamps.



MaxConnect plus p

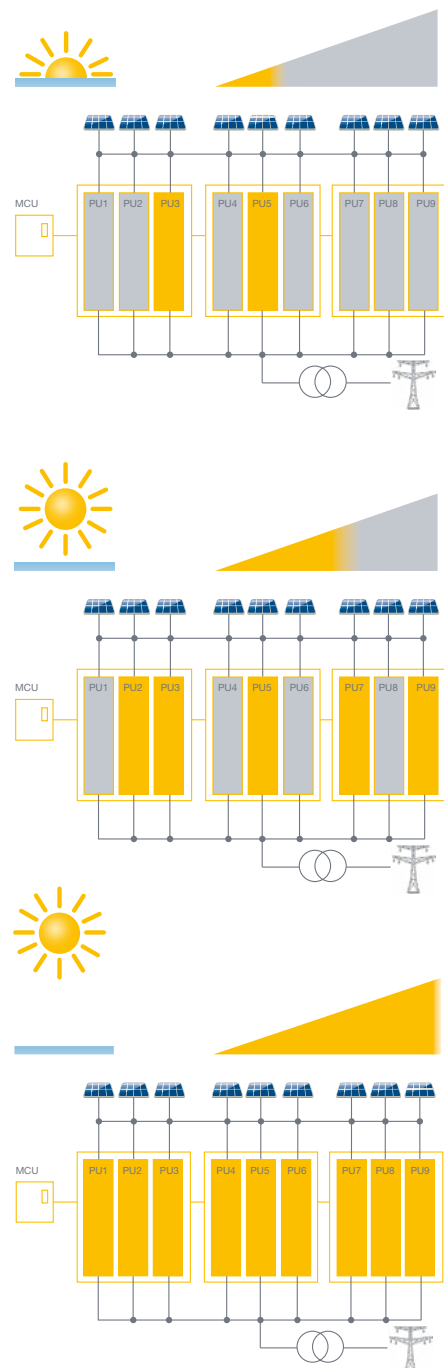
With its housing made of plastic and the MC4 plug-and-socket connections for connecting the cables, the MaxConnect plus p meets the requirements of protection class II and does not require any earthing. The device meets the French UTE standard specifying this protection class for components on the direct current side of PV power plants.

Application

The powerful TS-SV central inverters can be operated either with an MPP tracker (single-MPPT with partial-load optimisation and error management) or with three independent MPP trackers (multi-MPPT).

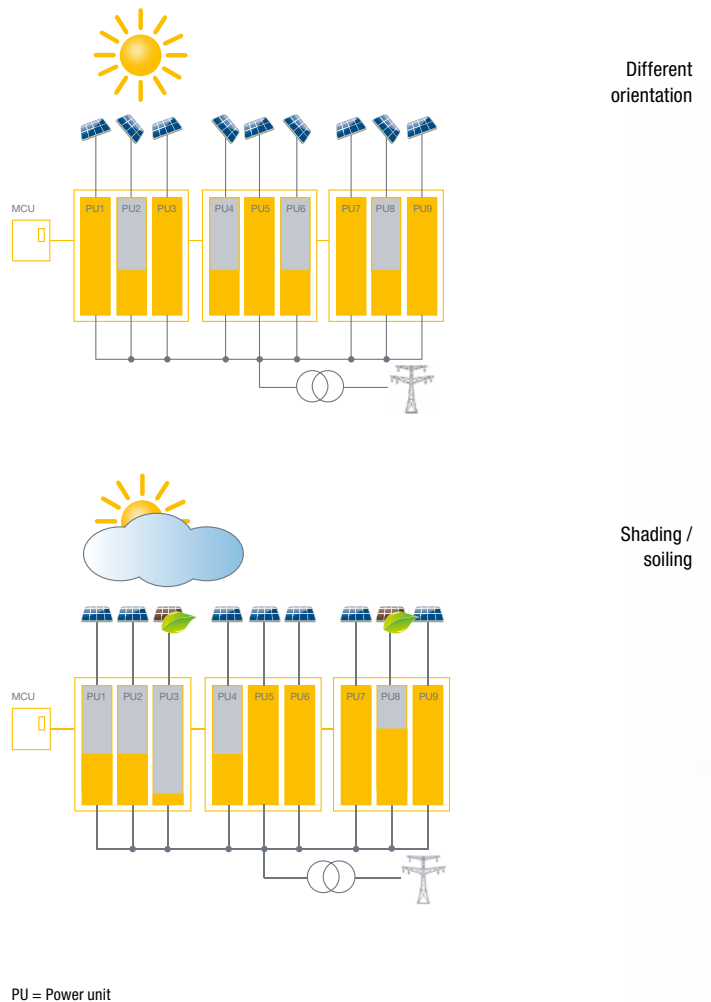
Single MPPT

Regarding homogeneous module fields, single-MPPT operation optimises the yield during periods of partial load by adding power units in response to power output. The parallel wiring of the PV generators and the redundant wiring of the power units increase operational reliability.



Multi MPPT

Multi-MPPT operation maximises the yield of each partial generator. Thanks to multi-tracking, each of the three power units per inverter always works at the best possible operation point - all partial generators are optimally used even if the surfaces face in different directions. For multi-MPPT operation, no DC fusing is required, additionally reducing the system costs.



PU = Power unit

Casing

The best choice for large-scale PV installations directly connected to the medium-voltage grid is the SolarMax Power Station and the SolarMax TS-SV Compact Station. The TS-SV inverters are the heart of the stations in this case, they can be operated either in single- or in multi-MPPT mode, and have a total joint PV output of up to 1.44 MW. The Power Station and the TS-SV Compact Station's high degree of efficiency, reliability, and durability maximise the yields.



Power Station

A SolarMax Power Station consists of as many as four TS-SV inverters, an MCU, a medium-voltage transformer, and medium-voltage switchgear. The installation of the Power Station into a container is made possible by the co-operation with our proven partners on site. This allows us to address your individual needs and meet local requirements. The complete solution is delivered completely assembled – the Power Station only needs to be connected.

TS-SV Compact Station

In a SolarMax TS-SV Compact Station, two TS-SV inverters can be accommodated, including MCU. A SolarMax 660TS-SV Compact Station delivers a rated output power of 660kW (2 x 330TS-SV), a 720TS-SV Compact Station delivers a rated output power of 720kW (2 x 360TS-SV). They are equipped with a light-weight, compact casing (2.2 x 2.9 x 2.0 m), and have a weight of approx. three tons with all equipment attached. Due to their low weight, the TS-SV Compact Station can be transported and installed easily. It is connected easily to a medium-voltage compact station via plug&play.

MaxComm for system monitoring

MaxWeb

The MaxWeb xp data logger forms the core of the web-based monitoring system; it enables multimedia communication with the photovoltaic plant and sends information via the internet to wherever you wish to receive it. Max-Remote enables remote-controlled performance reduction by the operator.



MaxMonitoring

The cost-free app visualises the performance data of the photovoltaic system and of individual inverters on site.

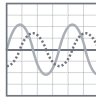
MaxTalk

User-friendly PC software for on site communication and for local system monitoring.



Smart operation and communication

All important data and settings are readable on the master control unit (MCU). The integrated data logger saves statistics on yields, peak outputs, and operating hours for as long as ten years. The inverters can be connected into the MaxComm communications system via the RS485 or Ethernet interfaces. Contacts for remote-controlled shut-downs and status messages have also been integrated. The AC and DC circuit breakers, which are accessible from the outside, allow easy disconnection from the PV generator and from the grid.



Grid management

The TS-SV inverters meet the requirements of the BDEW medium-voltage guideline. They are capable of supporting the grid actively with reactive power and of remaining on-line in case of short-term grid failures. The web based MaxWeb data logger facilitates comfortable inverter monitoring and remote control (e.g. to reduce the inverter output).



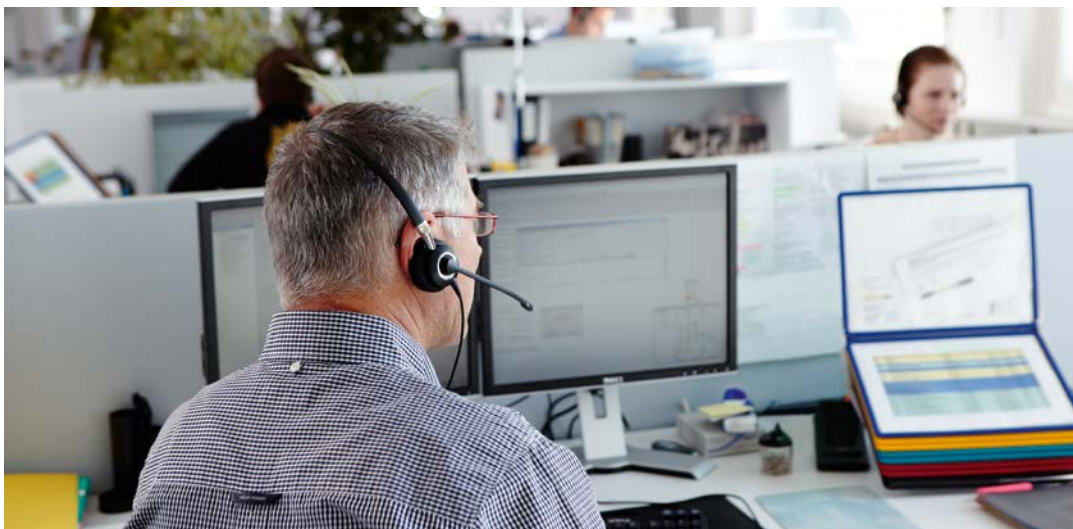
Comprehensive warranty

The central inverters of the TS-SV series are characterised by a two-year standard warranty that can be extended to five years at no charge by registering on our website. Altogether, you can take the warranty and the all-round carefree package MaxControl for up to 25 years. MaxControl (alarm system, device and system monitoring, data analysis) includes an availability guarantee: if the availability of the inverter (per year) is lower than 97%, we will provide you with a flat yield loss payment.



Competent after-sales service

And if a TS-SV inverter should at any time fail to operate as usual, our Service Centre will provide straight-forward and competent assistance. If the solution cannot be found quickly, our service technicians will come to the site right away. We also support our partners by providing them with practical training at regular intervals.



Specifications



		SolarMax 330/660/990/1320 TS-SV	SolarMax 360/720/1080/1440 TS-SV
Input values	MPP voltage range	450...800 V	510...800V
	Maximum DC voltage	900 V	900 V
	Maximum DC current	720/1440/2160/2880 A	720/1440/2160/2880 A
	Number of MPP-Trackers	1 or 3/6/9/12	1 or 3/6/9/12
	Connection type	threaded bolts M8	
	Overvoltage Category	2	
Output values	Rated output power	330/660/990/1320kW ¹⁾	360/720/1080/1440kW ²⁾
	Maximum apparent output power	340/680/1020/1360 kVA	370/740/1110/1480 kVA
	Nominal mains voltage	3 x 280 V	3 x 320 V
	Maximum AC current	700/1400/2100/2800 A	666/1332/1998/2664 A
	Mains nominal frequency / range	50 Hz / 45 Hz...55 Hz (60 Hz / 55 Hz...65 Hz upon request)	
	Power factor cos(φ)	Adjustable from 0.8 overexcited to 0.8 underexcited	
	Distortion factor at rated output power	< 3%	
	Connection type	3 x 3 threaded bolts M8	
	Grid connection	Three-phase (without a neutral conductor)	
	Overvoltage Category	2	
Efficiency	Max. efficiency	98%	98%
	European efficiency ³⁾	97.2%	97.4%
Power input	Own consumption (night)	< 7/14/21/28 W	
Ambient conditions	Protection type compliant with EN 60529	IP20	
	Ambient temperature range	-20 °C...+50 °C	
	Ambient temperature range for rated power output	-20 °C...+45 °C	
	Relative humidity	0...98% (no condensation)	
	Maximum altitude above sea level	2000m (without derating)	
	Noise emissions	< 65dBA (↔ 1.5 m)	
	Degree of soiling	PD2	
Configuration	Display (in MCU)	Graphic LC display with backlight and status LED	
	Data logger (in MCU)	Data logger for energy yield, peak output, and operating duration for the last 31 days, 12 months, and 10 years	
	Galvanic insulation	No galvanic isolation: direct connection to MV transformer	
Standards & guidelines	EMC	EN 61000-6-2/EN 61000-6-4	
	Standard / guideline compliance	G59/2 / BDEW MV Guideline / PPC Guide / RD 661	
	Device safety	IEC/EN 62109-1	
Interfaces	Data communication (in MCU)	RS485 / Ethernet via two RJ45 sockets	
	Status signalling contact (in MCU)	Potential-free terminal contact pair (configurable function)	
	Alarm input (in MCU)	Terminal contact pair for connection to MaxConnect plus	
	Inverter shut-off 1	Two terminal contact pairs (can be chain-linked via several SolarMax TS-SV)	
	Inverter shut-off 2 (in MCU)	Terminal contact pair	
Weight & dimensions	Weight	990/1980/2970/3960 kg	
	Dimensions in mm (W x H x D)	1/2/3/4 x 1200 x 1970 x 800 mm	
Warranty	Standard warranty	2 years (free extension to 5 years after registration)	
	Warranty extensions	to 10, 15, 20, or 25 years	

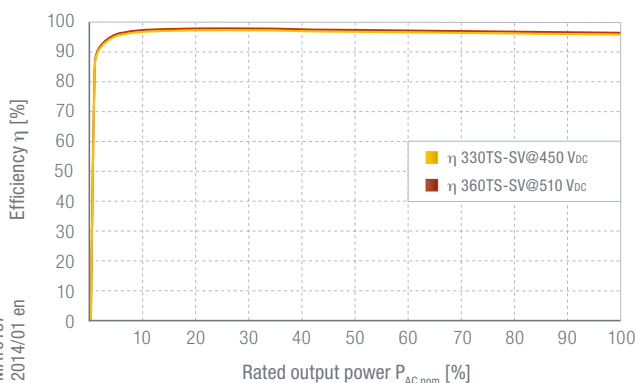
¹⁾ at cos(φ) = 1, UAC = 280 V

All rights, amendments and errors reserved.

²⁾ at cos(φ) = 1, UAC = 320 V

³⁾ in single-MPPT operation with active partial load optimisation (see operating instructions parameter configuration with MaxTalk 2 Pro)

Efficiency SolarMax TS-SV ³⁾



Appendix C Battery Datasheet

stored energy solutions for a demanding world

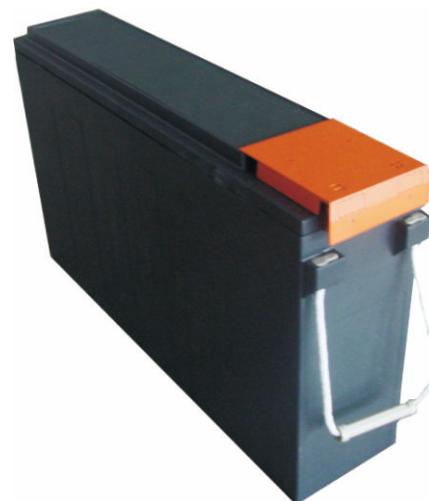
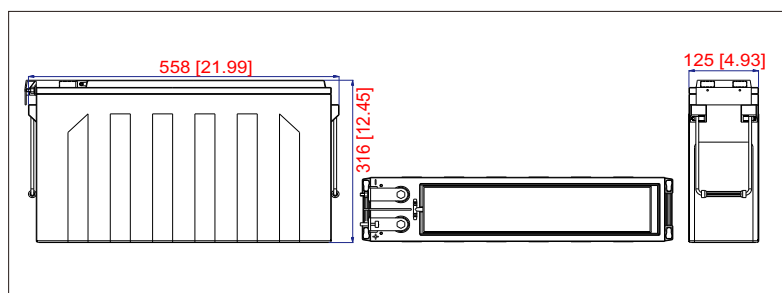
Narada

Model: **AG12V200**

AcmeG Series

The AcmeG range front access gel batteries is designed based on the Acme series. Using the polymer gel electrolyte with real front access structure. The state of the art internal and external design ensures AcmeG the high reliability and makes the installations quite simple and safe when placed on a standard relay rack tray or in a closed cabinet. The design float life is 12 years at 20°C(68°F).

Dimensions—mm [inch]



Specifications

Battery Model	AG12V200
Nominal Voltage	12V
Rated Capacity	200Ah - 20A for 10h to 1.80V/cell @25°C(77°F)
Typical Weight	61.0 kg
Internal Resistance	Approx 3.50mΩ
Temperature Ranges	Operation (maximum): -40°C to 50°C(-40°F to 122°F) Operation (recommended): 15°C to 25°C(59°F to 77°F) Storage: -20°C to 40°C(-4°F to 104°F)
Float Voltage	2.25V/cell@25°C(77°F)
Recommended Maximum Charging Current Limit	50 A
Equalize and Cycle Service	2.35V~2.40V/cell@25°C(77°F)
Self Discharge	The residual capacity is above 90% after 90 days storage(25°C/77°F)
Terminal	M6 Female
Terminal Hardware Torque	8 ± 1.0Nm
Container Material	ABS (V0 optional)

Constant Current Discharge Characteristics Units: Amperes (25°C, 77°F)

End voltage per cell	5MIN	15MIN	30MIN	45MIN	1HR	2HR	3HR	4HR	5HR	6HR	8HR	10HR	12HR	20HR	24HR
1.60V	554	348	229	168	138	78.8	55.9	44.1	37.5	32.0	25.3	20.4	17.5	10.9	9.13
1.67V	519	342	226	168	136	78.1	55.6	44.1	37.2	31.7	25.1	20.3	17.4	10.8	9.05
1.70V	501	336	222	167	135	77.7	55.4	44.0	37.1	31.6	25.0	20.2	17.3	10.7	9.01
1.75V	475	321	213	164	134	77.1	55.1	43.8	36.9	31.2	24.8	20.1	17.2	10.6	8.93
1.80V	427	295	201	156	130	75.5	54.2	43.3	36.3	30.7	24.6	20.0	17.1	10.6	8.89
1.83V	389	276	191	150	128	73.5	53.2	42.8	35.5	29.9	24.3	19.8	16.9	10.5	8.84
1.85V	370	263	187	144	124	71.5	52.3	42.2	35.0	29.6	24.0	19.6	16.8	10.5	8.79

Discharge Data with Constant Power Units: Watts per cell (25°C, 77°F)

End voltage per cell	5MIN	15MIN	30MIN	45MIN	1HR	2HR	3HR	4HR	5HR	6HR	8HR	10HR	12HR	20HR	24HR
1.60V	967	642	411	330	281	166	118	91.9	77.3	65.8	50.5	43.3	37.5	22.9	19.1
1.67V	914	625	408	328	279	166	118	91.5	77.0	65.6	50.4	43.2	37.4	22.8	19.0
1.70V	881	613	406	327	278	165	117	91.3	76.7	65.5	50.3	43.2	37.3	22.7	19.0
1.75V	814	590	398	322	274	164	117	90.8	76.2	65.1	49.9	43.1	37.2	22.4	18.9
1.80V	748	558	386	312	267	161	114	89.7	75.1	64.6	49.6	42.9	37.0	22.1	18.9
1.83V	726	528	374	304	260	156	112	88.6	74.0	63.9	49.4	42.6	36.7	21.9	18.8
1.85V	712	504	366	297	254	153	109	87.5	72.7	63.1	49.1	42.5	36.5	21.6	18.8

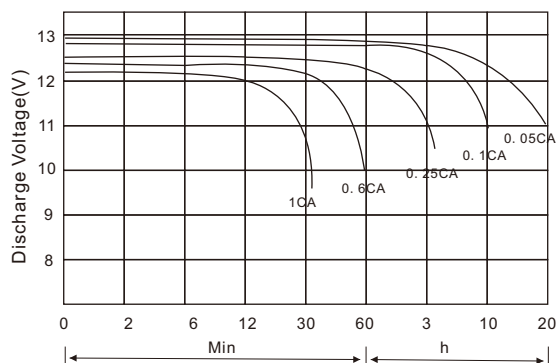
stored energy solutions for a demanding world

NARADA

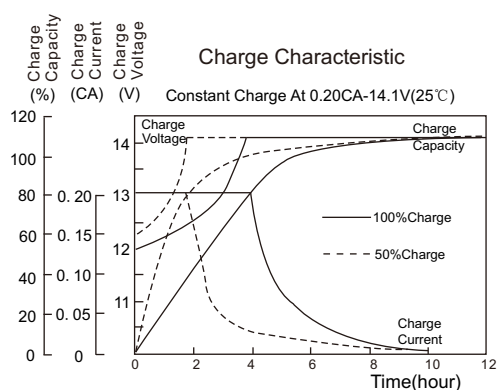
Model: **AG12V200**

AcmeG Series

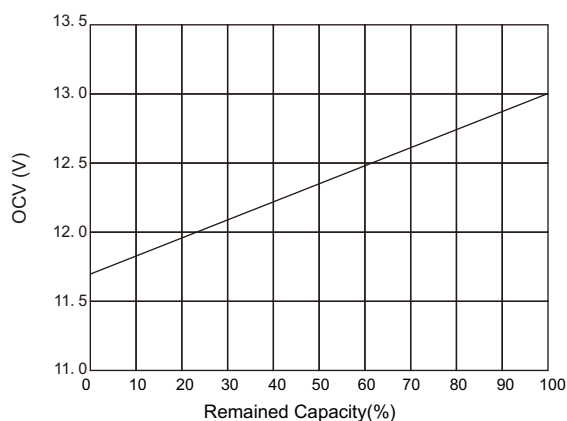
Terminal Voltage(V) vs. Discharge Time (25°C, 77°F)



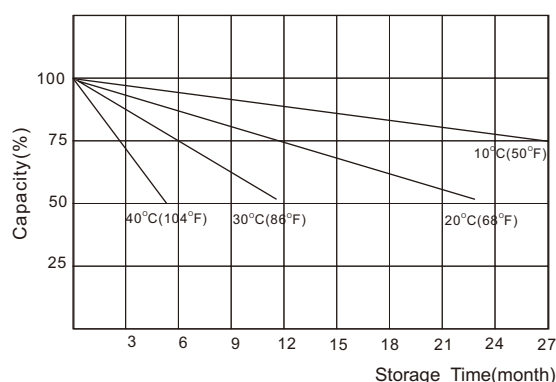
Battery Voltage vs. Charge Time



Relationship of OCV vs. State of Charge



Capacity Retention Characteristic



Charging Procedures

Application	Charge Voltage (V/Cell)			Max. Charge Current
	Temperature	Set Point	Allowable Range	
Cycle	25°C	2.40	2.35~2.40	0.25C
Standby	25°C	2.25	2.23~2.27	

Discharge Current VS. Discharge Voltage

Final Discharge Voltage V/Cell	1.80	1.70	1.55	1.30
Discharge Current (A)	0.2C ≥ (A)	0.2C < (A) < 0.5C	0.5C < (A) < 1.0C	(A) > 1.0C

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Appendix D PVsyst Simulation PDFs

PVsyst - Simulation report

Grid-Connected System

Project: GRID CONNECTED WITHOUT SHADING

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 410 kWp

George Town - India



Project: GRID CONNECTED WITHOUT SHADING

Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:38
with V7.4.7

Project summary

Geographical Site

George Town

India

Situation

Latitude 13.09 °N

Longitude 80.28 °E

Altitude 24 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

George Town

Meteonorm 8.1 (1996-2015) - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Fixed plane

Tilt/Azimuth 15 / 20 °

Near Shadings

No Shadings

User's needs

Unlimited load (grid)

System information

PV Array

Nb. of modules

630 units

Pnom total

410 kWp

Inverters

Nb. of units

1 unit

Pnom total

360 kWac

Pnom ratio

1.138

Results summary

Produced Energy 635493 kWh/year Specific production 1552 kWh/kWp/year Perf. Ratio PR 82.29 %

Table of contents

Project and results summary	2
General parameters, PV Array Characteristics, System losses	3
Main results	4
Loss diagram	5
Predef. graphs	6
Single-line diagram	7



Project: GRID CONNECTED WITHOUT SHADING

Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:38
with V7.4.7

General parameters

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Orientation

Fixed plane

Tilt/Azimuth 15 / 20 °

Sheds configuration

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

No Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer

AE Solar

Model

AE 650ME-132

(Original PVsyst database)

Unit Nom. Power

650 Wp

Number of PV modules

630 units

Nominal (STC)

410 kWp

Modules

35 string x 18 In series

At operating cond. (50°C)

Pmpp

375 kWp

U mpp

617 V

I mpp

607 A

Total PV power

Nominal (STC)

410 kWp

Total

630 modules

Module area

1957 m²

Cell area

1834 m²

Inverter

Manufacturer

SolarMax

Model

Solarmax 360TS-SV Multi MPPT

(Original PVsyst database)

Unit Nom. Power

360 kWac

Number of inverters

1 unit

Total power

360 kWac

Operating voltage

510-800 V

Pnom ratio (DC:AC)

1.14

Power sharing within this inverter

Total inverter power

Total power

360 kWac

Number of inverters

1 unit

Pnom ratio

1.14

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const)

20.0 W/m²K

Uv (wind)

0.0 W/m²K/m/s

DC wiring losses

Global array res.

17 mΩ

Loss Fraction

1.5 % at STC

Module Quality Loss

Loss Fraction

-0.8 %

Module mismatch losses

Loss Fraction

2.0 % at MPP

Strings Mismatch loss

Loss Fraction

0.2 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000



Project: GRID CONNECTED WITHOUT SHADING

Variant: New simulation variant

PVsyst V7.4.7

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with V7.4.7

Main results

System Production

Produced Energy

635493 kWh/year

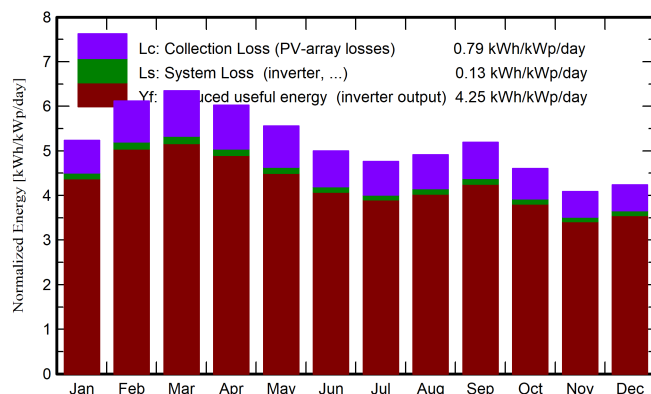
Specific production

1552 kWh/kWp/year

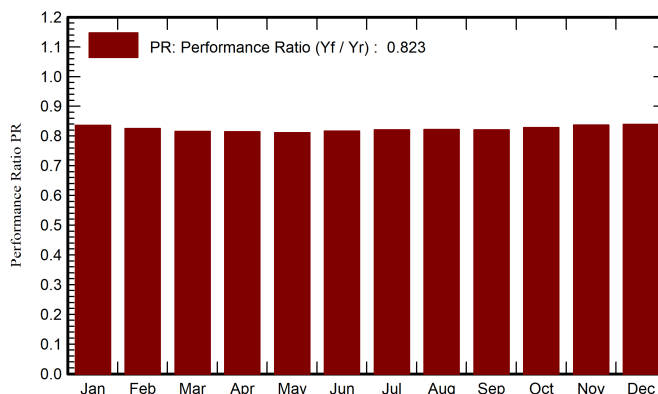
Perf. Ratio PR

82.29 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m²	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	ratio
January	144.2	66.32	25.11	162.4	158.9	57283	55621	0.836
February	156.4	60.47	26.21	171.4	168.0	59671	57892	0.825
March	189.7	74.76	28.44	196.7	192.8	67700	65663	0.815
April	184.7	81.64	30.16	180.7	176.8	62060	60229	0.814
May	182.6	91.33	32.17	172.3	168.1	58891	57205	0.811
June	162.3	84.49	31.21	150.0	146.1	51582	50112	0.816
July	158.0	91.51	30.64	147.6	143.6	51017	49581	0.821
August	157.1	89.73	29.68	152.3	148.7	52768	51244	0.822
September	153.7	78.09	28.85	155.7	152.2	53891	52315	0.820
October	134.9	74.42	27.86	142.8	139.9	49923	48451	0.828
November	112.4	62.00	25.92	122.7	120.0	43285	42018	0.837
December	117.8	63.02	25.21	131.4	128.5	46512	45164	0.840
Year	1853.9	917.78	28.47	1885.9	1843.6	654583	635493	0.823

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

PR Performance Ratio



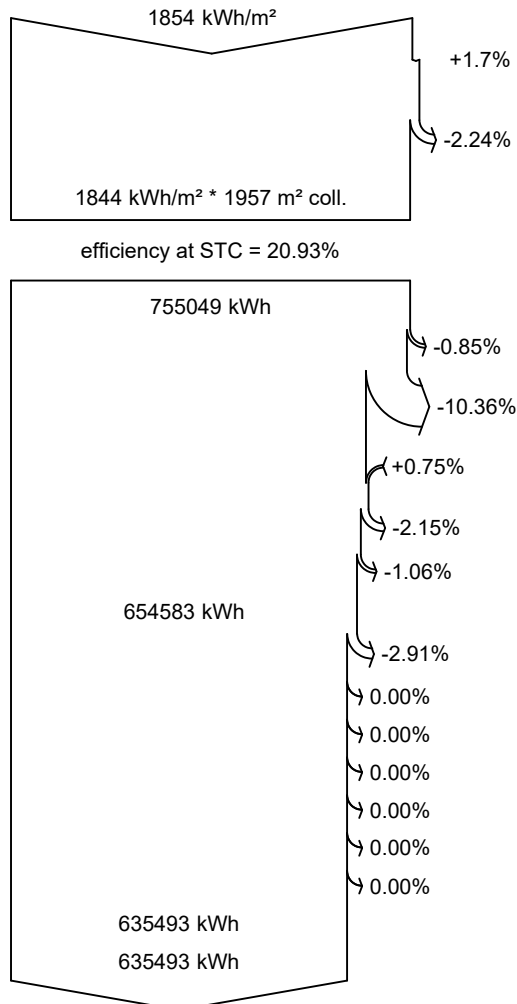
Project: GRID CONNECTED WITHOUT SHADING

Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
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with V7.4.7

Loss diagram



Global horizontal irradiation

Global incident in coll. plane

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Mismatch loss, modules and strings

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Energy injected into grid

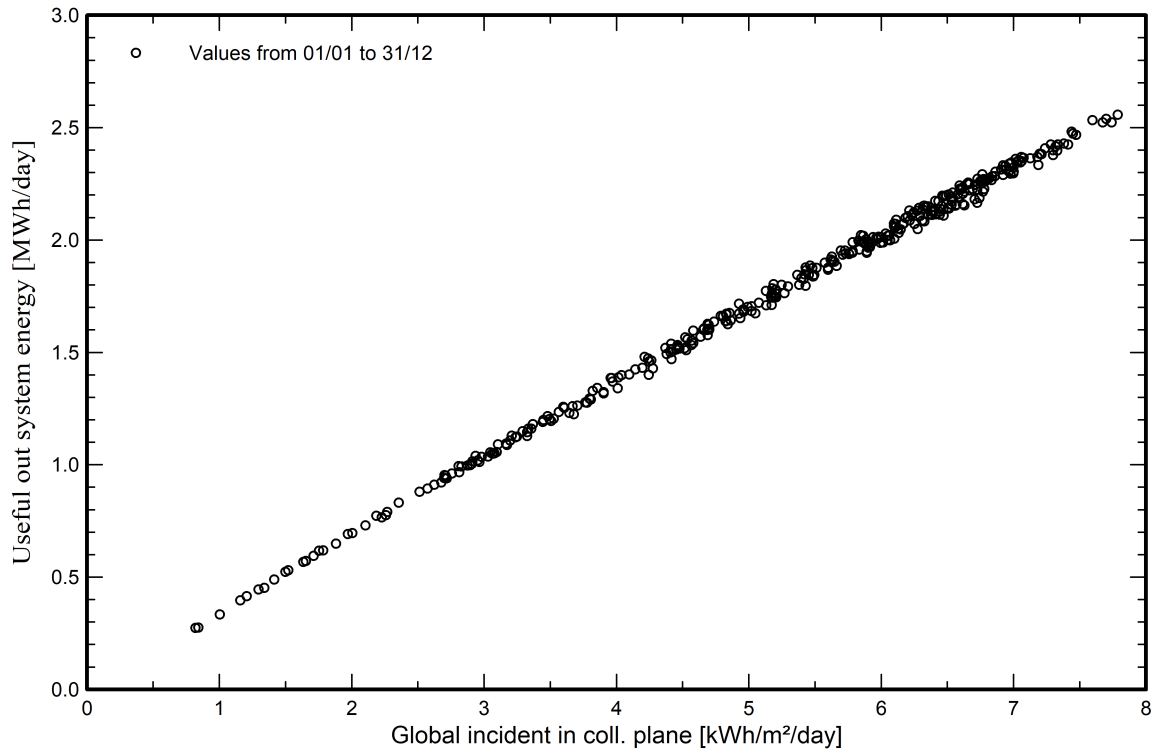


PVsyst V7.4.7

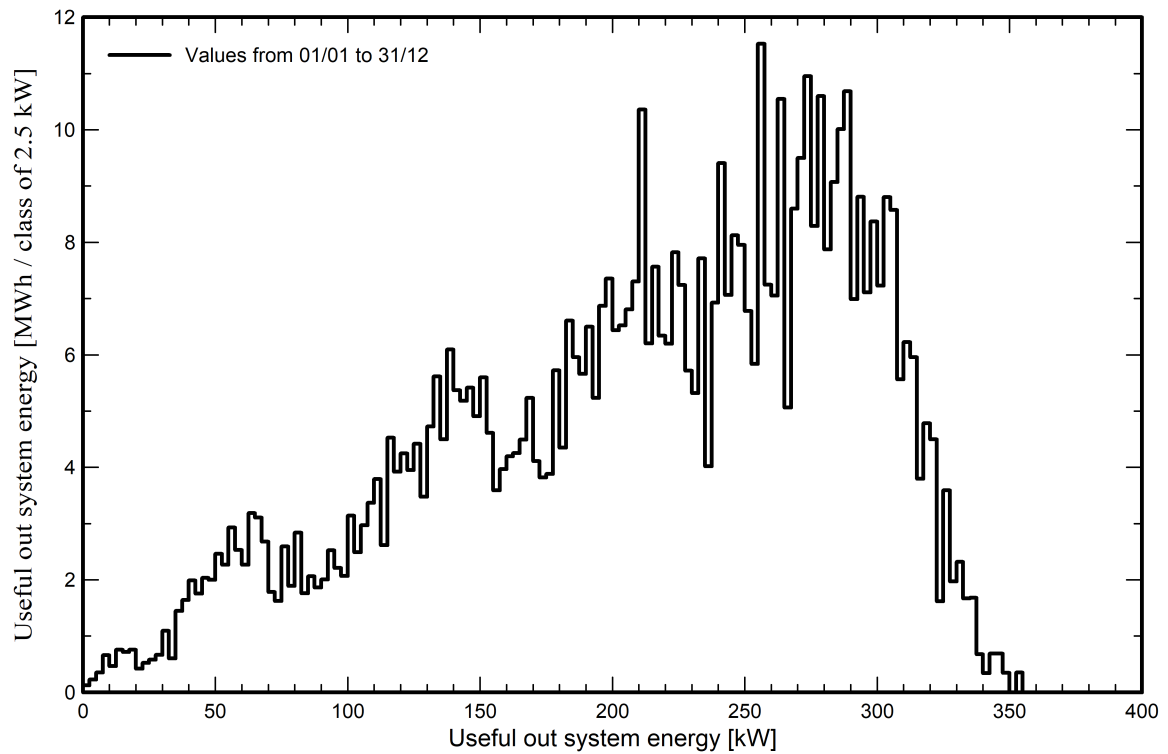
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with V7.4.7

Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

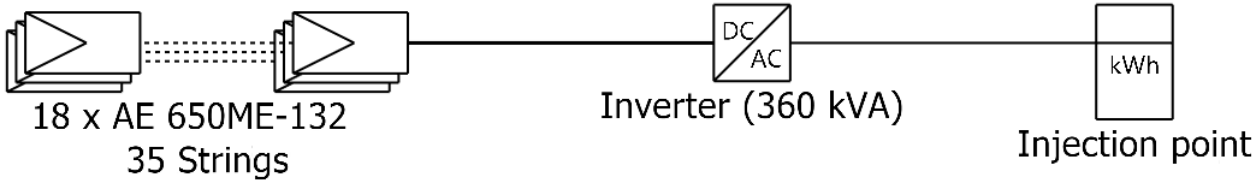




PVsyst V7.4.7

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with V7.4.7

Single-line diagram



PV module	AE 650ME-132
Inverter	Solarmax 360TS-SV Multi MPPT
String	18 x AE 650ME-132

GRID CONNECTED WITHOUT S
HADING

VC0 : New simulation variant

25/03/26

PVsyst - Simulation report

Grid-Connected System

Project: GRID CONNECTED WITH SHADING

Variant: New simulation variant

Sheds on ground

System power: 410 kWp

George Town - India



Project: GRID CONNECTED WITH SHADING

Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:45
with V7.4.7

Project summary

Geographical Site

George Town

India

Situation

Latitude 13.09 °N

Longitude 80.28 °E

Altitude 24 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

George Town

Meteonorm 8.1 (1996-2015) - Synthetic

System summary

Grid-Connected System

PV Field Orientation

Fixed plane

Tilt/Azimuth 15 / 20 °

Sheds on ground

Near Shadings

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

System information

PV Array

Nb. of modules

630 units

Pnom total

410 kWp

Inverters

Nb. of units

1 unit

Pnom total

360 kWac

Pnom ratio

1.138

Results summary

Produced Energy 622518 kWh/year Specific production 1520 kWh/kWp/year Perf. Ratio PR 80.61 %

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Project and results summary	2
General parameters, PV Array Characteristics, System losses	3
Near shading definition - Iso-shadings diagram	4
Main results	5
Loss diagram	6
Predef. graphs	7
Single-line diagram	8



Project: GRID CONNECTED WITH SHADING

Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:45
with V7.4.7

General parameters

Grid-Connected System

PV Field Orientation

Orientation

Fixed plane
Tilt/Azimuth 15 / 20 °

Sheds on ground

Sheds configuration

Nb. of sheds 66 units

Sizes

Sheds spacing 5.65 m
Collector width 2.63 m
Ground Cov. Ratio (GCR) 46.5 %
Top inactive band 0.02 m
Bottom inactive band 0.02 m

Shading limit angle

Limit profile angle 12.5 °

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer AE Solar
Model AE 650ME-132
(Original PVsyst database)

Unit Nom. Power 650 Wp
Number of PV modules 630 units
Nominal (STC) 410 kWp
Modules 35 string x 18 In series

At operating cond. (50°C)

Pmpp 375 kWp
U mpp 617 V
I mpp 607 A

Total PV power

Nominal (STC) 410 kWp
Total 630 modules
Module area 1957 m²
Cell area 1834 m²

Inverter

Manufacturer SolarMax
Model Solarmax 360TS-SV Multi MPPT
(Original PVsyst database)

Unit Nom. Power 360 kWac
Number of inverters 1 unit
Total power 360 kWac
Operating voltage 510-800 V
Pnom ratio (DC:AC) 1.14
Power sharing within this inverter

Total inverter power

Total power 360 kWac
Number of inverters 1 unit
Pnom ratio 1.14

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 17 mΩ
Loss Fraction 1.5 % at STC

Module Quality Loss

Loss Fraction -0.8 %

Module mismatch losses

Loss Fraction 2.0 % at MPP

Strings Mismatch loss

Loss Fraction 0.2 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000



Project: GRID CONNECTED WITH SHADING

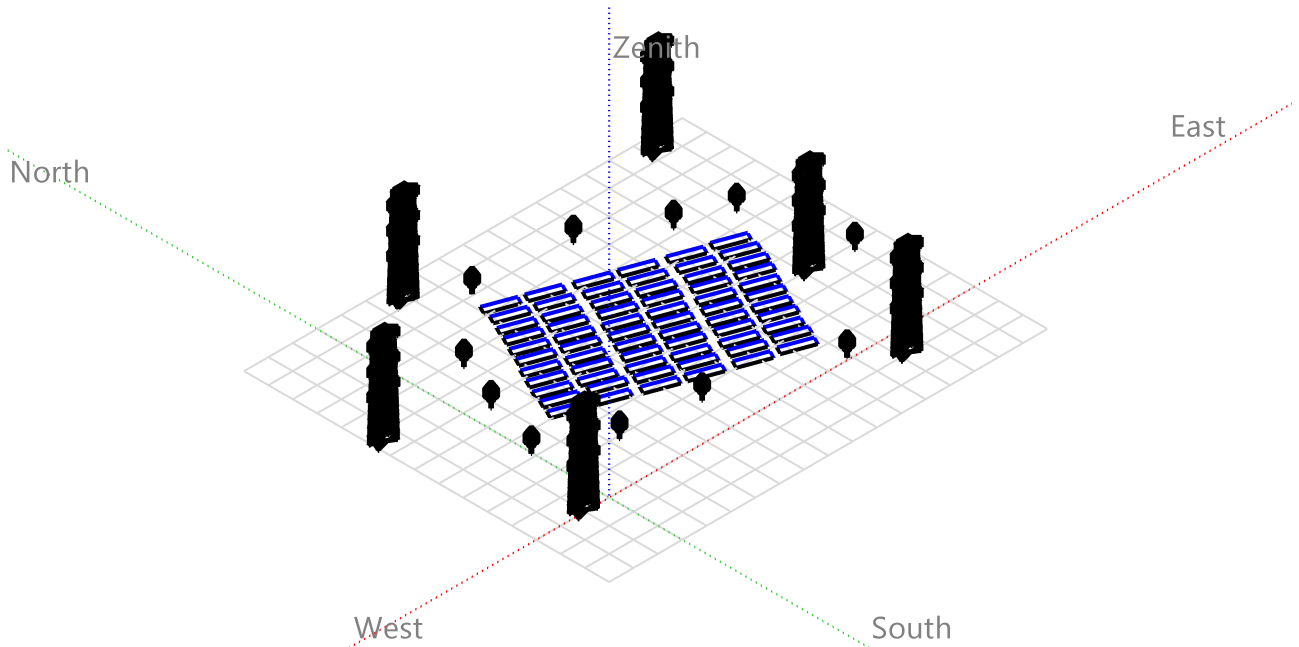
Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:45
with V7.4.7

Near shadings parameter

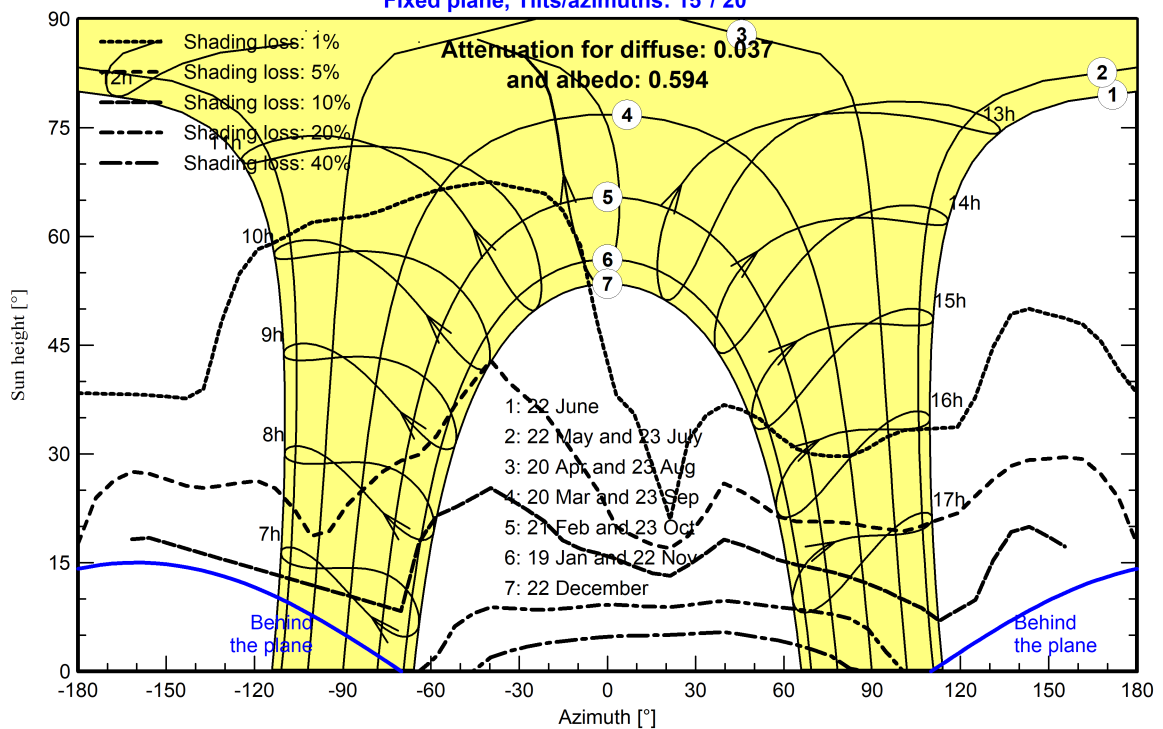
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 15°/ 20°





Project: GRID CONNECTED WITH SHADING

Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:45
with V7.4.7

Main results

System Production

Produced Energy

622518 kWh/year

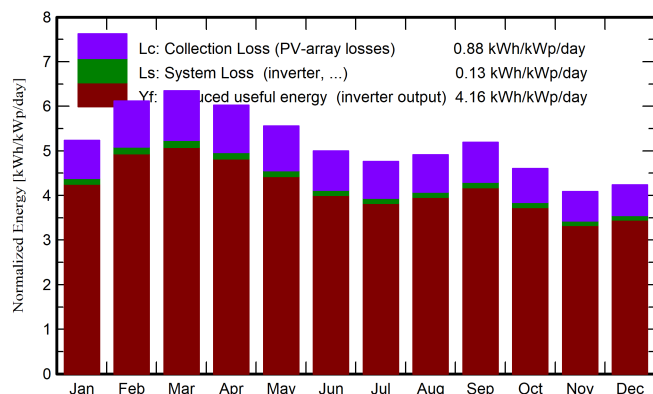
Specific production

1520 kWh/kWp/year

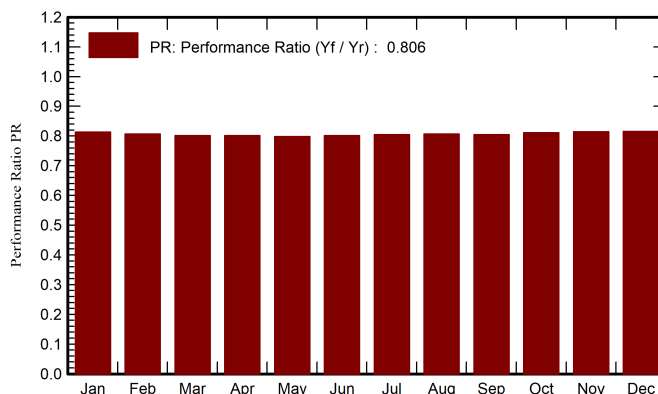
Perf. Ratio PR

80.61 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	144.2	66.32	25.11	162.4	154.4	55703	54090	0.813
February	156.4	60.47	26.21	171.4	164.2	58356	56619	0.807
March	189.7	74.76	28.44	196.7	189.4	66534	64532	0.801
April	184.7	81.64	30.16	180.7	173.9	61078	59278	0.801
May	182.6	91.33	32.17	172.3	165.3	57941	56285	0.798
June	162.3	84.49	31.21	150.0	143.5	50692	49248	0.802
July	158.0	91.51	30.64	147.6	140.8	50055	48645	0.805
August	157.1	89.73	29.68	152.3	146.0	51827	50330	0.807
September	153.7	78.09	28.85	155.7	149.2	52861	51316	0.805
October	134.9	74.42	27.86	142.8	136.9	48862	47421	0.811
November	112.4	62.00	25.92	122.7	116.8	42149	40914	0.815
December	117.8	63.02	25.21	131.4	124.6	45150	43841	0.815
Year	1853.9	917.78	28.47	1885.9	1804.8	641208	622518	0.806

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

PR Performance Ratio



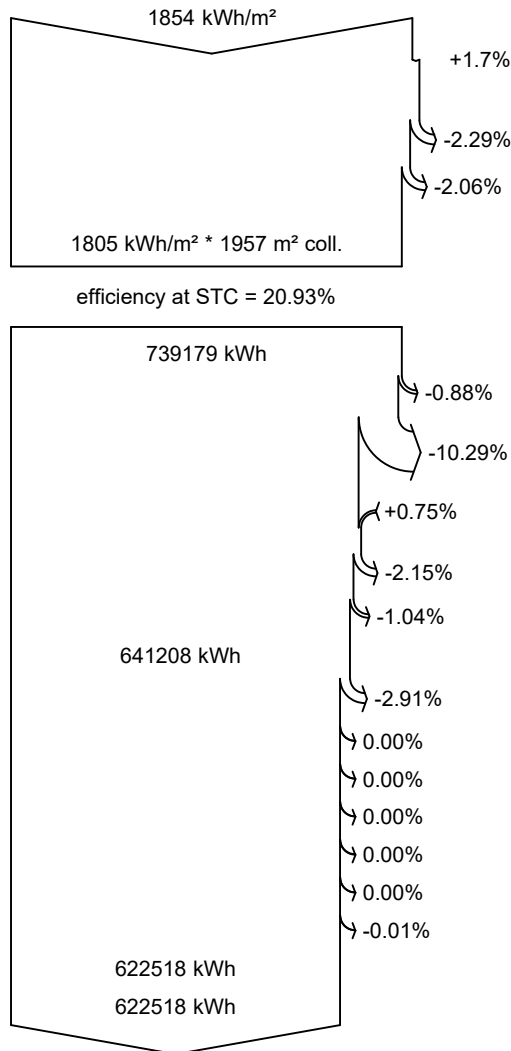
Project: GRID CONNECTED WITH SHADING

Variant: New simulation variant

PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:45
with V7.4.7

Loss diagram



Global horizontal irradiation

Global incident in coll. plane

Near Shadings: irradiance loss

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Mismatch loss, modules and strings

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Energy injected into grid

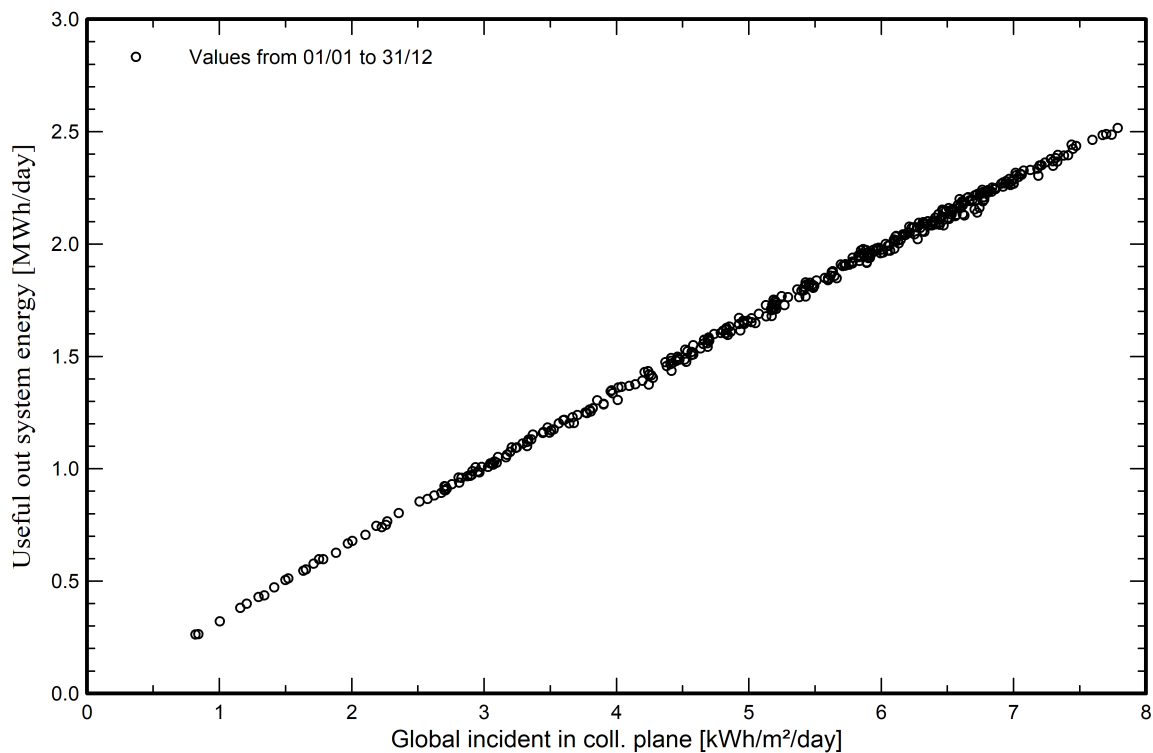


PVsyst V7.4.7

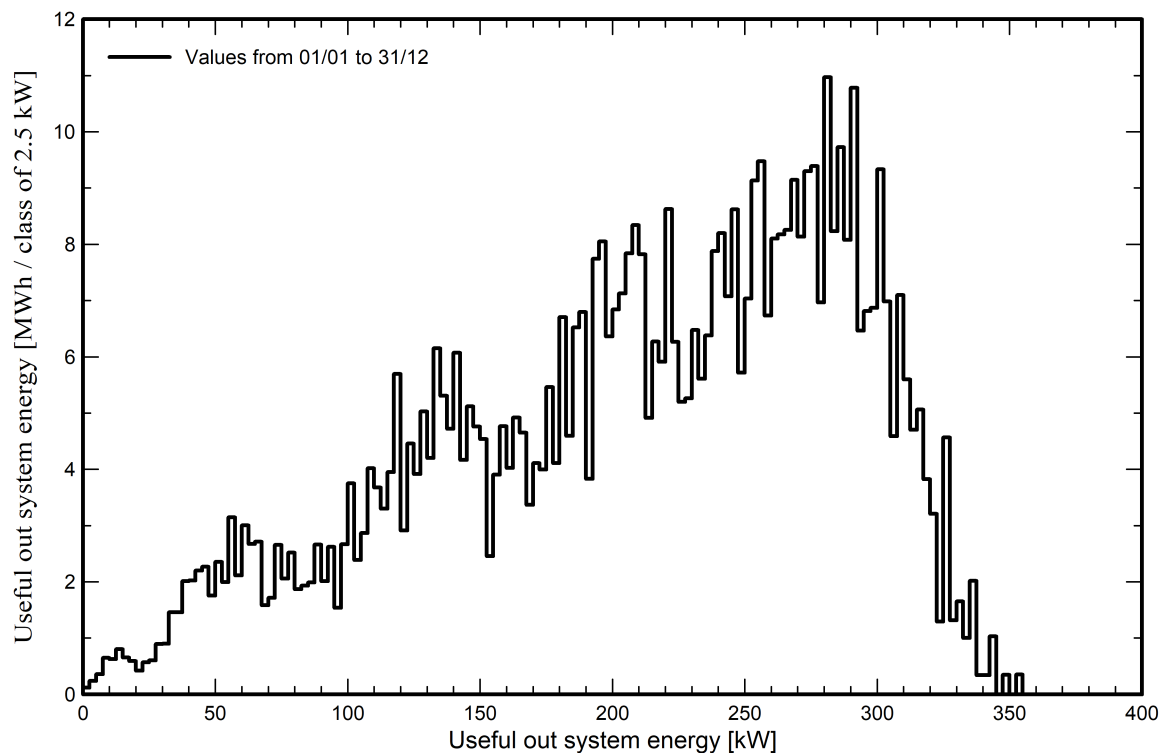
VC0, Simulation date:
25/03/26 20:45
with V7.4.7

Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

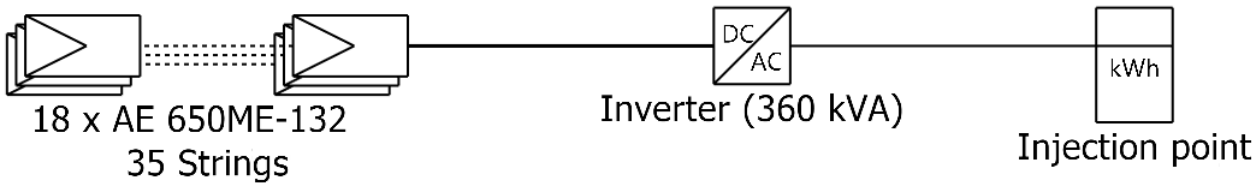




PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:45
with V7.4.7

Single-line diagram



PV module	AE 650ME-132
Inverter	Solarmax 360TS-SV Multi MPPT
String	18 x AE 650ME-132

GRID CONNECTED WITH SHADING

VC0 : New simulation variant

25/03/26

PVsyst - Simulation report

Standalone system

Project: standalone without shading 24-03-2026

Variant: New simulation variant

Standalone system with batteries

System power: 410 kWp

George Town - India

**PVsyst V7.4.7**

VC0, Simulation date:
25/03/26 20:53
with V7.4.7

Project summary**Geographical Site****George Town**

India

Situation

Latitude 13.09 °N

Longitude 80.28 °E

Altitude 24 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

George Town

Meteonorm 8.1 (1996-2015) - Synthetic

System summary**Standalone system****PV Field Orientation**

Fixed plane

Tilt/Azimuth 15 / 20 °

Standalone system with batteries**User's needs**

Fixed constant load

75.0 kW

Global

657 MWh/Year

System information**PV Array**

Nb. of modules

630 units

Pnom total

410 kWp

Battery pack

Technology

Lead-acid, sealed, Gel

Nb. of units

3540 units

Voltage

240 V

Capacity

35400 Ah

Results summary

Useful energy from solar 598274 kWh/year

Specific production 1461 kWh/kWp/year

Perf. Ratio PR 77.47 %

Missing Energy 58726 kWh/year

Available solar energy 626300 kWh/year

Solar Fraction SF 91.06 %

Excess (unused) 12145 kWh/year

Table of contents

Project and results summary	2
General parameters, PV Array Characteristics, System losses	3
Main results	4
Loss diagram	5
Predef. graphs	6

**PVsyst V7.4.7**

VC0, Simulation date:
25/03/26 20:53
with V7.4.7

General parameters**Standalone system****PV Field Orientation****Orientation**

Fixed plane

Tilt/Azimuth 15 / 20 °

User's needs

Fixed constant load

75.0 kW

Global

657 MWh/Year

Standalone system with batteries**Sheds configuration**

No 3D scene defined

Models used

Transposition Perez

Diffuse Perez, Meteonorm

Circumsolar separate

PV Array Characteristics**PV module**

Manufacturer

AE Solar

Model

AE 650ME-132

(Original PVsyst database)

Unit Nom. Power

650 Wp

Number of PV modules

630 units

Nominal (STC)

410 kWp

Modules

35 string x 18 In series

At operating cond. (50°C)

Pmpp

375 kWp

U mpp

617 V

I mpp

607 A

Controller

Universal controller

Technology

MPPT converter

Temp coeff.

-5.0 mV/°C/Elem.

Converter

Maxi and EURO efficiencies

97.0 / 95.0 %

Total PV power

Nominal (STC)

410 kWp

Total

630 modules

Module area

1957 m²

Cell area

1834 m²**Battery**

Manufacturer

Narada

Model

AcmeG 12V 200

Technology

Lead-acid, sealed, Gel

Nb. of units

177 in parallel x 20 in series

Discharging min. SOC

20.0 %

Stored energy

7005.6 kWh

Battery Pack Characteristics

Voltage

240 V

Nominal Capacity

35400 Ah (C10)

Temperature

Fixed 20 °C

Battery Management control

Threshold commands as

SOC calculation

Charging

SOC = 0.90 / 0.75

approx.

265.8 / 253.4 V

Discharging

SOC = 0.20 / 0.45

approx.

236.4 / 245.7 V

Array losses**Thermal Loss factor**

Module temperature according to irradiance

Uc (const) 20.0 W/m²KUv (wind) 0.0 W/m²K/m/s**DC wiring losses**

Global array res.

17 mΩ

Loss Fraction

1.5 % at STC

Serie Diode Loss

Voltage drop

0.7 V

Loss Fraction

0.1 % at STC

Module Quality Loss

Loss Fraction -0.8 %

Module mismatch losses

Loss Fraction

2.0 % at MPP

Strings Mismatch loss

Loss Fraction

0.2 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000



PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:53
with V7.4.7

Main results

System Production

Useful energy from solar 598274 kWh/year
Available solar energy 626300 kWh/year
Excess (unused) 12145 kWh/year

Perf. Ratio PR 77.47 %
Solar Fraction SF 91.06 %

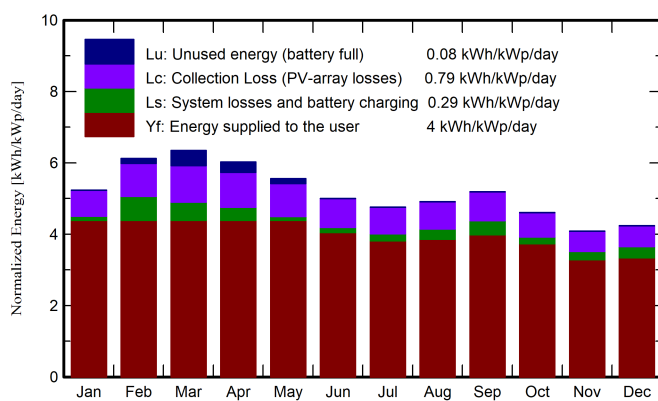
Loss of Load

Time Fraction 0.0 %
Missing Energy 58726 kWh/year

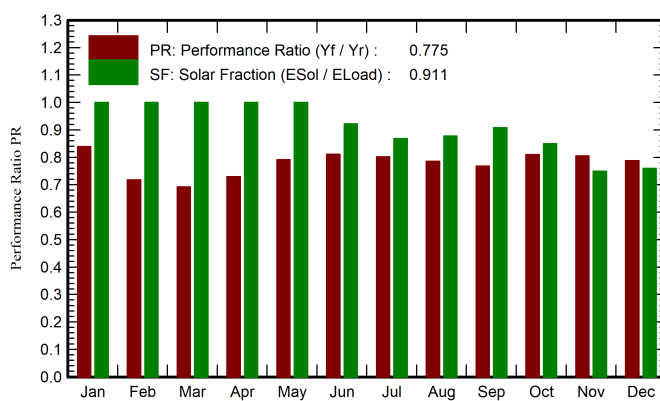
Battery aging (State of Wear)

Cycles SOW 96.5 %
Static SOW 91.7 %
Battery lifetime 12.0 years

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	GlobEff	E_Avail	EUnused	E_Miss	E_User	E_Load	SolFrac
	kWh/m ²	kWh/m ²	kWh	kWh	kWh	kWh	kWh	ratio
January	144.2	158.9	54882	0	0	55800	55800	1.000
February	156.4	168.0	57261	1497	0	50400	50400	1.000
March	189.7	192.8	65078	5389	0	55800	55800	1.000
April	184.7	176.8	59563	3530	0	54000	54000	1.000
May	182.6	168.1	56469	1723	0	55800	55800	1.000
June	162.3	146.1	49280	2	4213	49787	54000	0.922
July	158.0	143.6	48715	0	7358	48442	55800	0.868
August	157.1	148.7	50357	0	6804	48996	55800	0.878
September	153.7	152.2	51446	0	4994	49006	54000	0.908
October	134.9	139.9	47600	0	8402	47398	55800	0.849
November	112.4	120.0	41229	0	13562	40438	54000	0.749
December	117.8	128.5	44420	4	13394	42406	55800	0.760
Year	1853.9	1843.6	626300	12145	58726	598274	657000	0.911

Legends

GlobHor Global horizontal irradiation
GlobEff Effective Global, corr. for IAM and shadings
E_Avail Available Solar Energy
EUnused Unused energy (battery full)
E_Miss Missing energy

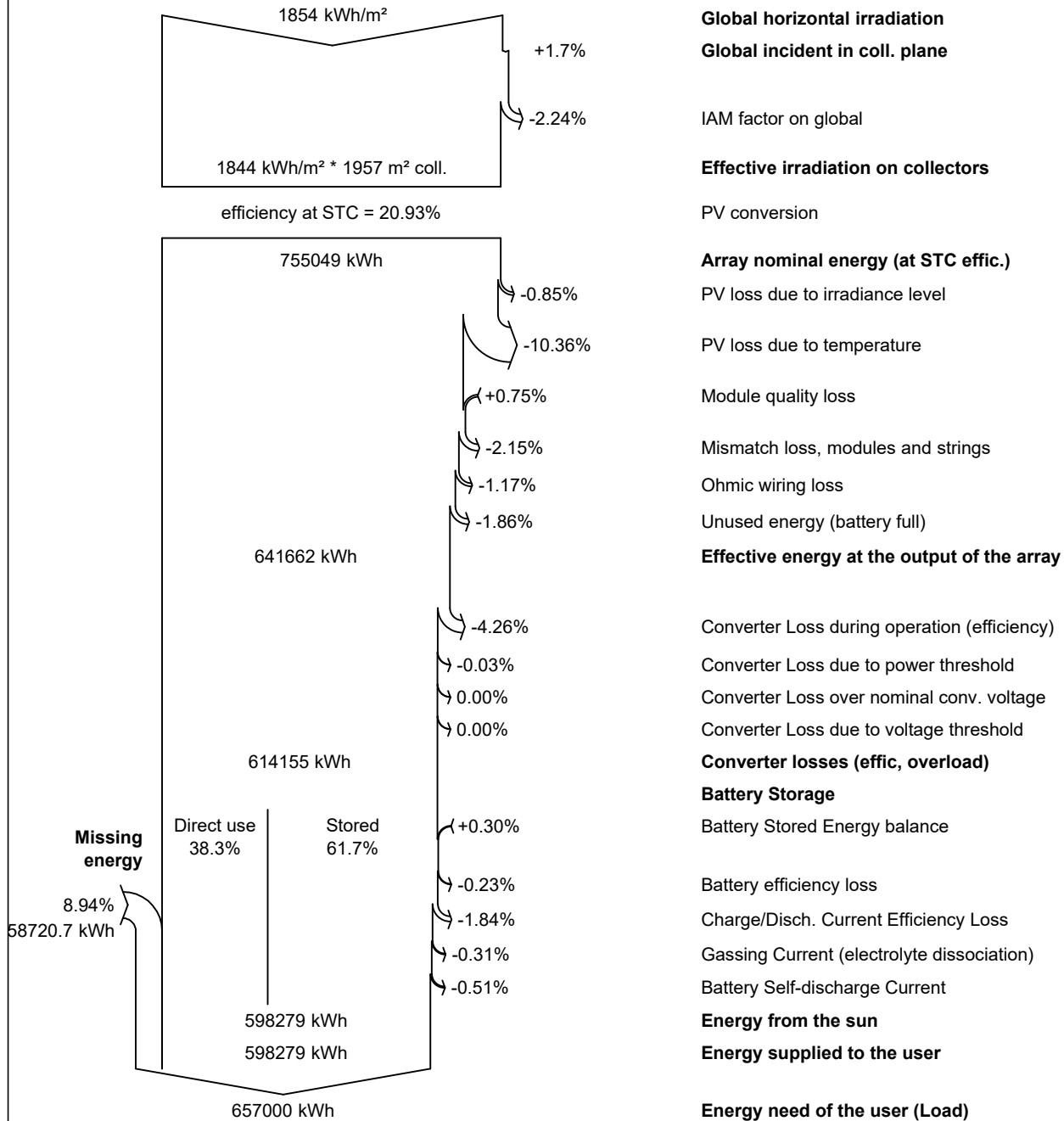
E_User Energy supplied to the user
E_Load Energy need of the user (Load)
SolFrac Solar fraction (EUsed / ELoad)



PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:53
with V7.4.7

Loss diagram



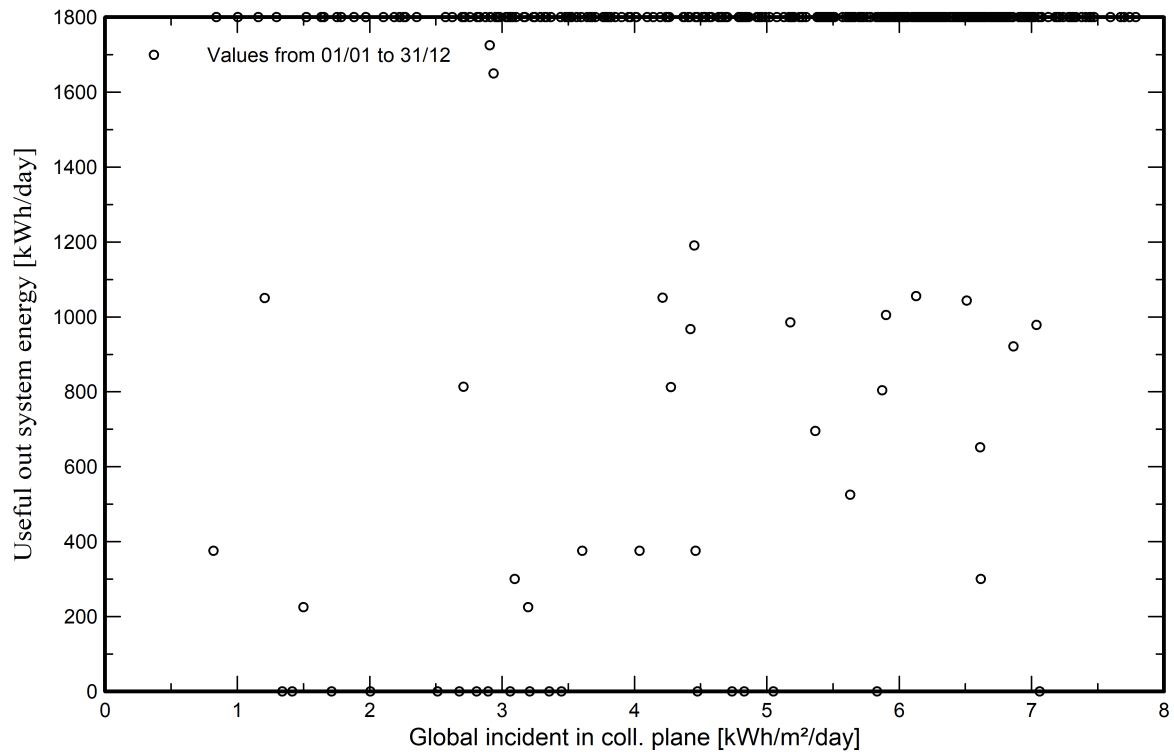


PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:53
with V7.4.7

Predef. graphs

Daily Input/Output diagram



PVsyst - Simulation report

Standalone system

Project: standalonw with shading 24-02-2026

Variant: New simulation variant

Standalone system with batteries

System power: 410 kWp

George Town - India



PVsyst V7.4.7

VC0, Simulation date:
25/03/26 21:05
with V7.4.7

Project summary

Geographical Site

George Town

India

Situation

Latitude 13.09 °N

Longitude 80.28 °E

Altitude 24 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

George Town

Meteonorm 8.1 (1996-2015) - Synthetic

System summary

Standalone system

PV Field Orientation

Fixed plane

Tilt/Azimuth 15 / 20 °

Standalone system with batteries

Near Shadings

Linear shadings : Fast (table)

User's needs

Fixed constant load

75.0 kW

Global

657 MWh/Year

System information

PV Array

Nb. of modules

630 units

Pnom total

410 kWp

Battery pack

Technology

Lead-acid, sealed, Gel

Nb. of units

3540 units

Voltage

240 V

Capacity

35400 Ah

Results summary

Useful energy from solar 590652 kWh/year

Specific production 1442 kWh/kWp/year

Perf. Ratio PR 76.48 %

Missing Energy 66348 kWh/year

Available solar energy 613087 kWh/year

Solar Fraction SF 89.90 %

Excess (unused) 7225 kWh/year

Table of contents

Project and results summary	2
General parameters, PV Array Characteristics, System losses	3
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**PVsyst V7.4.7**

VC0, Simulation date:
25/03/26 21:05
with V7.4.7

General parameters**Standalone system****PV Field Orientation****Orientation**

Fixed plane
Tilt/Azimuth 15 / 20 °

Near Shadings

Linear shadings : Fast (table)

Standalone system with batteries**Sheds configuration**

Nb. of sheds 66 units

Sizes

Sheds spacing 5.65 m
Collector width 2.63 m
Ground Cov. Ratio (GCR) 46.5 %
Top inactive band 0.02 m
Bottom inactive band 0.02 m

Shading limit angle

Limit profile angle 12.5 °

User's needs

Fixed constant load
75.0 kW
Global
657 MWh/Year

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

PV Array Characteristics**PV module**

Manufacturer AE Solar
Model AE 650ME-132
(Original PVsyst database)
Unit Nom. Power 650 Wp
Number of PV modules 630 units
Nominal (STC) 410 kWp
Modules 35 string x 18 In series

At operating cond. (50°C)

Pmpp 375 kWp
U mpp 617 V
I mpp 607 A

Controller

Universal controller
Technology MPPT converter
Temp coeff. -5.0 mV/°C/Elem.

Converter

Maxi and EURO efficiencies 97.0 / 95.0 %

Total PV power

Nominal (STC) 410 kWp
Total 630 modules
Module area 1957 m²
Cell area 1834 m²

Battery

Manufacturer Narada
Model AcmeG 12V 200
Technology Lead-acid, sealed, Gel
Nb. of units 177 in parallel x 20 in series
Discharging min. SOC 20.0 %
Stored energy 7005.6 kWh

Battery Pack Characteristics

Voltage 240 V
Nominal Capacity 35400 Ah (C10)
Temperature Fixed 20 °C

Battery Management control

Threshold commands as SOC calculation
Charging SOC = 0.90 / 0.75
approx. 265.8 / 253.4 V
Discharging SOC = 0.20 / 0.45
approx. 236.4 / 245.7 V



PVsyst V7.4.7

VC0, Simulation date:
25/03/26 21:05
with V7.4.7

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 17 mΩ
Loss Fraction 1.5 % at STC

Serie Diode Loss

Voltage drop 0.7 V
Loss Fraction 0.1 % at STC

Module Quality Loss

Loss Fraction -0.8 %

Module mismatch losses

Loss Fraction 2.0 % at MPP

Strings Mismatch loss

Loss Fraction 0.2 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

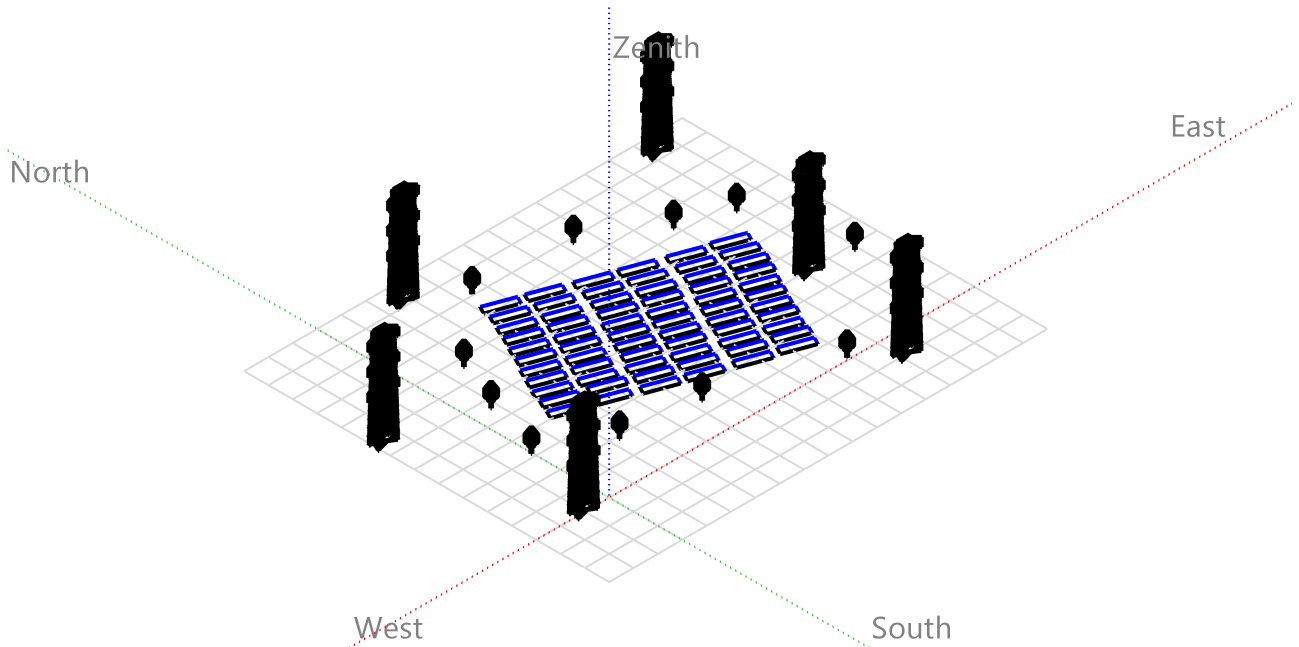


PVsyst V7.4.7

VC0, Simulation date:
25/03/26 21:05
with V7.4.7

Near shadings parameter

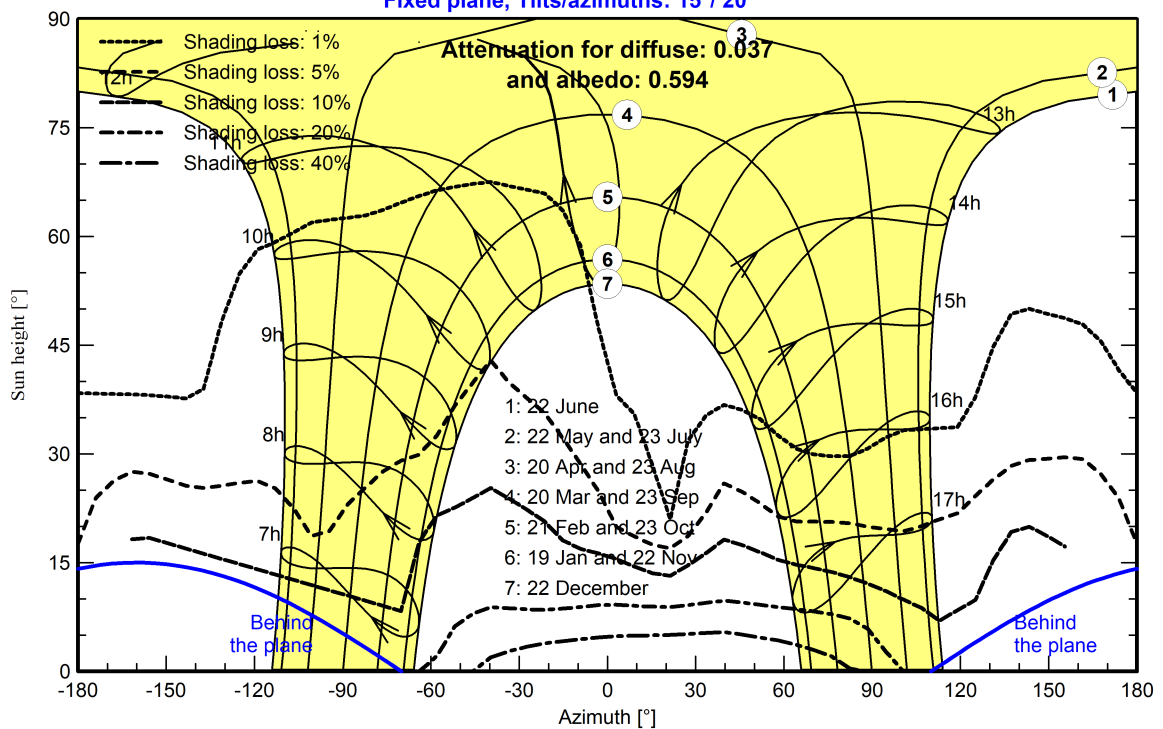
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 15°/ 20°





PVsyst V7.4.7

VC0, Simulation date:
25/03/26 21:05
with V7.4.7

Main results

System Production

Useful energy from solar 590652 kWh/year
Available solar energy 613087 kWh/year
Excess (unused) 7225 kWh/year

Loss of Load

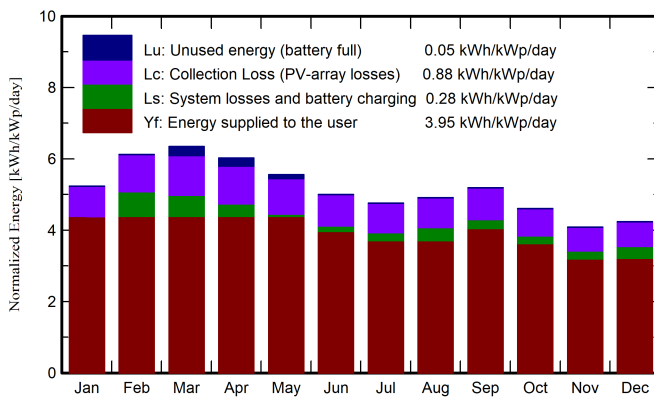
Time Fraction 0.0 %
Missing Energy 66348 kWh/year

Perf. Ratio PR 76.48 %
Solar Fraction SF 89.90 %

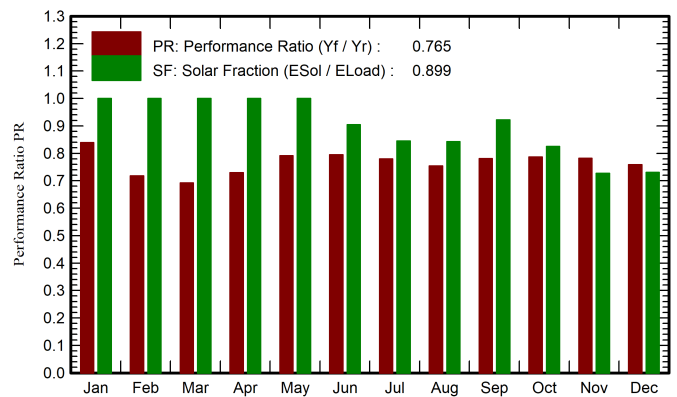
Battery aging (State of Wear)

Cycles SOW 96.5 %
Static SOW 91.7 %
Battery lifetime 12.0 years

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	GlobEff	E_Avail	EUnused	E_Miss	E_User	E_Load	SolFrac
	kWh/m ²	kWh/m ²	kWh	kWh	kWh	kWh	kWh	ratio
January	144.2	154.4	53347	0	0	55800	55800	1.000
February	156.4	164.2	55916	0	0	50400	50400	1.000
March	189.7	189.4	63868	3155	0	55800	55800	1.000
April	184.7	173.9	58578	2723	0	54000	54000	1.000
May	182.6	165.3	55522	1332	0	55800	55800	1.000
June	162.3	143.5	48408	2	5182	48818	54000	0.904
July	158.0	140.8	47775	4	8663	47137	55800	0.845
August	157.1	146.0	49438	0	8764	47036	55800	0.843
September	153.7	149.2	50445	0	4222	49778	54000	0.922
October	134.9	136.9	46570	0	9767	46033	55800	0.825
November	112.4	116.8	40122	2	14745	39255	54000	0.727
December	117.8	124.6	43097	7	15003	40797	55800	0.731
Year	1853.9	1804.8	613087	7225	66348	590652	657000	0.899

Legends

GlobHor Global horizontal irradiation
GlobEff Effective Global, corr. for IAM and shadings
E_Avail Available Solar Energy
EUnused Unused energy (battery full)
E_Miss Missing energy

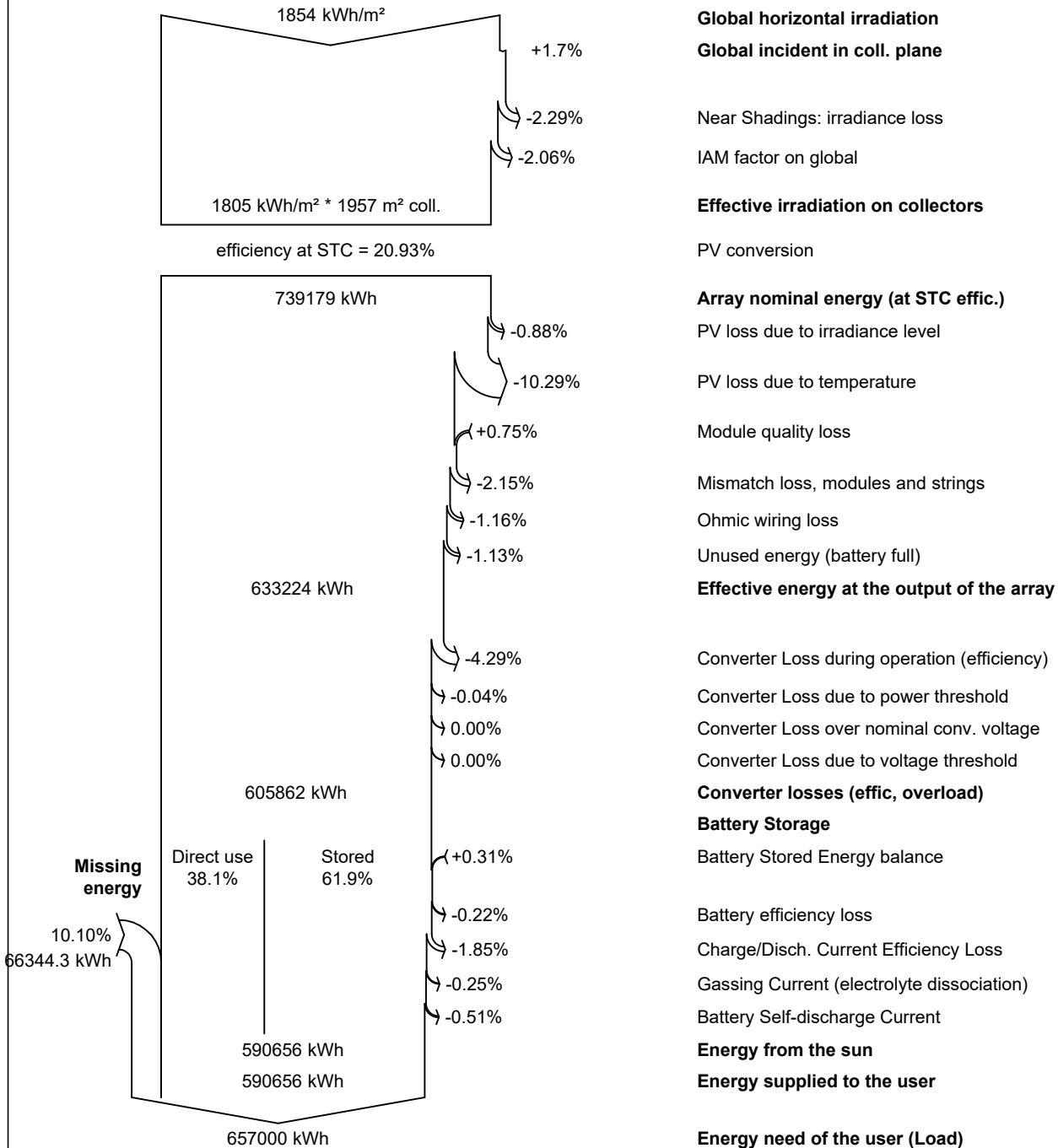
E_User Energy supplied to the user
E_Load Energy need of the user (Load)
SolFrac Solar fraction (EUsed / ELoad)



PVsyst V7.4.7

VC0, Simulation date:
25/03/26 21:05
with V7.4.7

Loss diagram



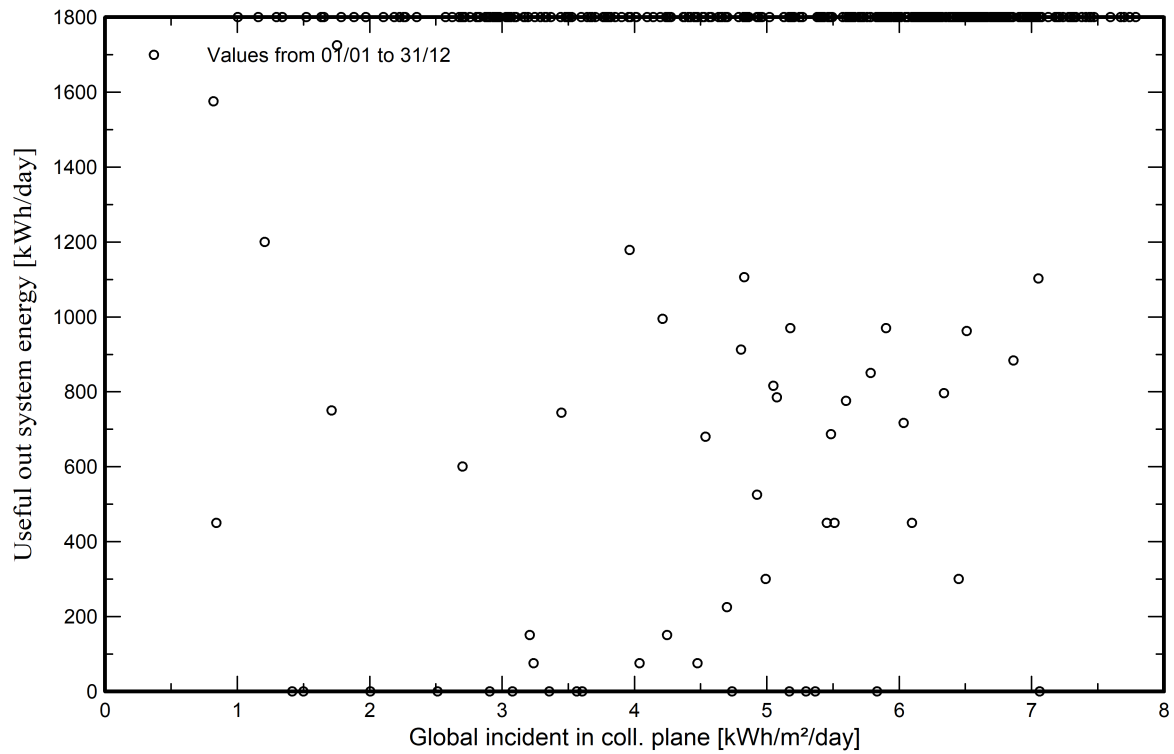


PVsyst V7.4.7

VC0, Simulation date:
25/03/26 21:05
with V7.4.7

Predef. graphs

Daily Input/Output diagram



Appendix E AI/ML MATLAB Codes

GRID CONNECTED WITHOUT SHADING

```
%% ===== GRID CONNECTED PV =====

clc; clear; close all;

%% === Load Dataset (GRID CONNECTED) ===

genData = readtable("C:\Users\gudit\Downloads\HOURLY DATA\GRID CONNECTED WITHOUT
SHADING_Project_VCO_HourlyRes_2.CSV");

disp(genData)

genData.date = datetime(genData.date,'InputFormat','dd-MM-yyyy HH:mm:ss');

% Valid operating condition

genData = genData(genData.GlobInc > 0 & genData.E_Grid > 0, :);

disp(genData)

%% === Features & Target ===

X = table2array(genData(:, ...
    {'GlobHor','WindVel','TArray','T_Amb','GlobInc'}));

y = genData.E_Grid; % Grid injected AC power

%% === STANDARDIZATION (VERY IMPORTANT) ===

X = normalize(X); % <-- FIX FOR WARNING

%% === Train-Test Split ===

cv = cvpartition(height(genData),'HoldOut',0.2);

XTrain = X(training(cv),:); yTrain = y(training(cv));

XTest = X(test(cv),:); yTest = y(test(cv));

%% === ML Models ===

models = {
    'Regression Tree', @() fitrtree(XTrain,yTrain);
```

```

'Random Forest', @() fitrensemble(XTrain,yTrain,'Method','Bag');
'SVR', @() fitrsvm(XTrain,yTrain,'Standardize',true);
'Gaussian Process', @() fitrgp(XTrain,yTrain,'Standardize',true);
'Neural Network', @() fitrnet(XTrain,yTrain);
};

numModels = size(models,1);
metrics = zeros(numModels,4);
yPreds = zeros(length(yTest),numModels);

%% === Train & Evaluate ===
for i = 1:numModels
    modelName = models{i,1};
    fprintf('\nTraining Model: %s\n', modelName);

    model = models{i,2}();
    yPred = predict(model,XTest);

    rmse = sqrt(mean((yTest-yPred).^2));
    mae = mean(abs(yTest-yPred));
    r2 = 1 - sum((yTest-yPred).^2)/sum((yTest-mean(yTest)).^2);
    mape = mean(abs((yTest-yPred)./yTest))*100;

    metrics(i,:) = [rmse mae r2 mape];
    yPreds(:,i) = yPred;

    fprintf('%s → RMSE: %.2f | MAE: %.2f | R²: %.4f | MAPE: %.2f %%\n', ...
        modelName, rmse, mae, r2, mape);
end

%% === Plot 1: Actual vs Predicted (First 100 Samples) ===
figure;

```

```

plot(yTest(1:100),'k','LineWidth',1.5); hold on;

colors = lines(numModels);

for i = 1:numModels
    plot(yPreds(1:100,i),'--','Color',colors(i,:), 'LineWidth',1.2);
end

legend(['Actual', models(:,1)], 'Location','northwest');
xlabel('Sample Index');
ylabel('AC Power (W)');
title('Grid-Connected Without shading PV: Actual vs Predicted AC Power');
grid on;

%% === Plot 2: Model Performance Comparison ===

figure;

metricNames = {'RMSE','MAE','R2 Score','MAPE'};

for j = 1:4
    subplot(2,2,j);
    bar(metrics(:,j));
    set(gca,'XTickLabel',models(:,1),'XTickLabelRotation',30);
    ylabel(metricNames{j});
    title(metricNames{j});
    grid on;
end

```


GRID CONNECTED WITH SHADING

```
%% ===== GRID CONNECTED PV =====
```

```
clc; clear; close all;
```

```
%% === Load Dataset (GRID CONNECTED) ===
```

```
genData = readtable("C:\Users\gudit\Downloads\HOURLY DATA\GRID CONNECTED WITH  
SHADING_Project_VCO_HourlyRes_2.CSV");
```

```
disp(genData)
```

```
genData.date = datetime(genData.date,'InputFormat','dd-MM-yyyy HH:mm:ss');
```

```
% Valid operating condition
```

```
genData = genData(genData.GlobInc > 0 & genData.E_Grid > 0, :);
```

```
disp(genData)
```

```
%% === Features & Target ===
```

```
X = table2array(genData(:, ...  
    {'GlobHor','WindVel','TArray','T_Amb','GlobInc'}));
```

```
y = genData.E_Grid; % Grid injected AC power
```

```
%% === STANDARDIZATION (VERY IMPORTANT) ===
```

```
X = normalize(X); % <-- FIX FOR WARNING
```

```
%% === Train-Test Split ===
```

```
cv = cvpartition(height(genData),'HoldOut',0.2);
```

```
XTrain = X(training(cv),:); yTrain = y(training(cv));
```

```
XTest = X(test(cv),:); yTest = y(test(cv));
```

```
%% === ML Models ===
```

```
models = {
```

```
    'Regression Tree', @() fitrtree(XTrain,yTrain);
```

```

'Random Forest', @() fitrensemble(XTrain,yTrain,'Method','Bag');
'SVR', @() fitrsvm(XTrain,yTrain,'Standardize',true);
'Gaussian Process', @() fitrgp(XTrain,yTrain,'Standardize',true);
'Neural Network', @() fitrnet(XTrain,yTrain);
};

numModels = size(models,1);
metrics = zeros(numModels,4);
yPreds = zeros(length(yTest),numModels);

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for i = 1:numModels
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    fprintf('\nTraining Model: %s\n', modelName);

    model = models{i,2}();
    yPred = predict(model,XTest);

    rmse = sqrt(mean((yTest-yPred).^2));
    mae = mean(abs(yTest-yPred));
    r2 = 1 - sum((yTest-yPred).^2)/sum((yTest-mean(yTest)).^2);
    mape = mean(abs((yTest-yPred)./yTest))*100;

    metrics(i,:) = [rmse mae r2 mape];
    yPreds(:,i) = yPred;

    fprintf('%s → RMSE: %.2f | MAE: %.2f | R²: %.4f | MAPE: %.2f %%\n', ...
        modelName, rmse, mae, r2, mape);
end

%% === Plot 1: Actual vs Predicted (First 100 Samples) ===
figure;

```

```

plot(yTest(1:100),'k','LineWidth',1.5); hold on;

colors = lines(numModels);

for i = 1:numModels
    plot(yPreds(1:100,i),'--','Color',colors(i,:), 'LineWidth',1.2);
end

legend(['Actual', models(:,1)], 'Location','northwest');
xlabel('Sample Index');
ylabel('AC Power (W)');
title('Grid-Connected with shading PV: Actual vs Predicted AC Power');
grid on;

%% === Plot 2: Model Performance Comparison ===

figure;

metricNames = {'RMSE','MAE','R2 Score','MAPE'};

for j = 1:4
    subplot(2,2,j);
    bar(metrics(:,j));
    set(gca,'XTickLabel',models(:,1),'XTickLabelRotation',30);
    ylabel(metricNames{j});
    title(metricNames{j});
    grid on;
end

```

STANDALONE WITHOUT SHADING

```
%% ===== STANDALONE PV =====

clc; clear; close all;

%% === Load Dataset (STANDALONE) ===

genData = readtable("C:\Users\gudit\Downloads\HOURLY DATA\standalone without shading 24-03-2026_Project_VCO_HourlyRes_1.CSV");

genData.date = datetime(genData.date,'InputFormat','dd-MM-yyyy HH:mm:ss');

% Standalone condition: PV available

genData = genData(genData.GlobInc > 0, :);

disp(genData)

%% === Features & Target ===

X = table2array(genData(:, ...

    {'GlobHor','WindVel','TArray','T_Amb','GlobInc'}));

y = genData.E_solar; % Power supplied to load

%% === STANDARDIZATION (VERY IMPORTANT) ===

X = normalize(X);

%% === Train-Test Split (80% / 20%) ===

cv = cvpartition(height(genData),'HoldOut',0.2);

XTrain = X(training(cv),:); yTrain = y(training(cv));

XTest = X(test(cv),:); yTest = y(test(cv));

%% === ML Models ===

models = {
```

```

'Regression Tree', @() fitrtree(XTrain,yTrain);
'Random Forest', @() fitrensemble(XTrain,yTrain,'Method','Bag');
'SVR', @() fitrsvm(XTrain,yTrain,'Standardize',true);
'Gaussian Process', @() fitrgp(XTrain,yTrain,'Standardize',true);
'Neural Network', @() fitrnet(XTrain,yTrain);
};

numModels = size(models,1);
metrics = zeros(numModels,4);
yPreds = zeros(length(yTest),numModels);

%% === Train & Evaluate ===
for i = 1:numModels
    modelName = models{i,1};
    fprintf('\nTraining Model: %s\n', modelName);

    model = models{i,2}();
    yPred = predict(model,XTest);

    rmse = sqrt(mean((yTest-yPred).^2));
    mae = mean(abs(yTest-yPred));
    r2 = 1 - sum((yTest-yPred).^2)/sum((yTest-mean(yTest)).^2);

    % ---- SAFE MAPE (Standalone Fix) ----
    mape = mean(abs((yTest - yPred) ./ max(yTest,1))) * 100;

    metrics(i,:) = [rmse mae r2 mape];
    yPreds(:,i) = yPred;

    fprintf('%s → RMSE: %.2f | MAE: %.2f | R²: %.4f | MAPE: %.2f %%\n', ...
        modelName, rmse, mae, r2, mape);

```

end

%% === Plot 1: Actual vs Predicted (Test Set) ===

N = min(100,length(yTest));

figure;

plot(yTest(1:N),'k','LineWidth',1.5); hold on;

colors = lines(numModels);

for i = 1:numModels

plot(yPreds(1:N,i),'--','Color',colors(i,:),'LineWidth',1.2);

end

legend(['Actual', models(:,1)], 'Location','northwest');

xlabel('Sample Index');

ylabel('Load Power (W)');

title('Standalone PV: Actual vs Predicted Load Power');

grid on;

%% === Plot 2: Model Performance Comparison ===

figure;

metricNames = {'RMSE','MAE','R2 Score','MAPE'};

for j = 1:4

subplot(2,2,j);

bar(metrics(:,j));

set(gca,'XTickLabel',models(:,1),'XTickLabelRotation',30);

ylabel(metricNames{j});

title(metricNames{j});

grid on;

end

STANDALONE WITH SHADING

```
%% ===== STANDALONE PV =====

clc; clear; close all;

%% === Load Dataset (STANDALONE) ===

genData = readtable("C:\Users\gudit\Downloads\HOURLY DATA\standalonw with shading 24-02-2026_Project_VCO_HourlyRes_1.CSV");

genData.date = datetime(genData.date,'InputFormat','dd-MM-yyyy HH:mm:ss');

% Standalone condition: PV available

genData = genData(genData.GlobInc > 0, :);

disp(genData)

%% === Features & Target ===

X = table2array(genData(:, ...
    {'GlobHor','WindVel','TArray','T_Amb','GlobInc'}));

y = genData.E_solar; % Power supplied to load

%% === STANDARDIZATION (VERY IMPORTANT) ===

X = normalize(X);

%% === Train-Test Split (80% / 20%) ===

cv = cvpartition(height(genData),'HoldOut',0.2);

XTrain = X(training(cv),:); yTrain = y(training(cv));

XTest = X(test(cv),:); yTest = y(test(cv));

%% === ML Models ===

models = {
```

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'Regression Tree', @() fitrtree(XTrain,yTrain);
'Random Forest', @() fitrensemble(XTrain,yTrain,'Method','Bag');
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};

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for i = 1:numModels
    modelName = models{i,1};
    fprintf('\nTraining Model: %s\n', modelName);

    model = models{i,2}();
    yPred = predict(model,XTest);

    rmse = sqrt(mean((yTest-yPred).^2));
    mae = mean(abs(yTest-yPred));
    r2 = 1 - sum((yTest-yPred).^2)/sum((yTest-mean(yTest)).^2);

    % ---- SAFE MAPE (Standalone Fix) ----
    mape = mean(abs((yTest - yPred) ./ max(yTest,1))) * 100;

    metrics(i,:) = [rmse mae r2 mape];
    yPreds(:,i) = yPred;

    fprintf('%s → RMSE: %.2f | MAE: %.2f | R²: %.4f | MAPE: %.2f %%\n', ...
        modelName, rmse, mae, r2, mape);

```


end

%% === Plot 1: Actual vs Predicted (Test Set) ===

N = min(100,length(yTest));

figure;

plot(yTest(1:N),'k','LineWidth',1.5); hold on;

colors = lines(numModels);

for i = 1:numModels

plot(yPreds(1:N,i),'--','Color',colors(i,:),'LineWidth',1.2);

end

legend(['Actual', models(:,1)], 'Location','northwest');

xlabel('Sample Index');

ylabel('Load Power (W)');

title('Standalone PV: Actual vs Predicted Load Power');

grid on;

%% === Plot 2: Model Performance Comparison ===

figure;

metricNames = {'RMSE','MAE','R2 Score','MAPE'};

for j = 1:4

subplot(2,2,j);

bar(metrics(:,j));

set(gca,'XTickLabel',models(:,1),'XTickLabelRotation',30);

ylabel(metricNames{j});

title(metricNames{j});

grid on;

end



**SCAN THE QR CODE FOR THE
COMPLATE PROJECT DETAILS**

EEE- BATCH-7

ORIGINALITY REPORT

21 %

SIMILARITY INDEX

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INTERNET SOURCES

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PUBLICATIONS

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STUDENT PAPERS

PRIMARY SOURCES

1

Submitted to Engineers Australia

Student Paper

2 %

2

Submitted to Cranfield University

Student Paper

1 %

3

Proceedings of ISES World Congress 2007 (Vol I – Vol V), 2009.

Publication

<1 %

4

Submitted to Engineering Institute of Technology

Student Paper

<1 %

5

svcetedu.org

Internet Source

<1 %

6

AS Mathur, Vikas Kumar Rajput, Amita, BP Singh. "Techno-Economic Feasibility and Sensitivity Analysis of a Stand-Alone Solar Photovoltaic Power Plant with Battery Energy Storage Using PVSyst", Next Research, 2025

Publication

<1 %

7

Green Energy and Technology, 2015.

Publication

<1 %